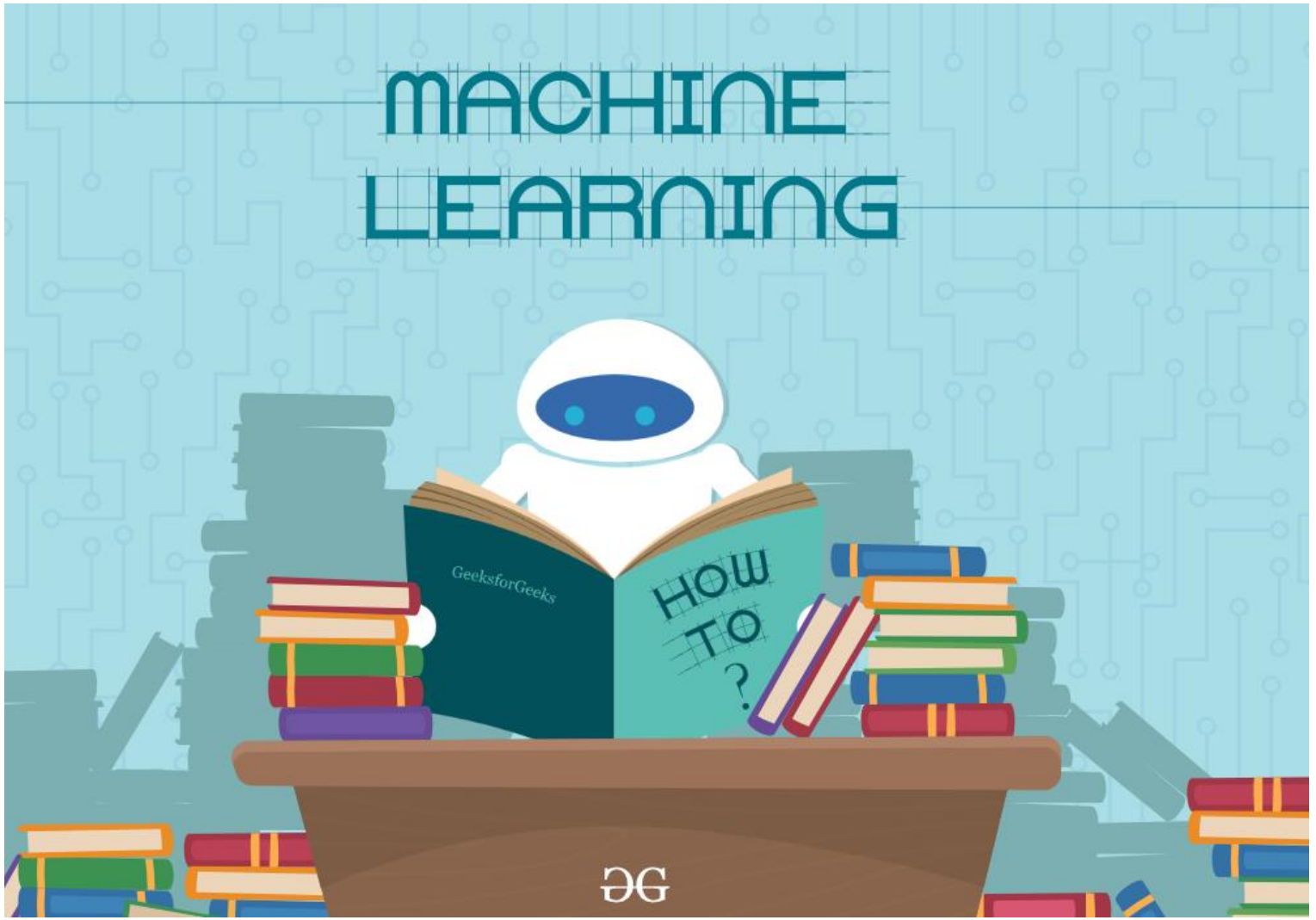
AI and Machine Learning

- AI Xin
- aixin@nus.edu.sg
- School of Computing
- National University of Singapore



Learning Objectives

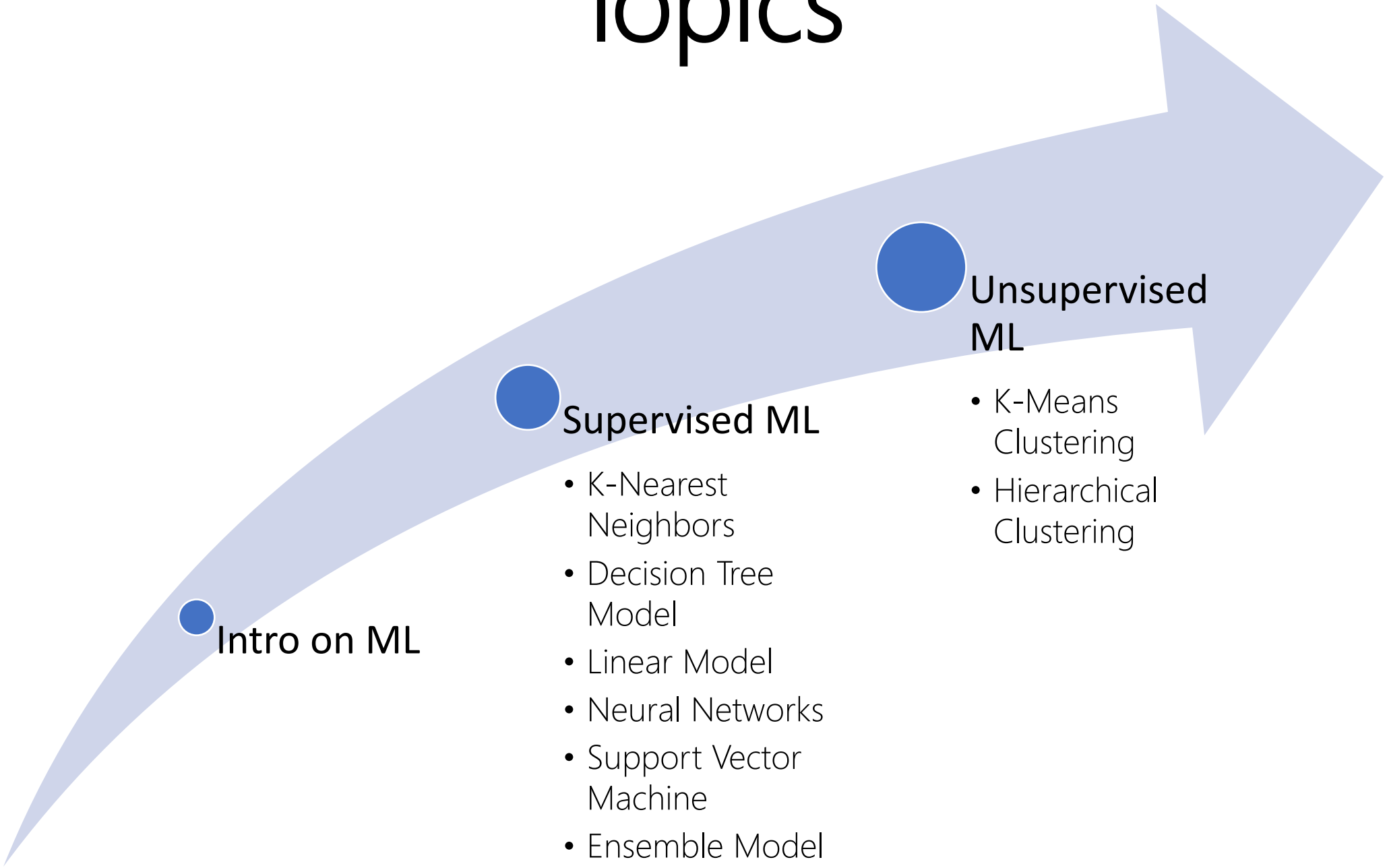
Upon completion of this course, you should be able to:

- Understand the machine learning process
- Collect data, prepare data, explore and understand the data
- Select, Build, Train and Evaluate the Machine Learning Models
- Tuning model hyperparameters to achieve the best performance
- Apply the machine learning models to solve the real-life problems

Assessment

- Quiz (Individual, 50%)
  - Quiz 1 (10%)
  - Quiz 2 (10%)
  - Quiz 3 (10%)
  - Quiz 4 (20%)
- Project (Group, 50%)
  - Create machine learning solution for:
    - a classification problem or
    - a regression problem

Topics

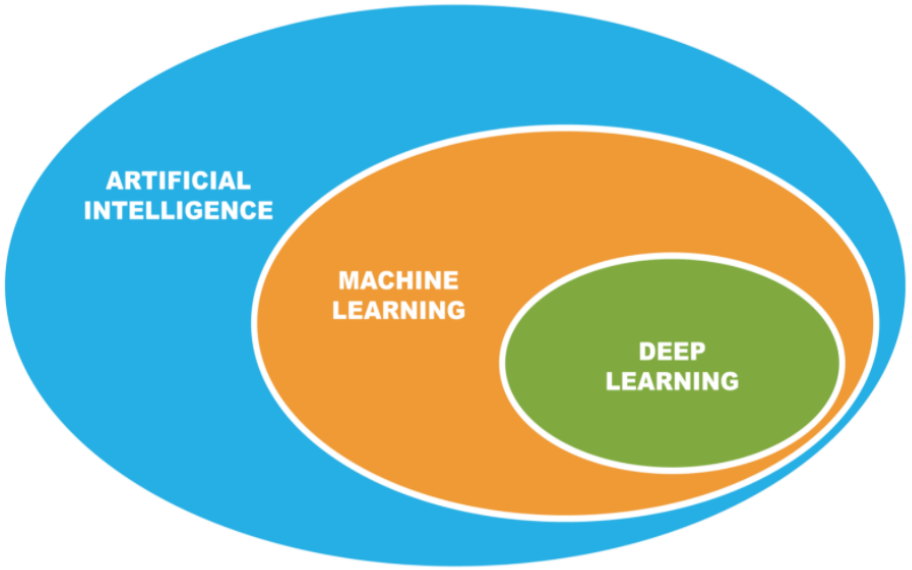

Intro on ML
Supervised ML
• K-Nearest Neighbors
• Decision Tree Model
• Linear Model
• Neural Networks
• Support Vector Machine
• Ensemble Model
Unsupervised ML
• K-Means Clustering
• Hierarchical Clustering

Part I: Introduction on Machine Learning

WHAT IS ARTIFICAL INTELLIGENCE

AI definition: the effort to automate intellectual tasks normally performed by humans

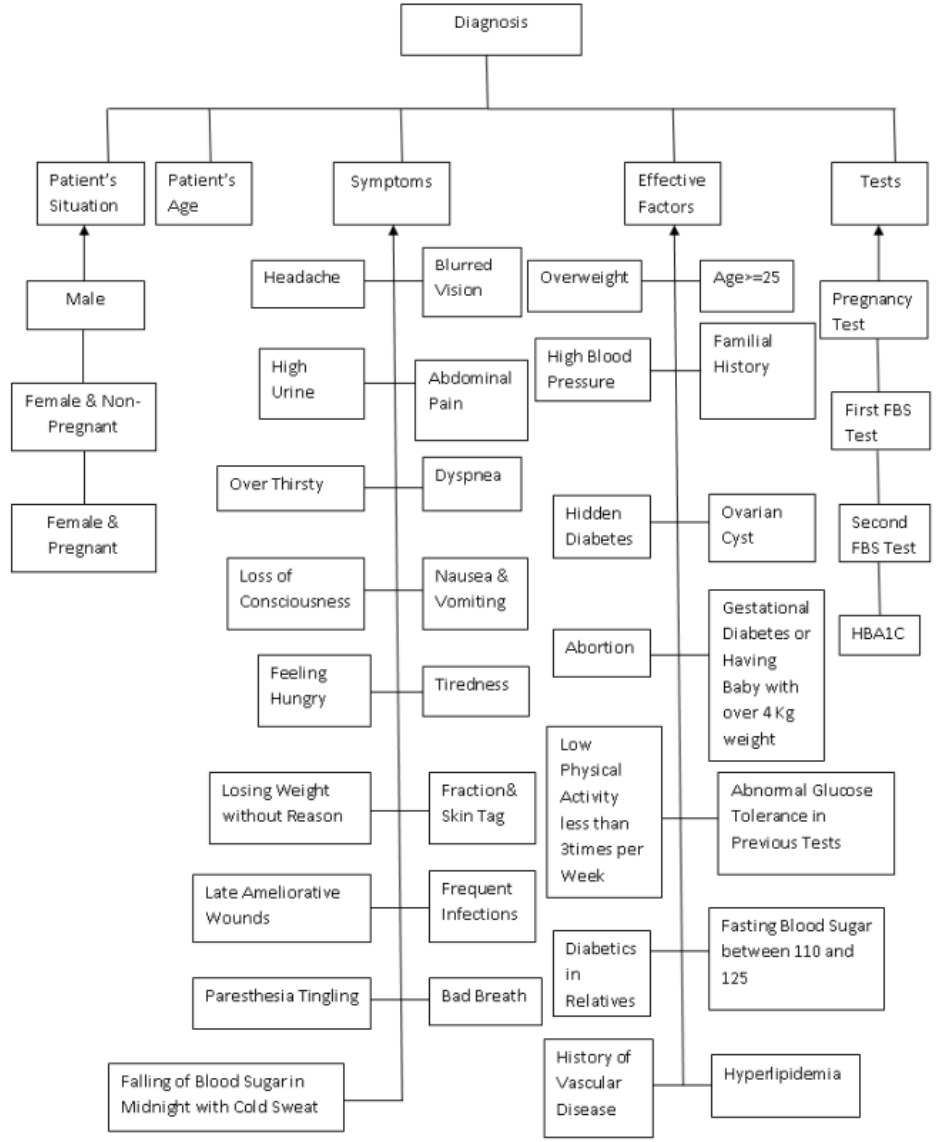
AI is a general field that encompasses machine learning and deep learning, and many other approaches which don't involve any learning.



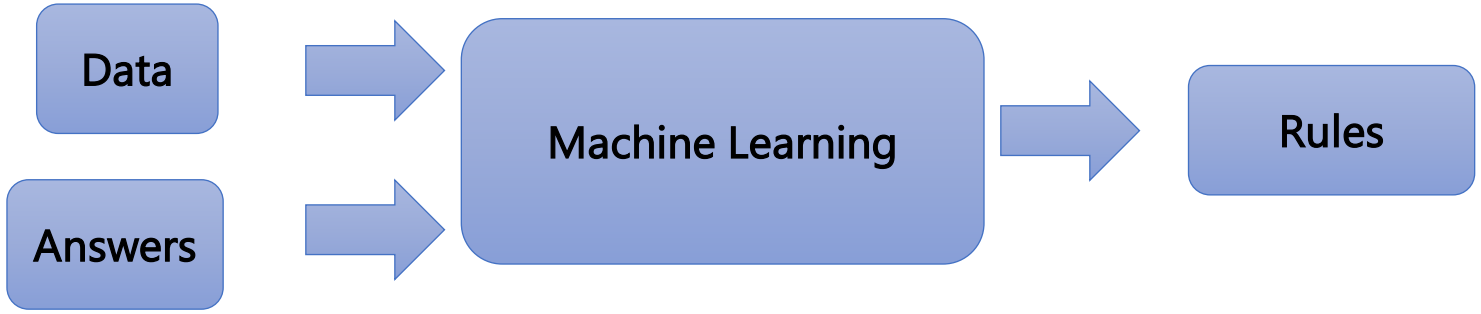
Symbolic AI (1950s- 1980s)



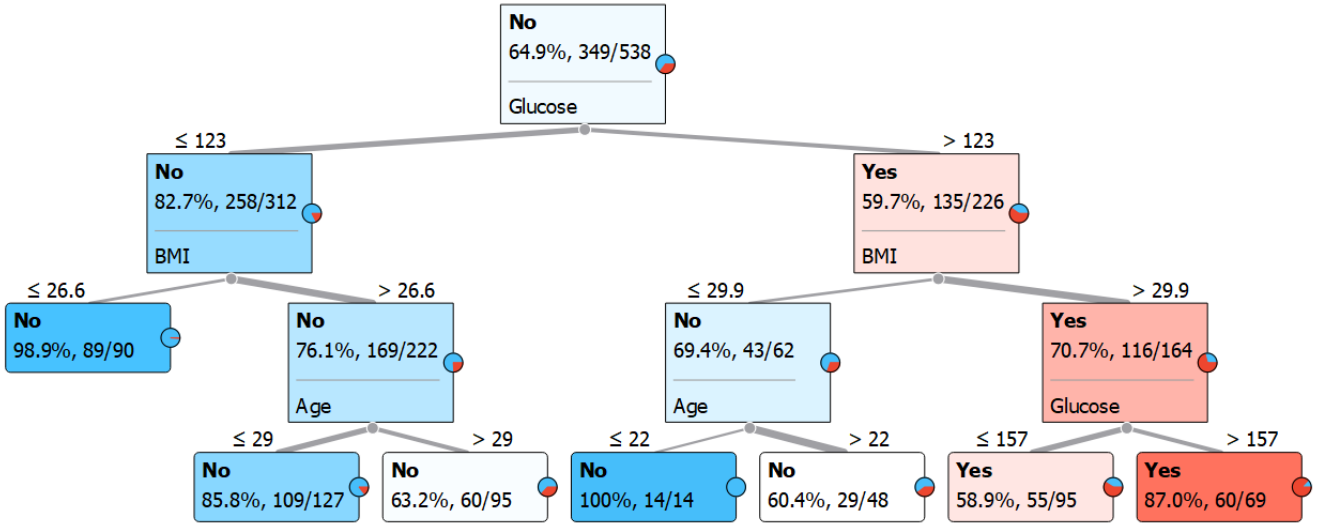




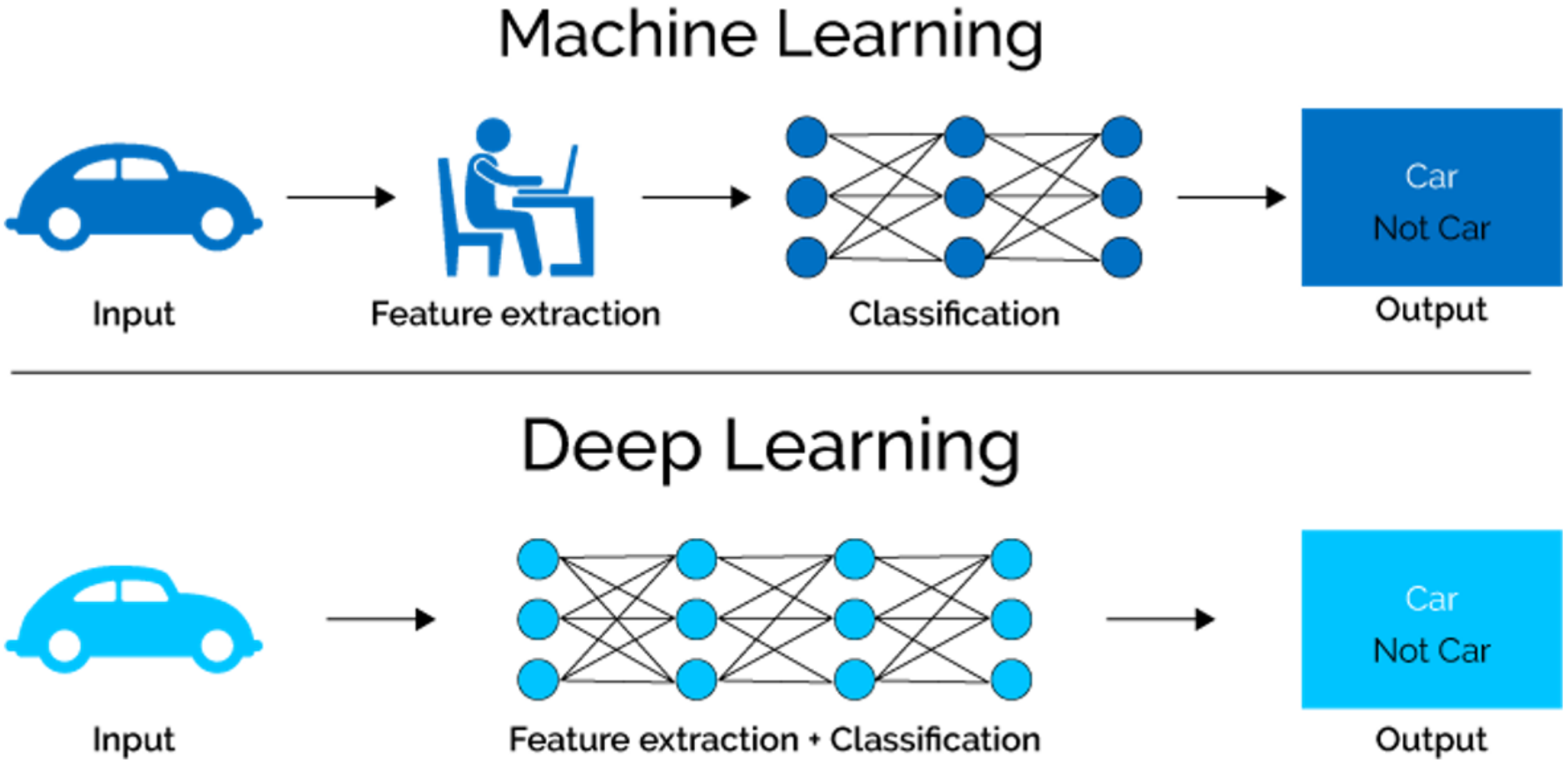
Machine Learning (1990s- now)



	Outcome	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
1	Yes	6	148	72	35	0	33.6	0.627	50
2	No	1	85	66	29	0	26.6	0.351	31
3	Yes	8	183	64	0	0	23.3	0.672	32
4	No	1	89	66	23	94	28.1	0.167	21
5	Yes	0	137	40	35	168	43.1	2.288	33
6	No	5	116	74	0	0	25.6	0.201	30
7	Yes	3	78	50	32	88	31.0		
8	No	10	115	0	0	0	35.3		
9	Yes	2	197	70	45	543	30.5		
10	Yes	8	125	96	0	0	0.0		
11	No	4	110	92	0	0	37.6		
12	Yes	10	168	74	0	0	38.0		
13	No	10	139	80	0	0	27.1		
14	Yes	1	189	60	23	846	30.1		
15	Yes	5	166	72	19	175	25.8		
16	Yes	7	100	0	0	0	30.0		
17	Yes	0	118	84	47	230	45.8		
18	Yes	7	107	74	0	0	29.6		
19	No	1	103	30	38	83	43.3		



Deep Learning (2010s- now)



What is Machine Learning?

What is Machine Learning?

- Machine Learning is the science (and art) of programming computers so they can learn from data.

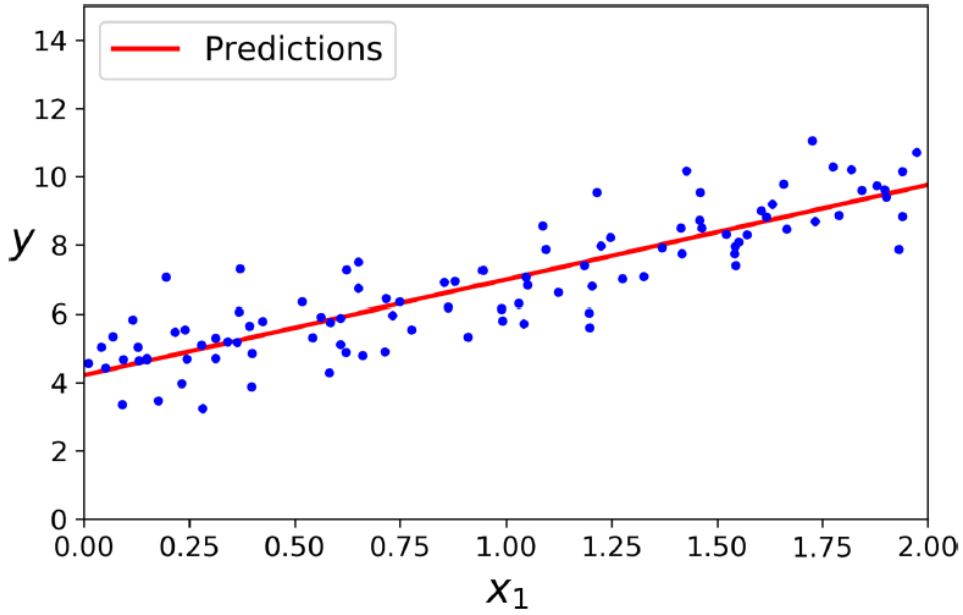
- A slightly more general definition:

  [Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.

  —Arthur Samuel, 1959

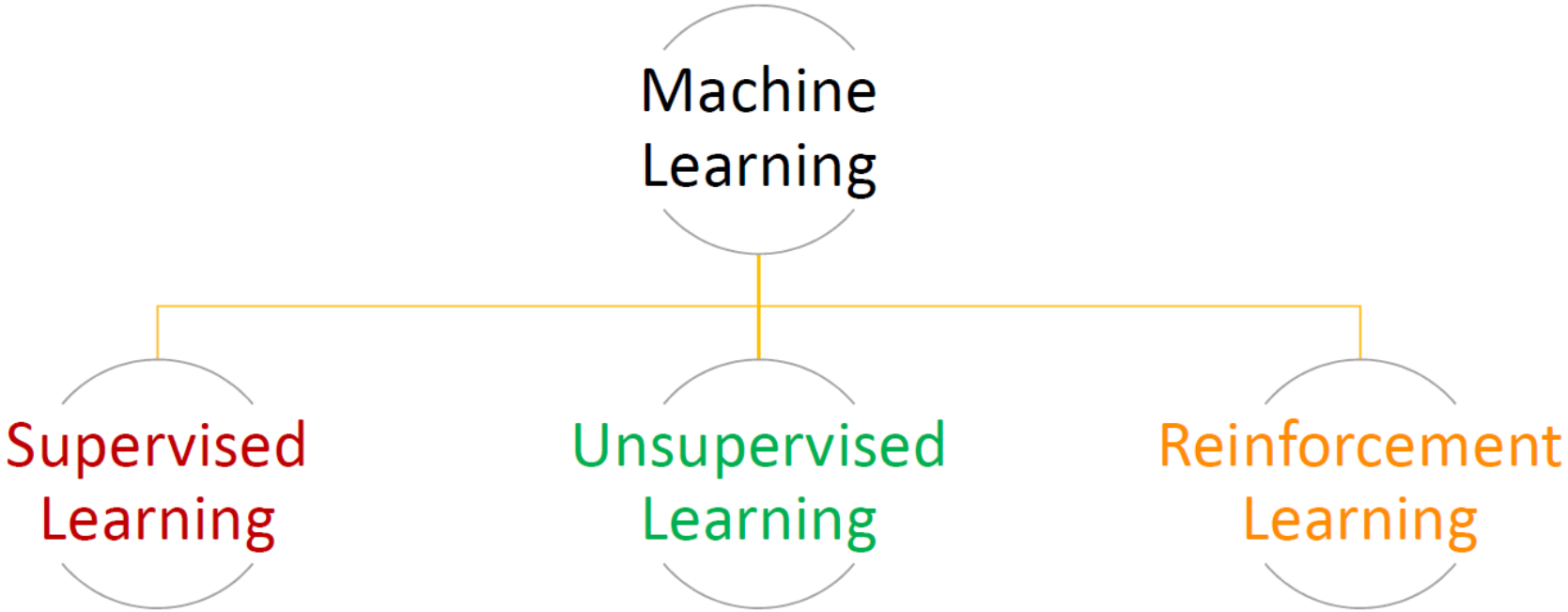
Machine Learning Examples

Linear Regression





Types of Machine Learning



Supervised vs. unsupervised

DATA DEFINITIONS: LABELED AND UNLABELED DATA

We do not know what type of iris flowers they are.

There is no labels.

No Target (labels) **Variables or Features**

?

	sepal length	sepal width	petal length	petal width
45	5.100	3.800	1.900	0.400
46	4.800	3.000	1.400	0.300
47	5.100	3.800	1.600	0.200
48	4.600	3.200	1.400	0.200
49	5.300	3.700	1.500	0.200
50	5.000	3.300	1.400	0.200
51	7.000	3.200	4.700	1.400
52	6.400	3.200	4.500	1.500
53	6.900	3.100	4.900	1.500

We collected the data from a known group of iris flower types.

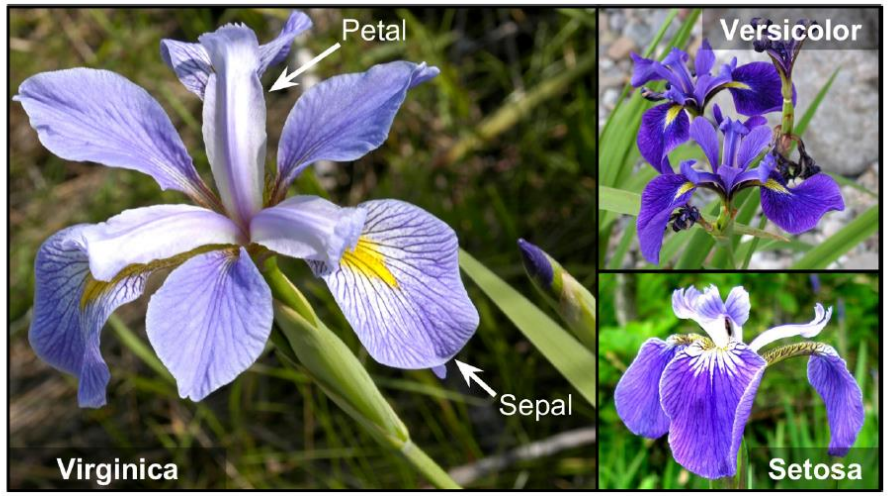
The iris column is the target or labels.

Hence labeled data.

Classification or Prediction techniques are used in these cases.

Target (labels) **Variables or Features**

	iris	sepal length	sepal width	petal length	petal width
43	Iris-setosa	4.400	3.200	1.300	0.200
44	Iris-setosa	5.000	3.500	1.600	0.600
45	Iris-setosa	5.100	3.800	1.900	0.400
46	Iris-setosa	4.800	3.000	1.400	0.300
47	Iris-setosa	5.100	3.800	1.600	0.200
48	Iris-setosa	4.600	3.200	1.400	0.200
49	Iris-setosa	5.300	3.700	1.500	0.200
50	Iris-setosa	5.000	3.300	1.400	0.200
51	Iris-versicolor	7.000	3.200	4.700	1.400
52	Iris-versicolor	6.400	3.200	4.500	1.500
53	Iris-versicolor	6.900	3.100	4.900	1.500
54	Iris-versicolor	5.500	2.300	4.000	1.300
55	Iris-versicolor	6.500	2.800	4.600	1.500



Classification vs. Regression in Supervised Learning

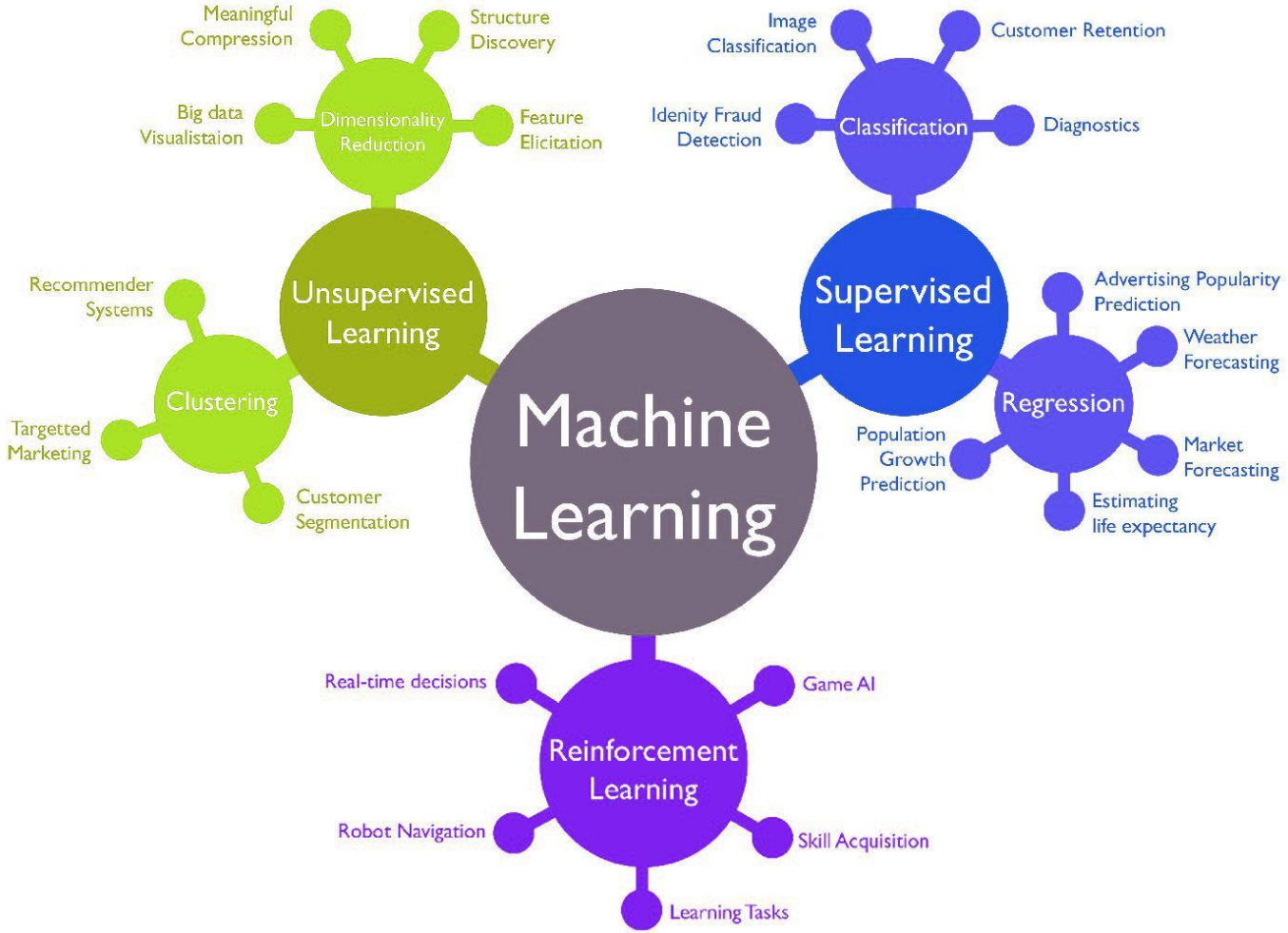
- Regression

month	town	flat_type	block	street_name	storey_range	floorAreaSqm	resale_price
2012-09	CHOA CHU KANG	4 ROOM	119	TECK WHYE LANE	04 TO 06	104	400000
2013-10	JURONG WEST	3 ROOM	510	JURONG WEST ST 52	13 TO 15	74	375000
2013-03	JURONG EAST	5 ROOM	284	TOH GUAN RD	07 TO 09	120	655000

- Classification

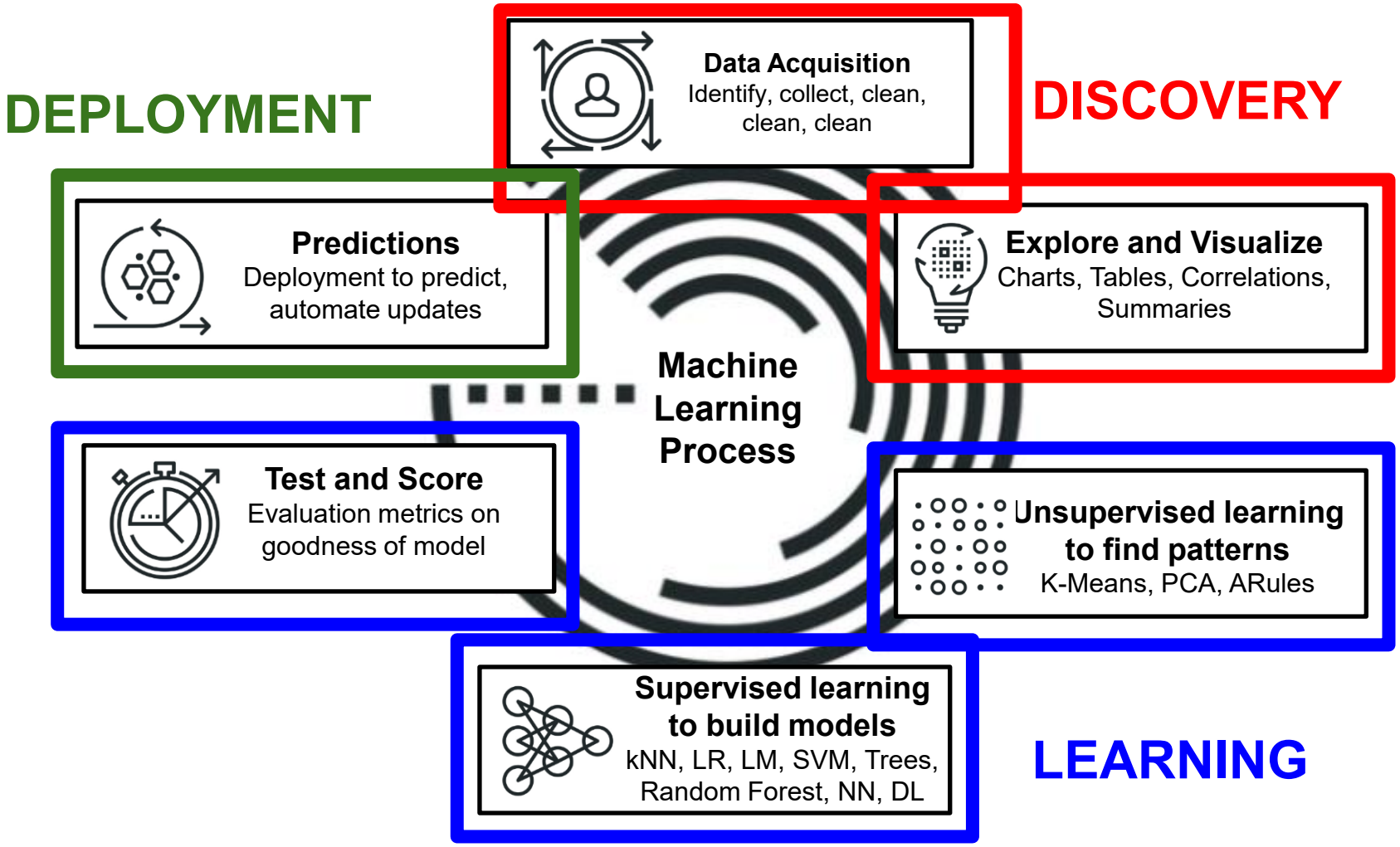
month	town	flat_type	block	street_name	storey_range	floorAreaSqm	resale_price
2012-09	CHOA CHU KANG	4 ROOM	119	TECK WHYE LANE	04 TO 06	104	Medium
2013-10	JURONG WEST	3 ROOM	510	JURONG WEST ST 52	13 TO 15	74	Low
2013-03	JURONG EAST	5 ROOM	284	TOH GUAN RD	07 TO 09	120	High

Machine Learning Approaches



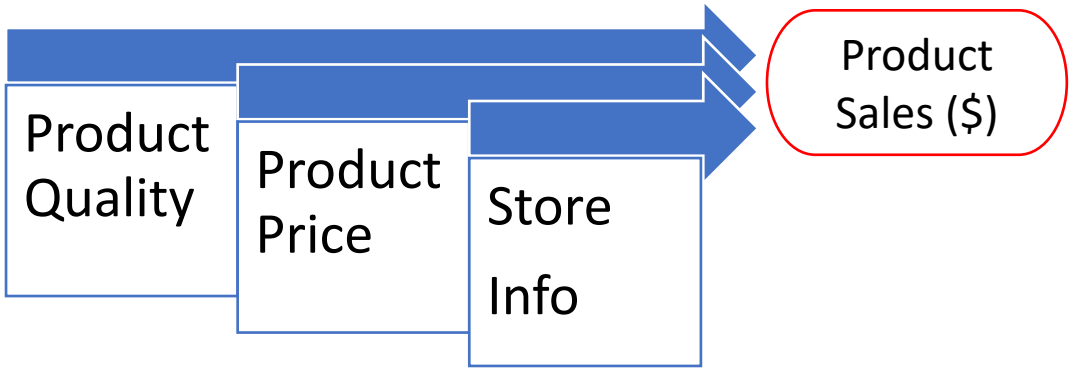
HTTPS://WWW.WORDSTREAM.COM/BLOG/WS/2017/07/28/MACHINE-LEARNING-APPLICATIONS, ABDUL WAHID

Machine Learning Process



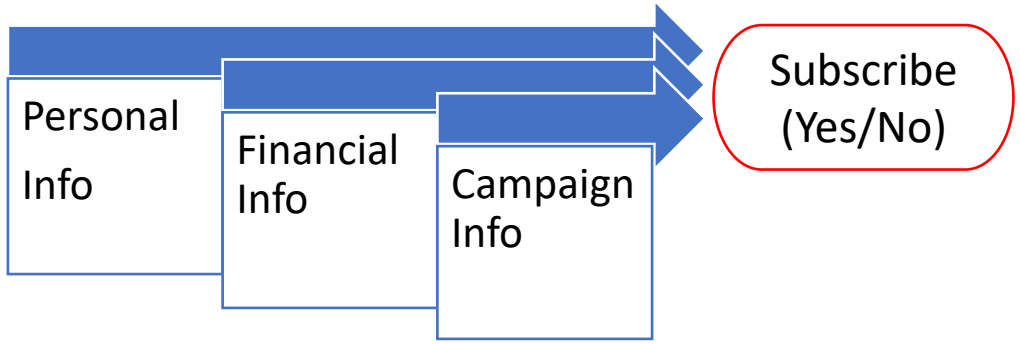
Datasets

Regression



Supermarket Sales Forecast

Classification



Bank Marketing Campaign

Data Dictionary

We will be using the supermarket data for regression task. The data (`supermarket.csv`) have been collected at various supermarket outlets and stores in different cities. The aim is to build a predictive model and predict the sales of each product at a particular outlet. Using this model, supermarket managment team will try to understand the properties of products and outlets which play a key role in increasing sales.

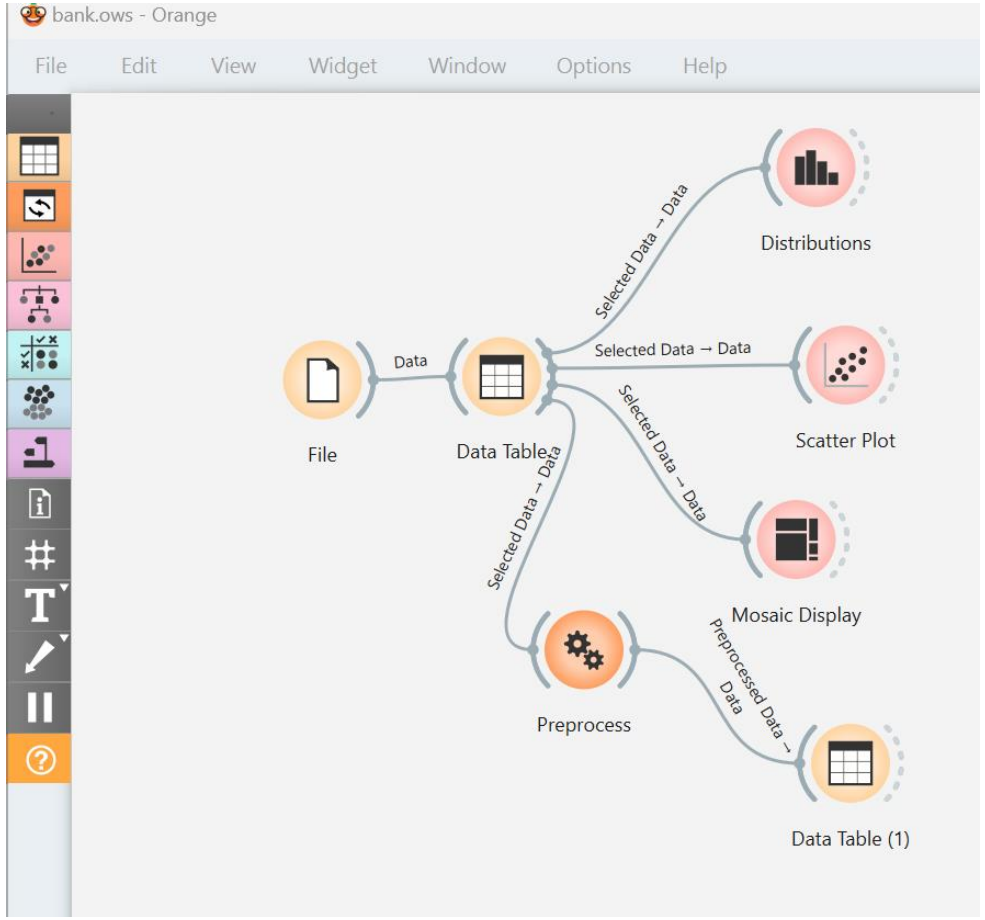
Detailed information (i.e. column description) is provided below.

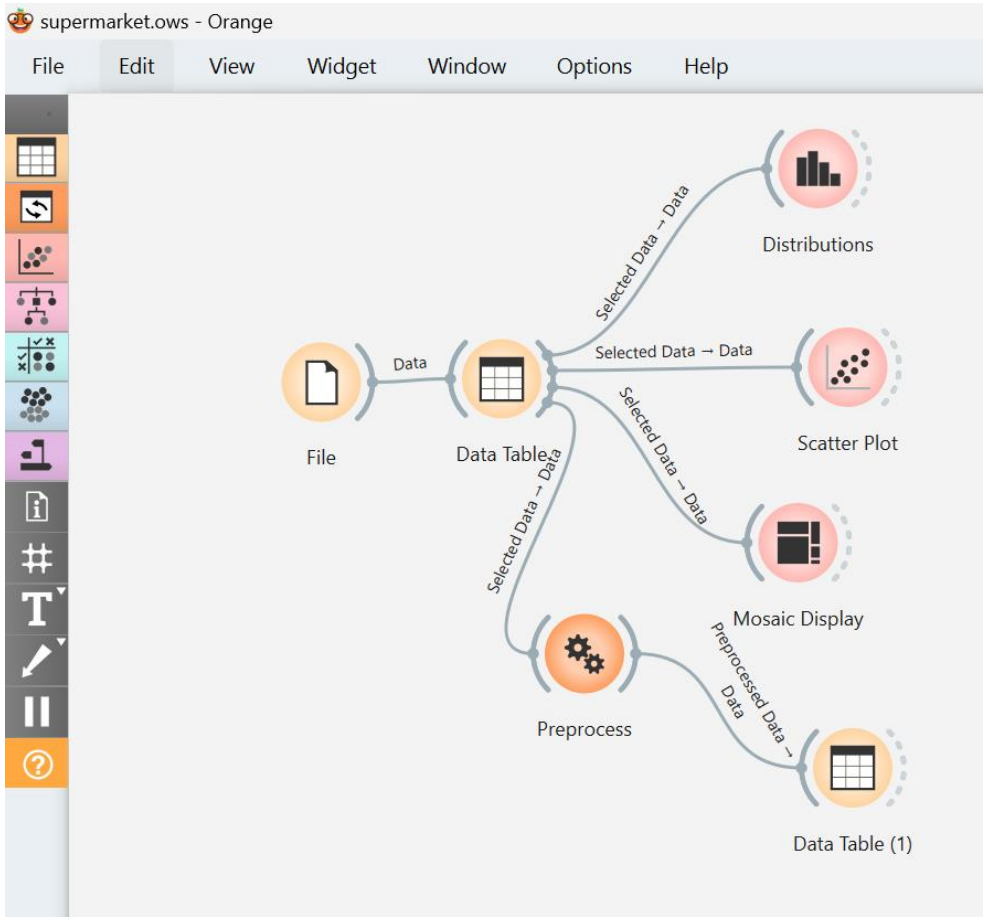
- **Item_Weight:** Weight of product
- **Item_Fat_Content:** Whether the product is low fat or not
- **Item_Visibility:** The % of total display area of all products in a store allocated to the particular product
- **Item_Type:** The category to which the product belongs
- **Item_MRP:** Maximum Retail Price (list price) of the product
- **Outlet_Identifier:** Unique store ID
- **Outlet_Establishment_Year:** The year in which store was established
- **Outlet_Size:** The size of the store in terms of ground area covered
- **Outlet_Location_Type:** The type of city in which the store is located
- **Outlet_Type:** Whether the outlet is just a grocery store or some sort of supermarket
- **Item_Outlet_Sales:** Sales of the product in the particular store. This is the TARGET variable.

We will be using the bank marketing campaign data for classification task. The dataset (`bank.csv`) gives you information about a marketing campaign done by a financial institution. Detailed information (i.e. column description) is provided below. We are trying to predict whether the customer/client will deposit the money into the bank or not?

- **age :** age in years
- **job :** type of job
- **marital :** marital status
- **education :** education background
- **default :** has credit in default?
- **balance :** Balance of the individual
- **housing :** has housing loan?
- **loan :** has personal loan?
- **contact :** contact communication type
- **day :** last contact day of the month
- **month :** last contact month of year
- **duration :** last contact duration, in seconds
- **campaign :** number of contacts performed during this campaign and for this client
- **pdays :** number of days that passed by after the client was last contacted from a previous campaign
- **previous :** number of contacts performed before this campaign and for this client
- **poutcome :** outcome of the previous marketing campaign
- **deposit :** has the client subscribed a term deposit? This is the TARGET variable

LAB: DATA EXPLORATION



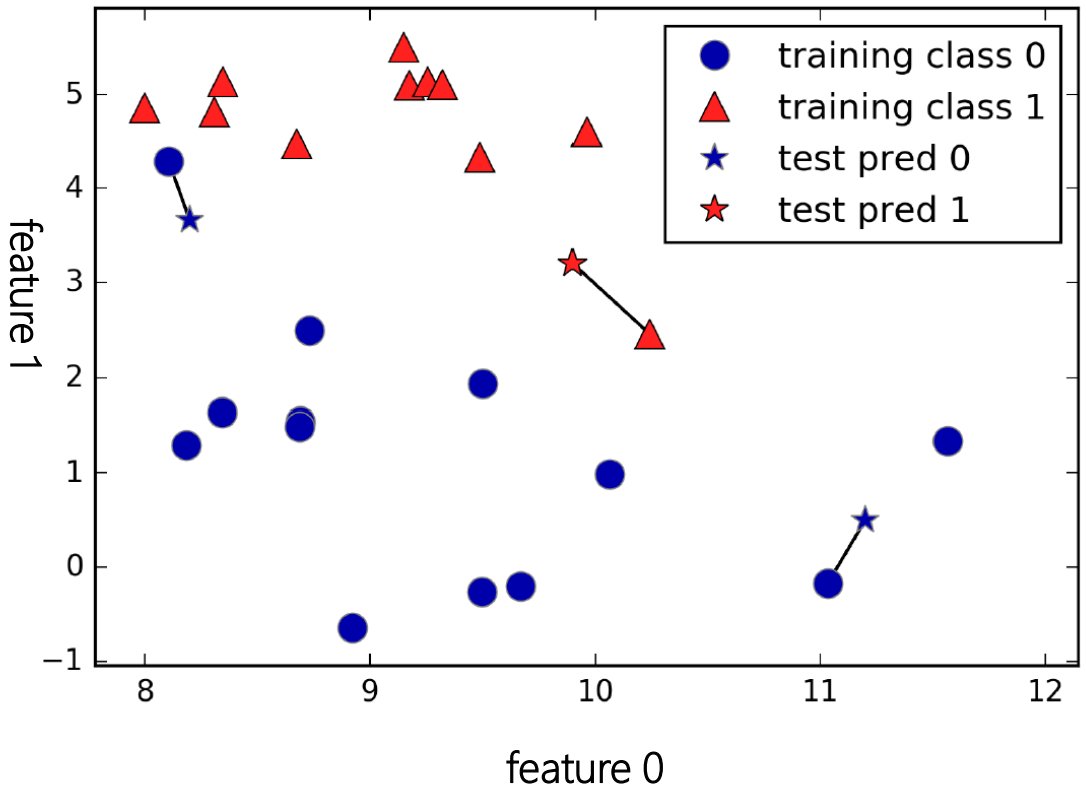


Part II: Supervised Machine Learning

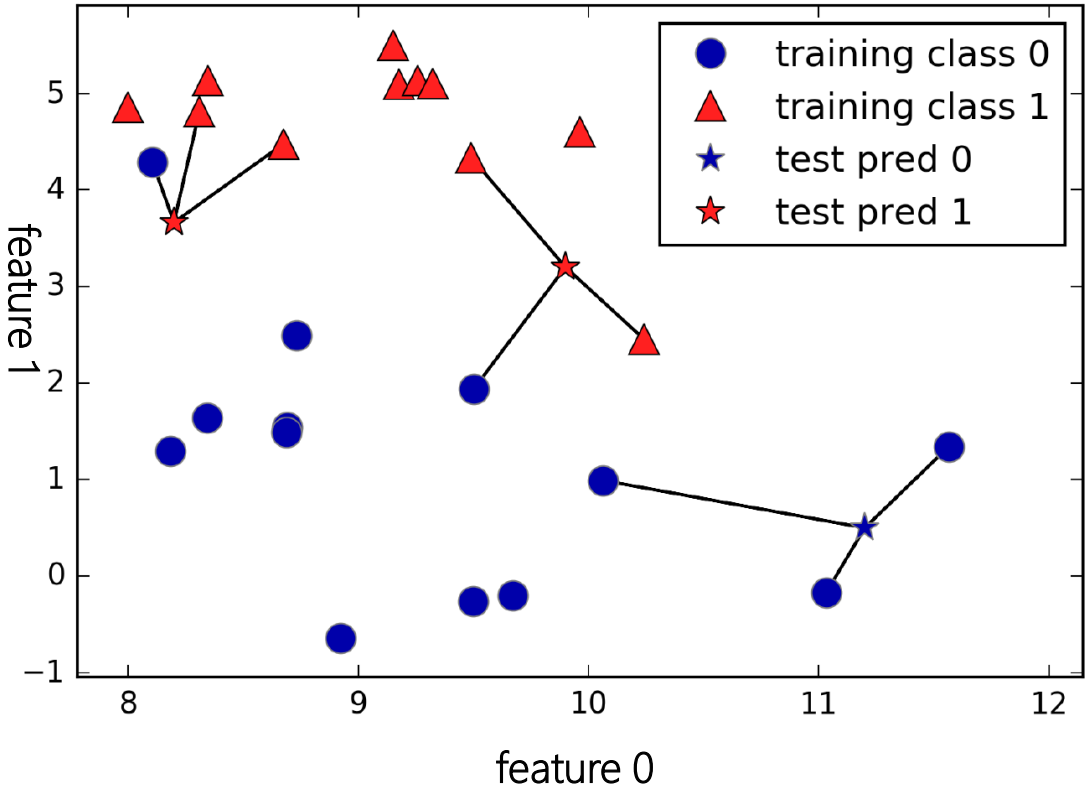
1. K-Nearest Neighbors

Classification

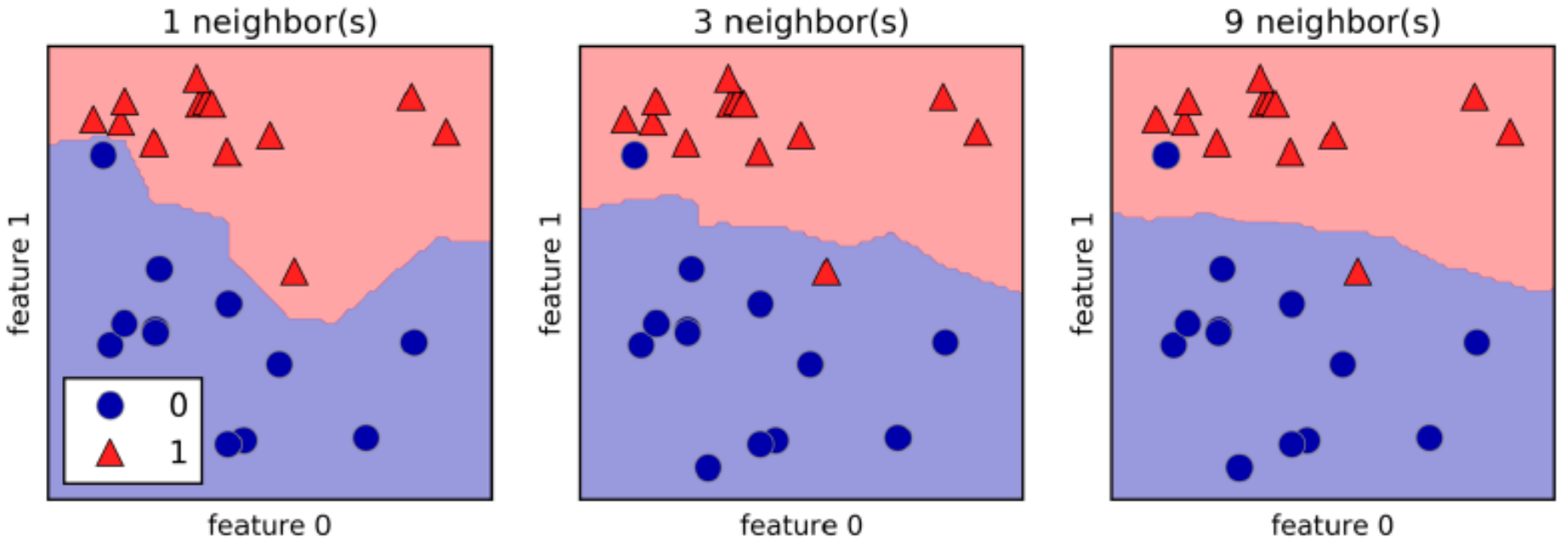
one-nearest-neighbor model



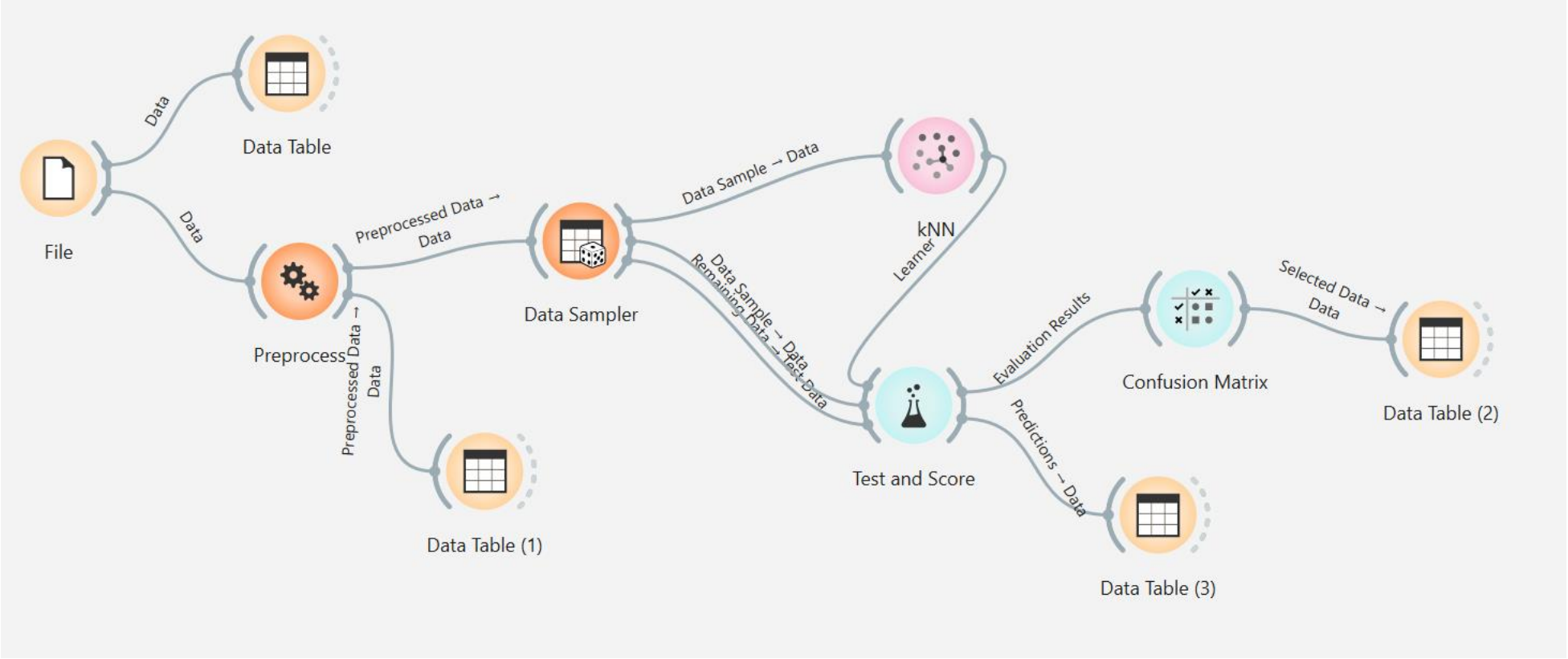
three-nearest-neighbor model



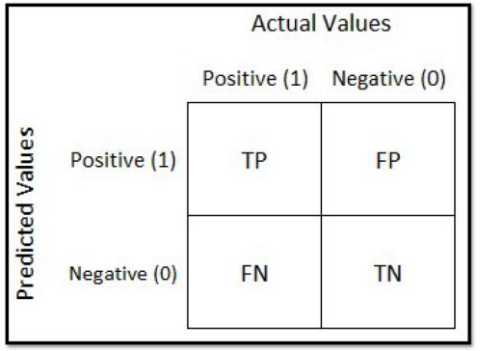
Decision boundaries created by the nearest neighbors model for different values of n_neighbors



LAB: KNN CLASSIFIER



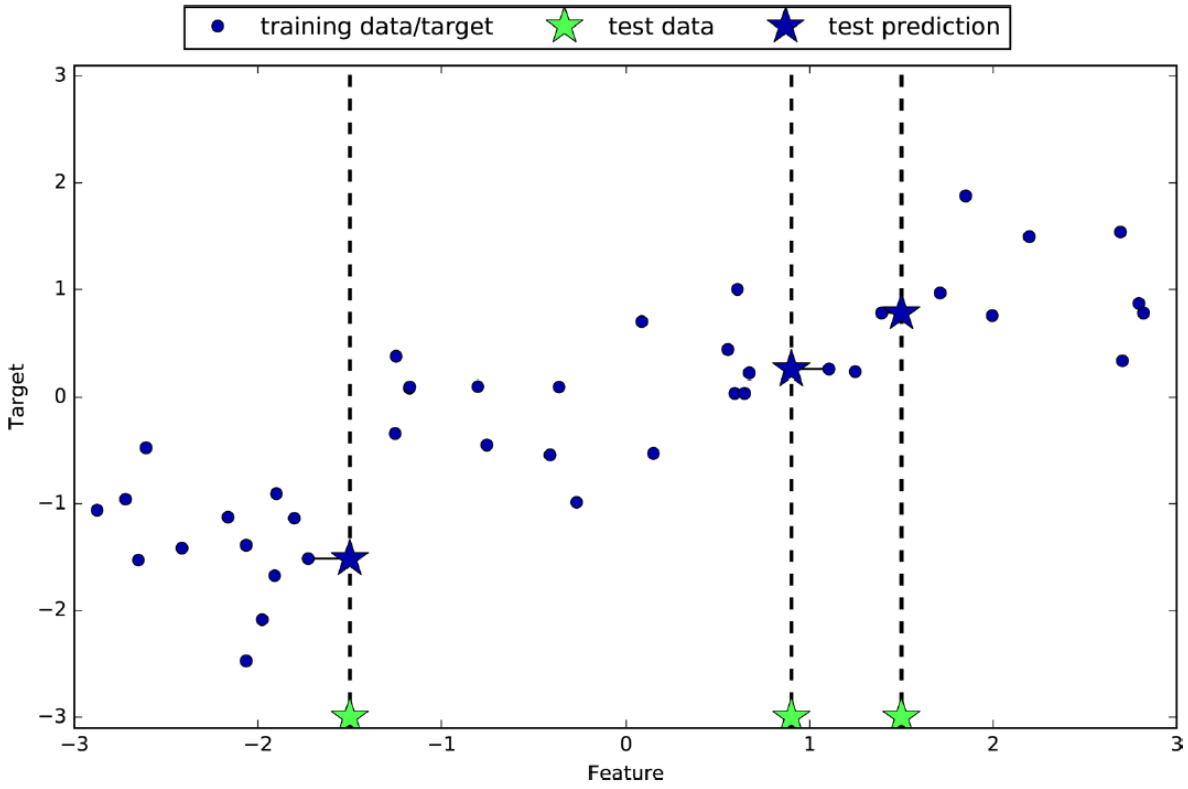
EVALUATE CLASSIFICATION MODEL



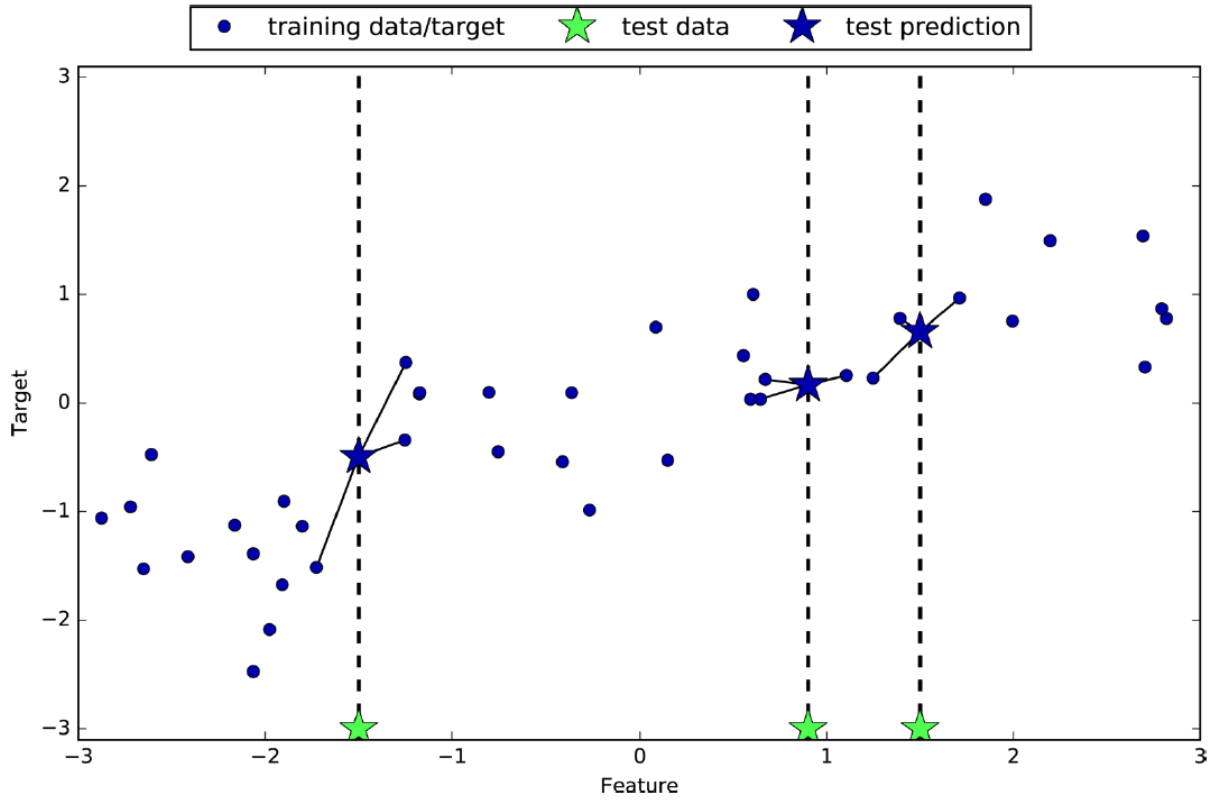
- An Example: Imagine a Machine Learning model is used to classify patients with (P) or without (N) diabetes
- Accuracy = (TP + TN) / (TP + FP + FN +TN)
- Precision = TP / (TP + FP)
  - the ratio of P patients correctly classified by the model out of all P patients classified by the model
- Recall = TP / (TP + FN)
  - the ratio of P patients correctly classified by the model out of all the actual P patients
- F1 score = 2 x (Precision x Recall) / (Precision + Recall)
  - Hamonic mean of precision and recall

Regression

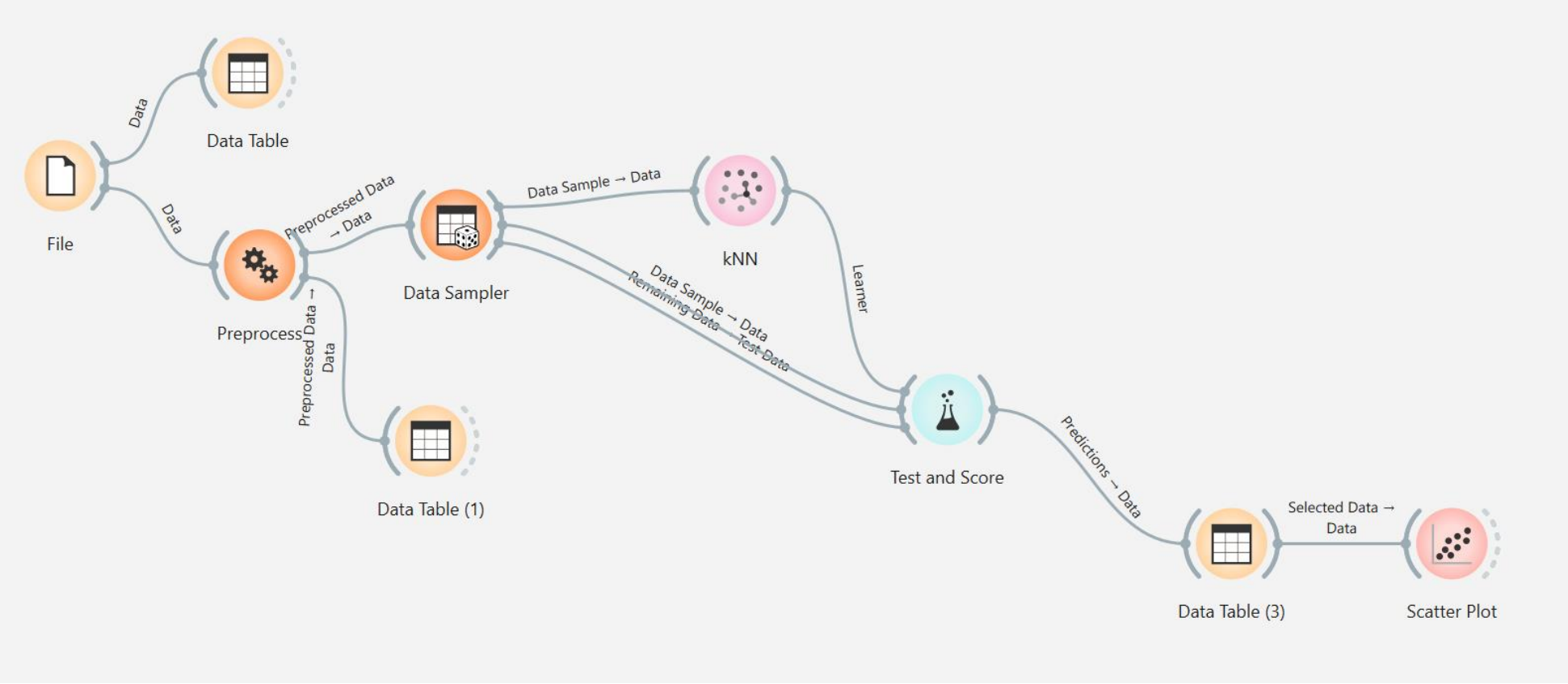
one-nearest-neighbor model



three-nearest-neighbor model



LAB: KNN REGRESSOR

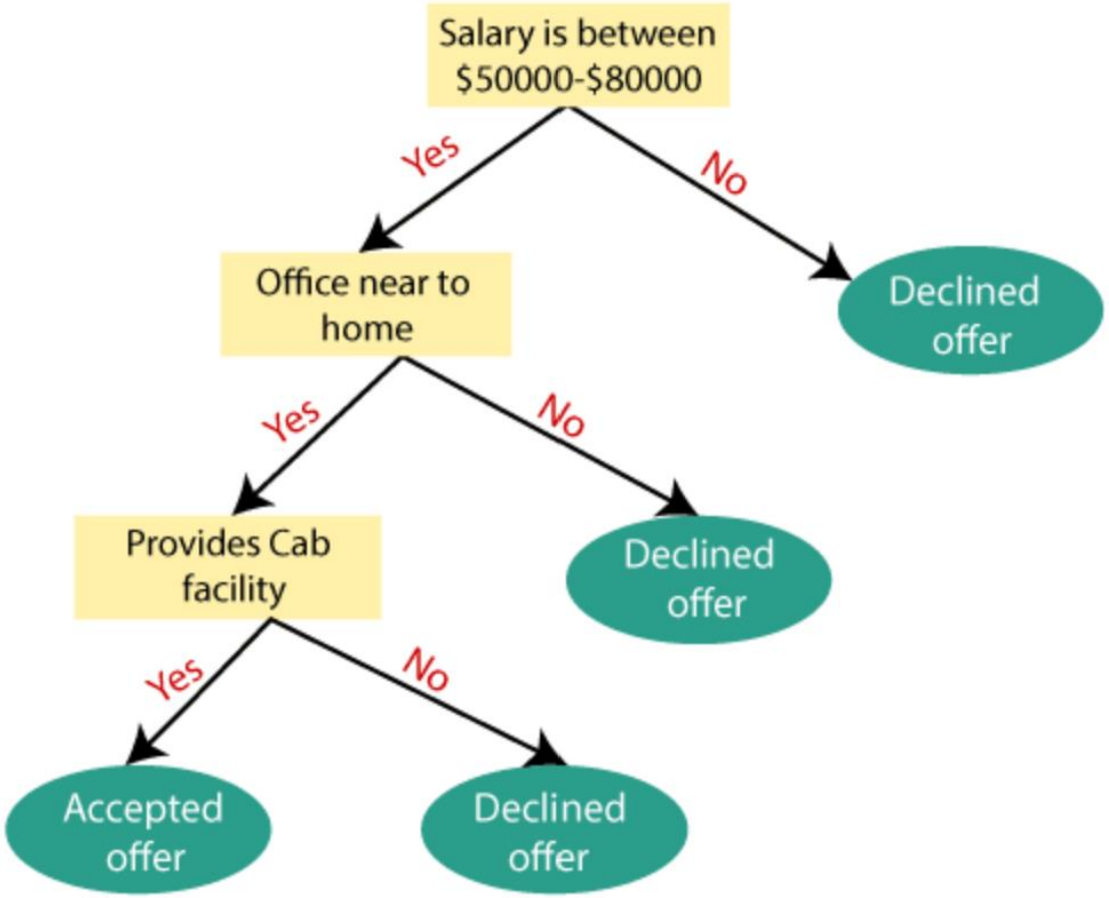


EVALUATE REGRESSION MODEL

- Errors: the lower the better
  - Mean Squared Error (MSE) = $\Sigma(y_i - y'_i)^2 / n$
  - Root Mean Squared Error (RMSE) = $\sqrt{MSE}$
  - Mean Absolute Error (MAE) = $\Sigma |y_i - y'_i| / n$
- R Squared Value (0 to 1): the higher the better
  - the percentage of variance from the samples can be explained through the model

2. Decision Tree Model

Decision Tree Model



- Intuitive and easy to interpret
- Require very little data preparation
- Don’t require feature scaling
- Easily deployed in rule-based system
- Build-in variable selection

Image source: https://www.mygreatlearning.com/blog/decision-tree-algorithm/

CART Algorithm

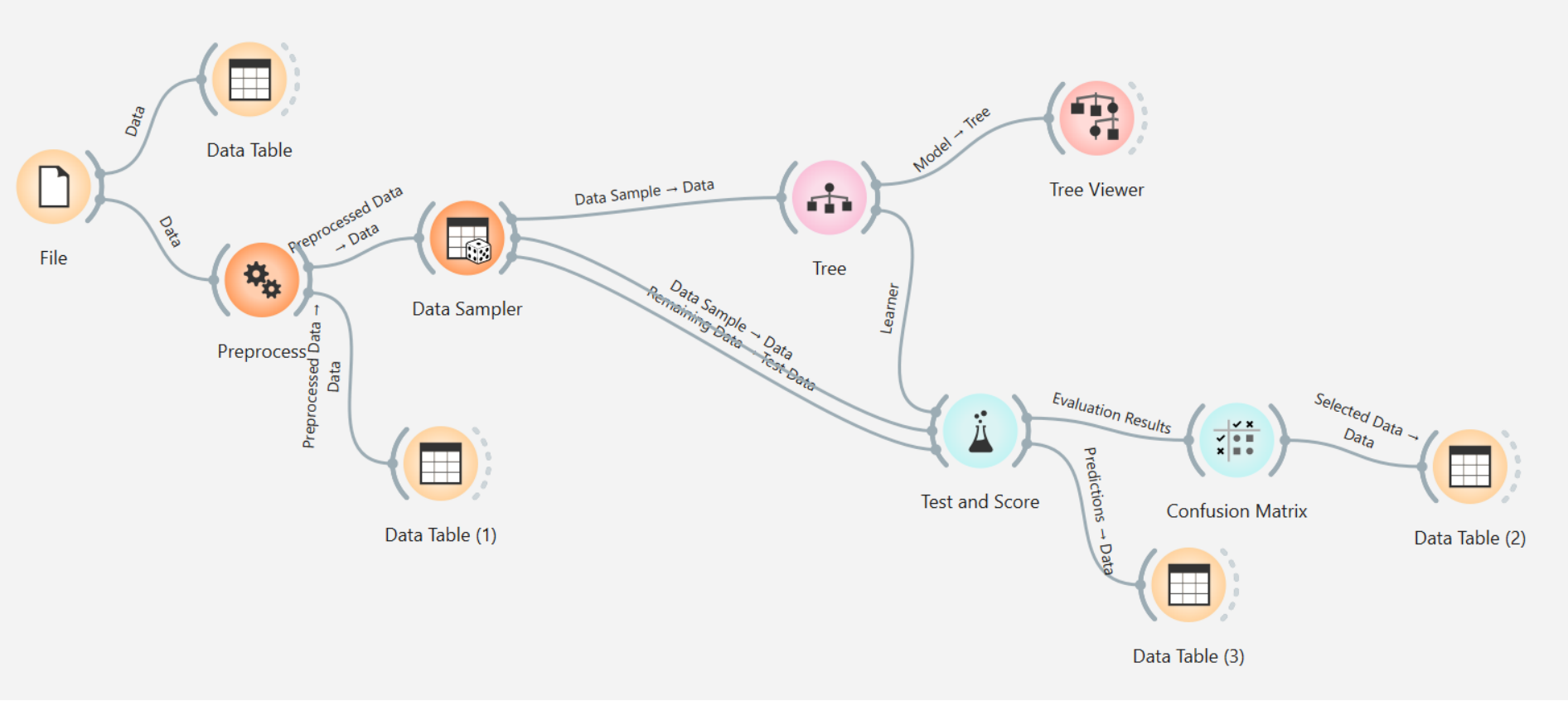
- CART: Classification and Regression Tree

- Split the data into two subgroups to make the decision nodes as pure as possible

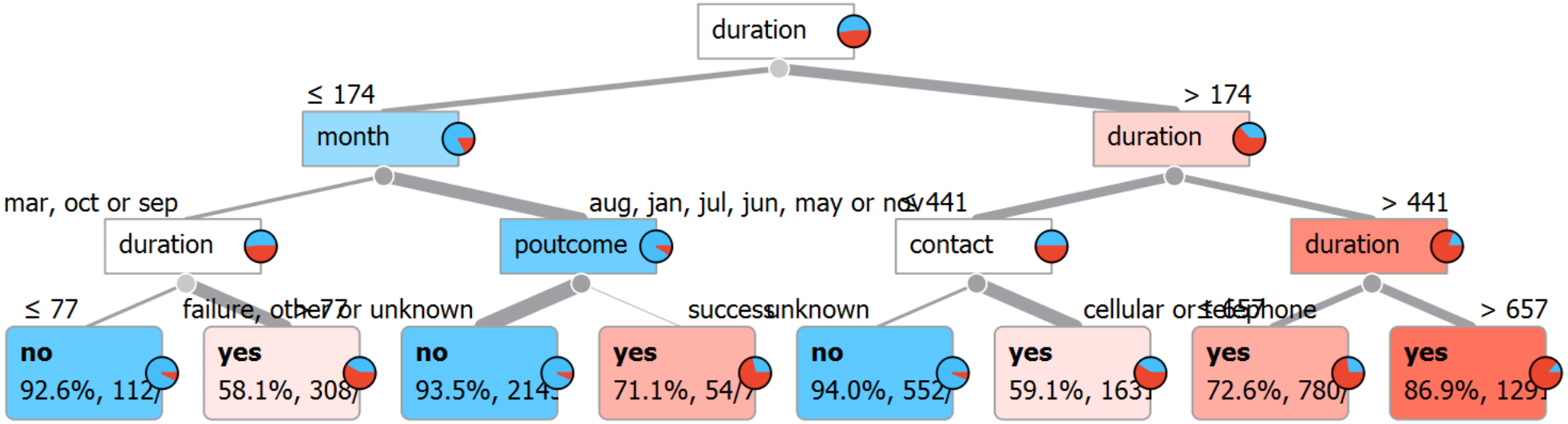
- How to measure the purity of a node?
  - Classification task:
    - e.g. Gini Impurity Index
    - the lower the Gini, the purer the node

  - Regression task:
    - e.g. Mean Squared Error
    - the lower the MSE, the purer the node

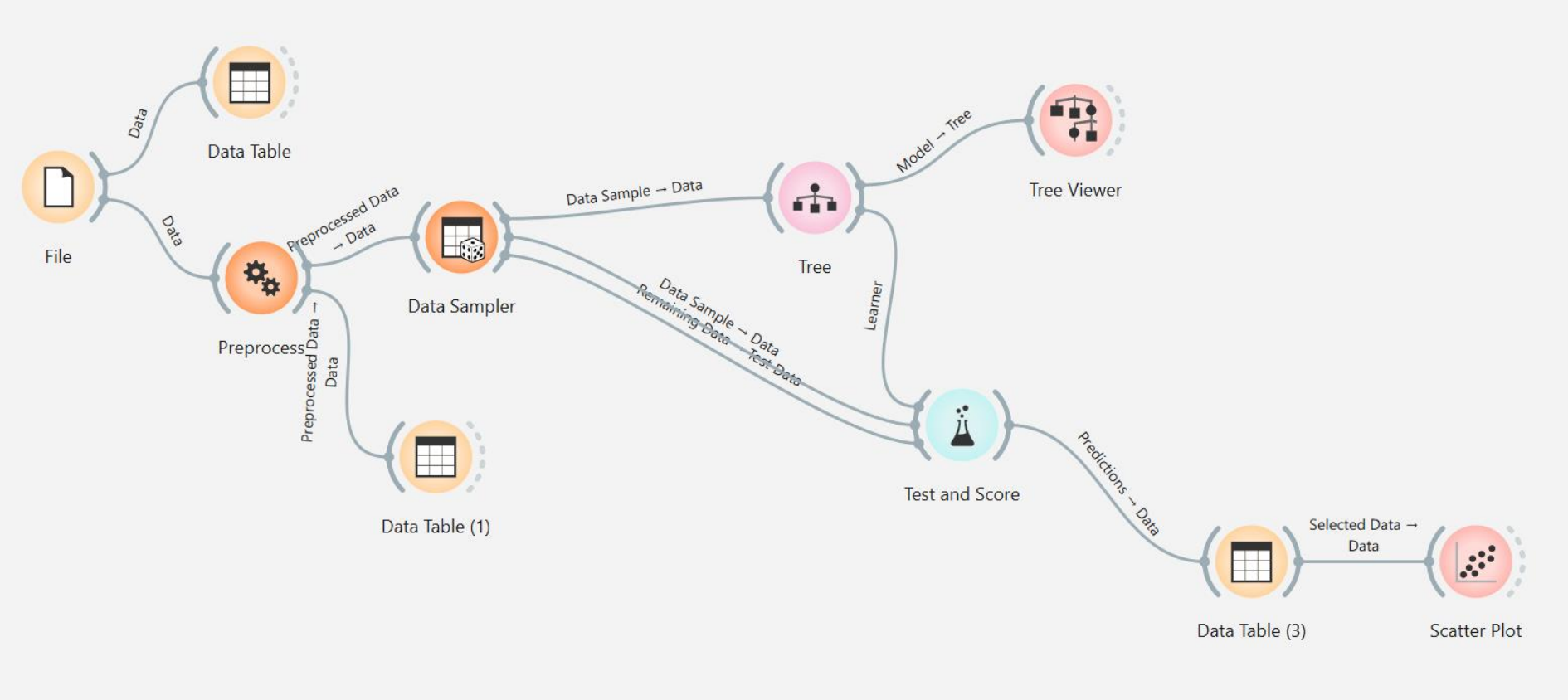
LAB: TREE CLASSIFIER



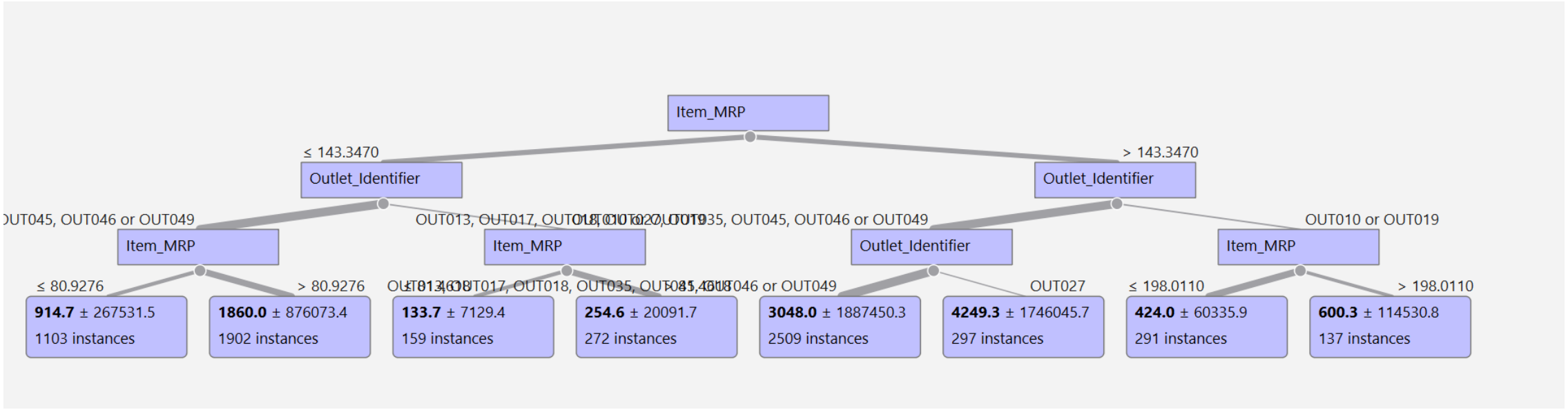
Classification



LAB: TREE REGRESSOR



Regression


Item_MRP
≤ 143.3470
> 143.3470
Outlet_Identifier
Outlet_Identifier
UT045, OUT046 or OUT049
Item_MRP
Item_MRP
Outlet_Identifier
Item_MRP
≤ 80.9276
> 80.9276
> 198.0110
OUT027
914.7 ± 267531.5
1103 instances
1860.0 ± 876073.4
1902 instances
133.7 ± 7129.4
159 instances
254.6 ± 20091.7
272 instances
3048.0 ± 1887450.3
2509 instances
4249.3 ± 1746045.7
297 instances
424.0 ± 60335.9
291 instances
600.3 ± 114530.8
137 instances

The Steps for CART

1. For every input feature
   - identify all possible binary split points
   - choose the best split point with the highest reduction in impurity/error

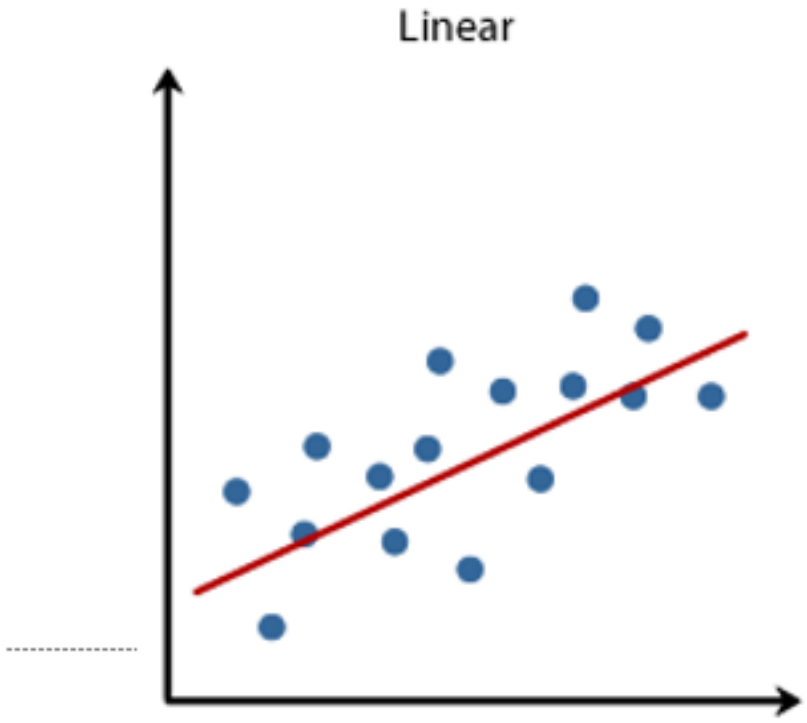
2. Rank the best splits and choose the feature that has the highest reduction in impurity/error

3. Divide the data into subgroups defined by the split

4. Continue the splitting process until:
   - All the nodes are 100% pure/error free or
   - Stopping condition is met
     1. Max tree depth
     2. Min samples at leaf node
     3. Min samples a node must have before it can split
     4. Others

3. Linear Model

Linear Regression



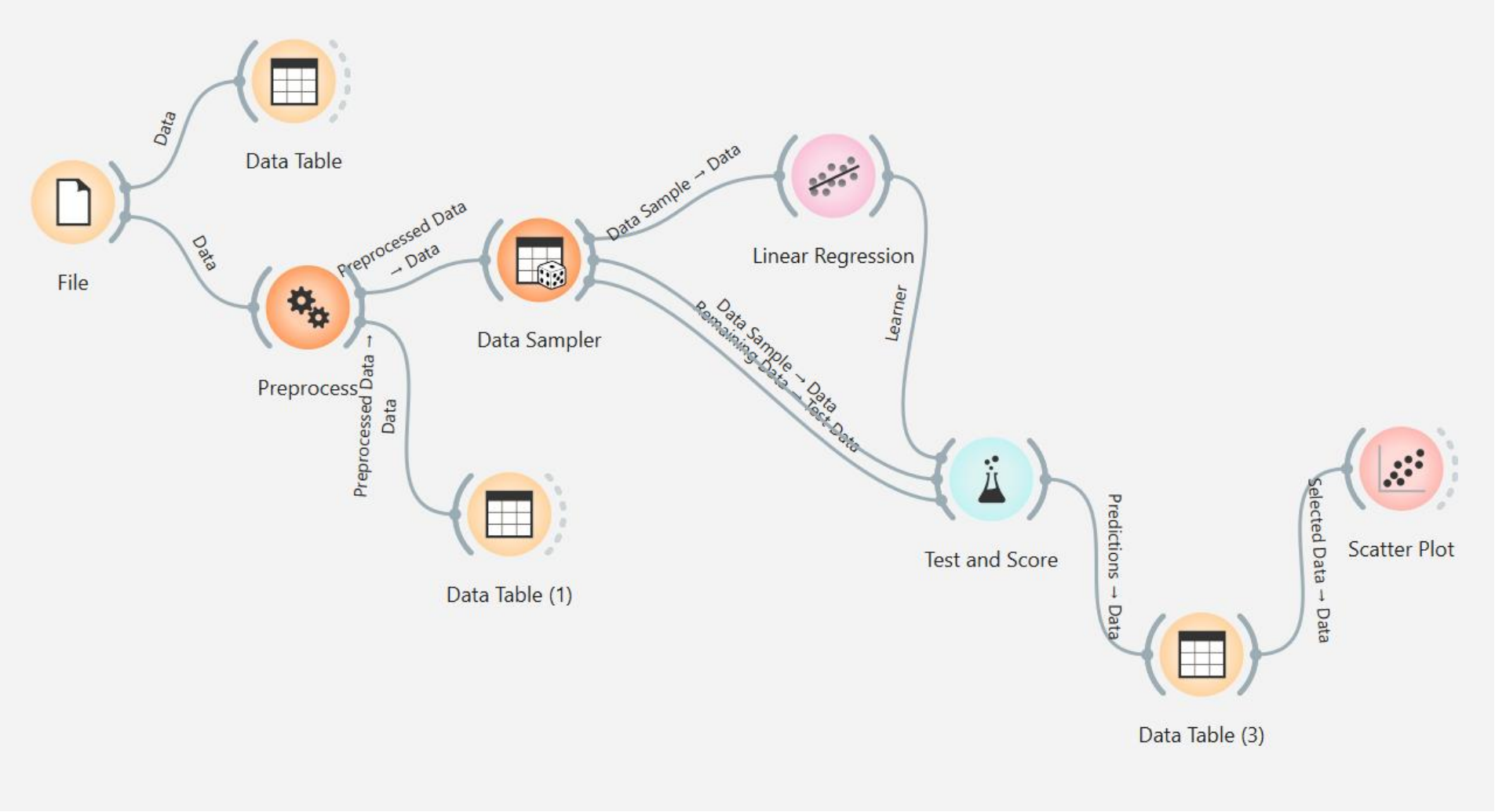
Simple Linear Regression

$$y = b_0 + b_1 x_1$$

Multiple Linear Regression

$$y = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_n x_n$$

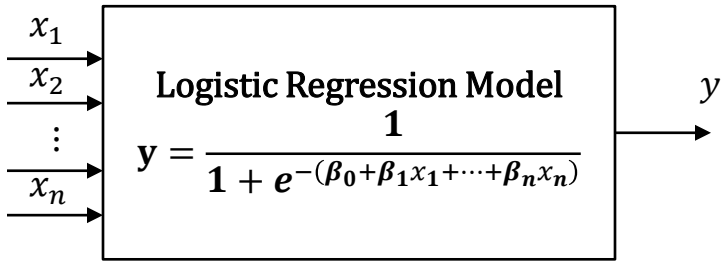
LAB: LINEAR REGRESSOR



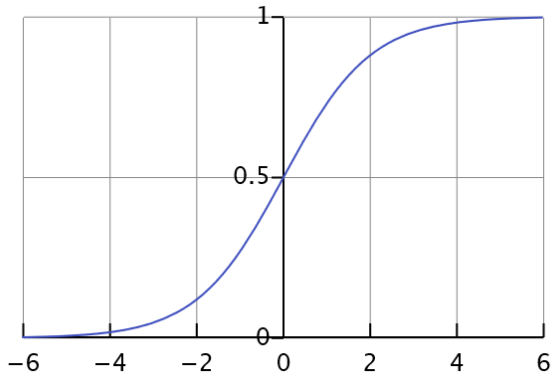
Logistic Regression

- Logistic regression model is represented in terms of logistic function as

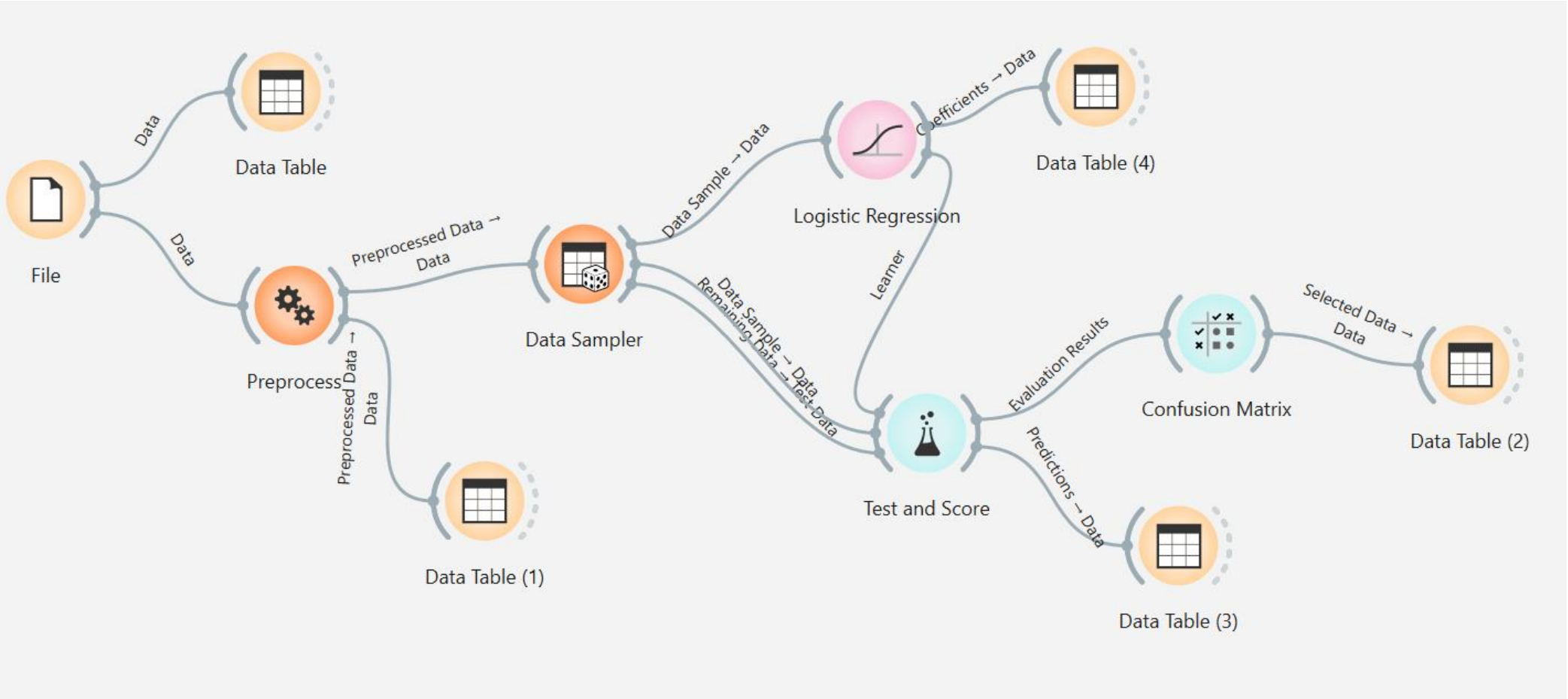
$$\mathbf{y} = \frac{\mathbf{1}}{\mathbf{1} + \boldsymbol{e}^{-(\boldsymbol{\beta_0}+\boldsymbol{\beta_1}x_1+\cdots+\boldsymbol{\beta_n}x_n)}}$$



- Where x_i is input to the model and β_i is coefficients estimated by the model
- The probability formula is the function known as logistic curve
- Decision boundary is often set at 0.5

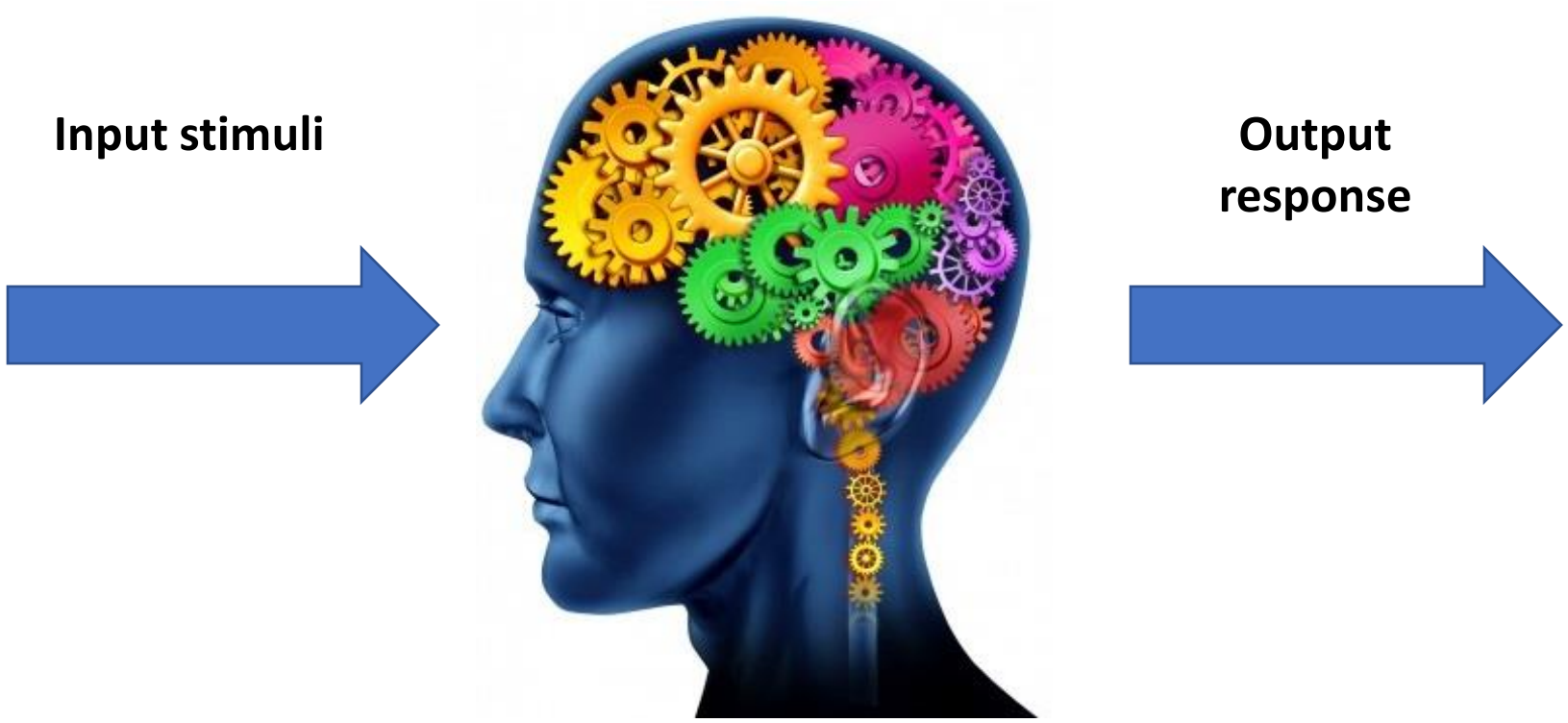


LAB: LOGISTIC CLASSIFIER

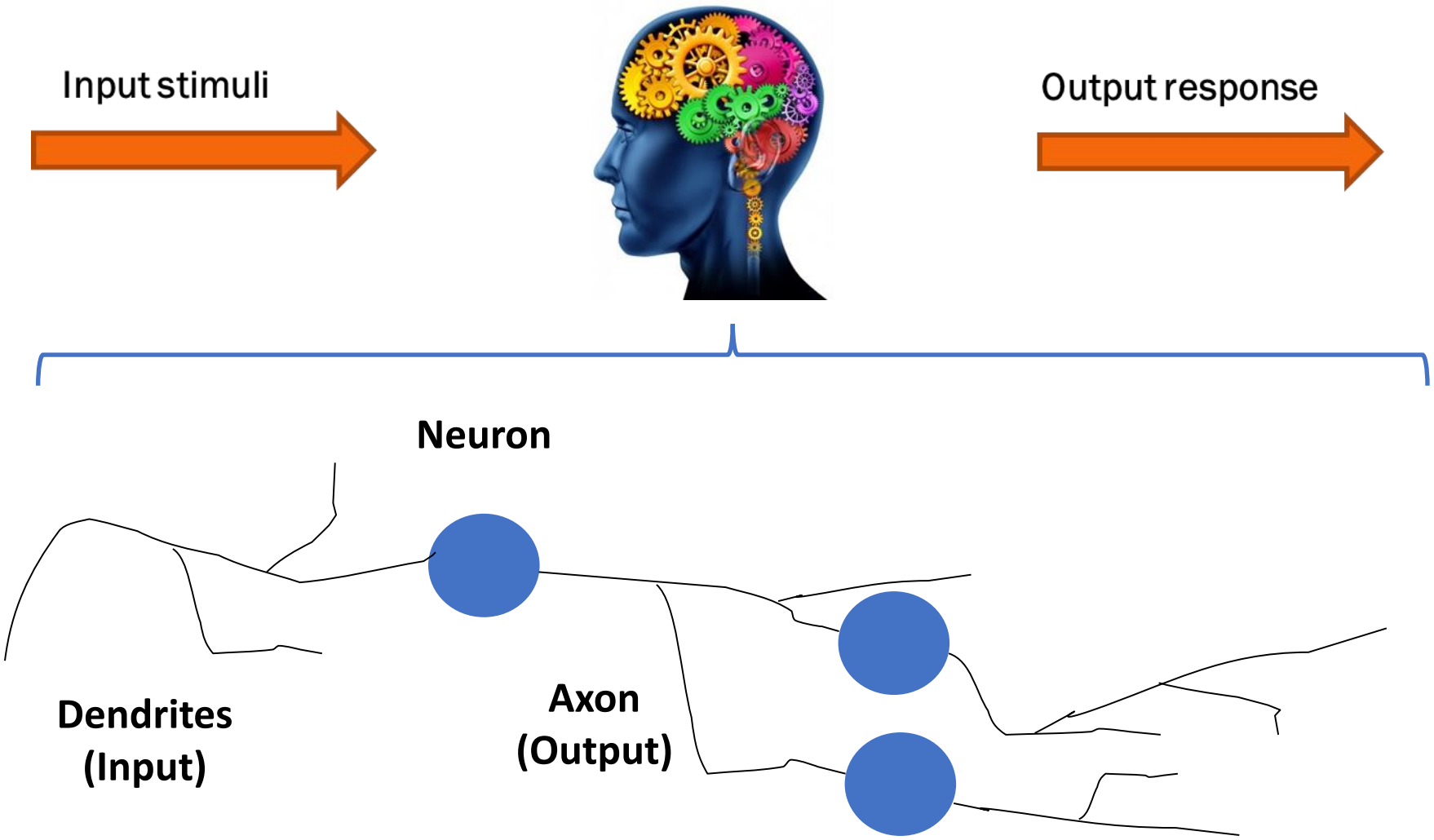


4. Neural Networks

Human Brain vs Artificial Neural Networks

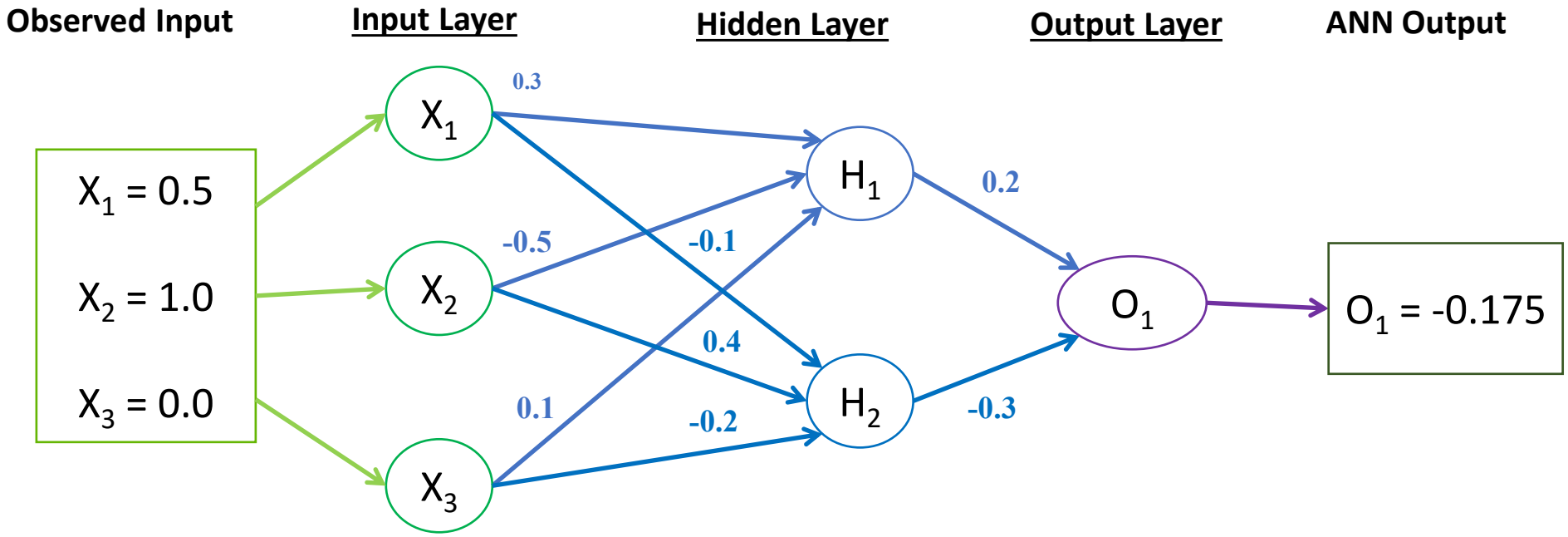


Human Brain vs Artificial Neural Networks

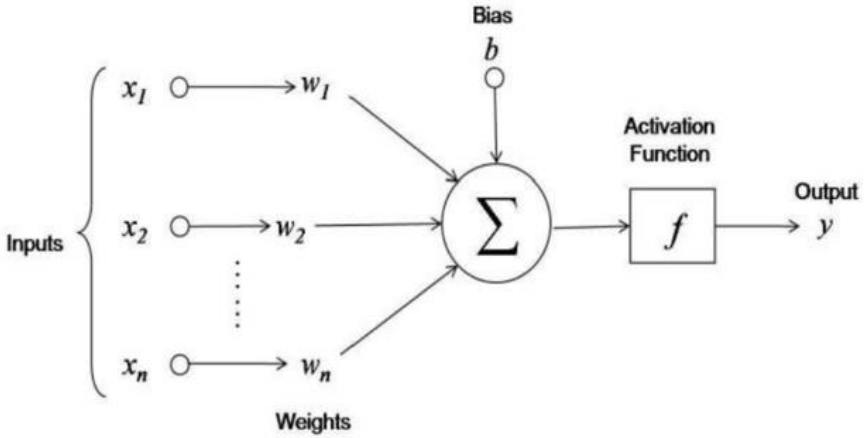


Human Brain vs Artificial Neural Networks

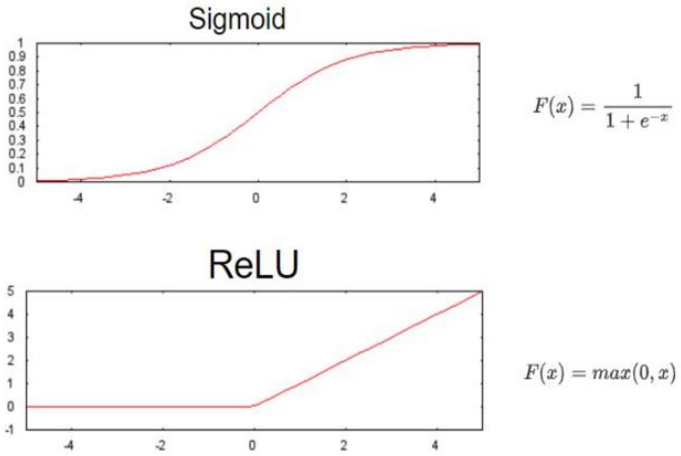




WHAT DOES A NEURON DO?



Activation Functions



Step 1: sum of weights

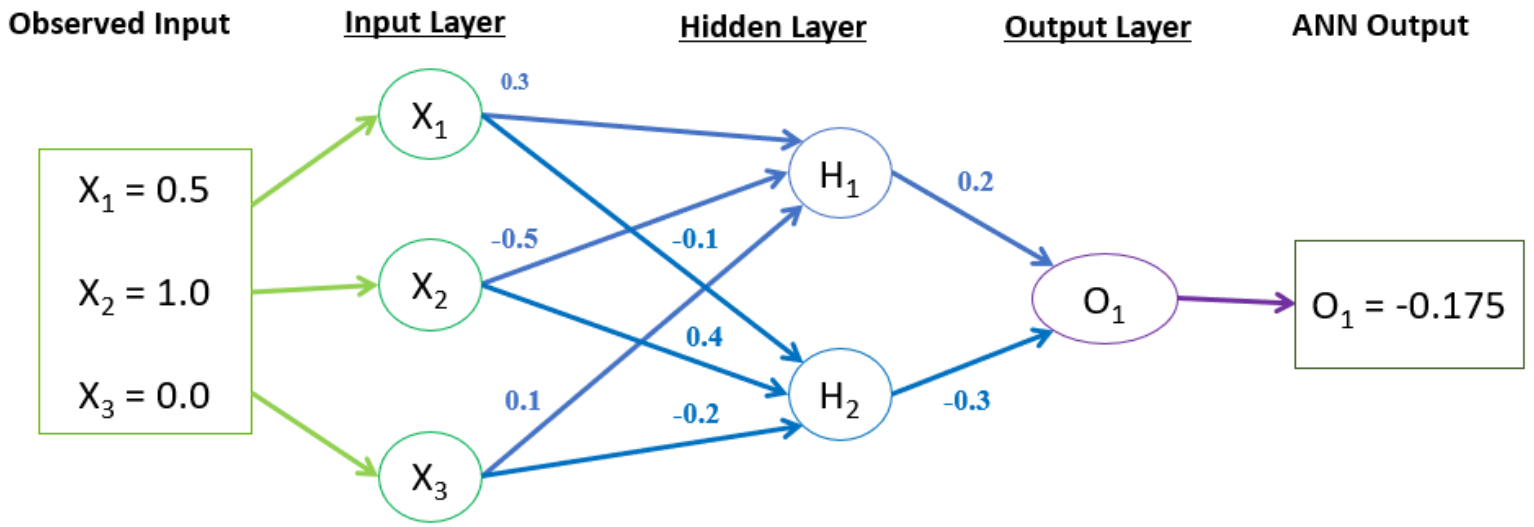
$$w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

Step 2: apply activation function

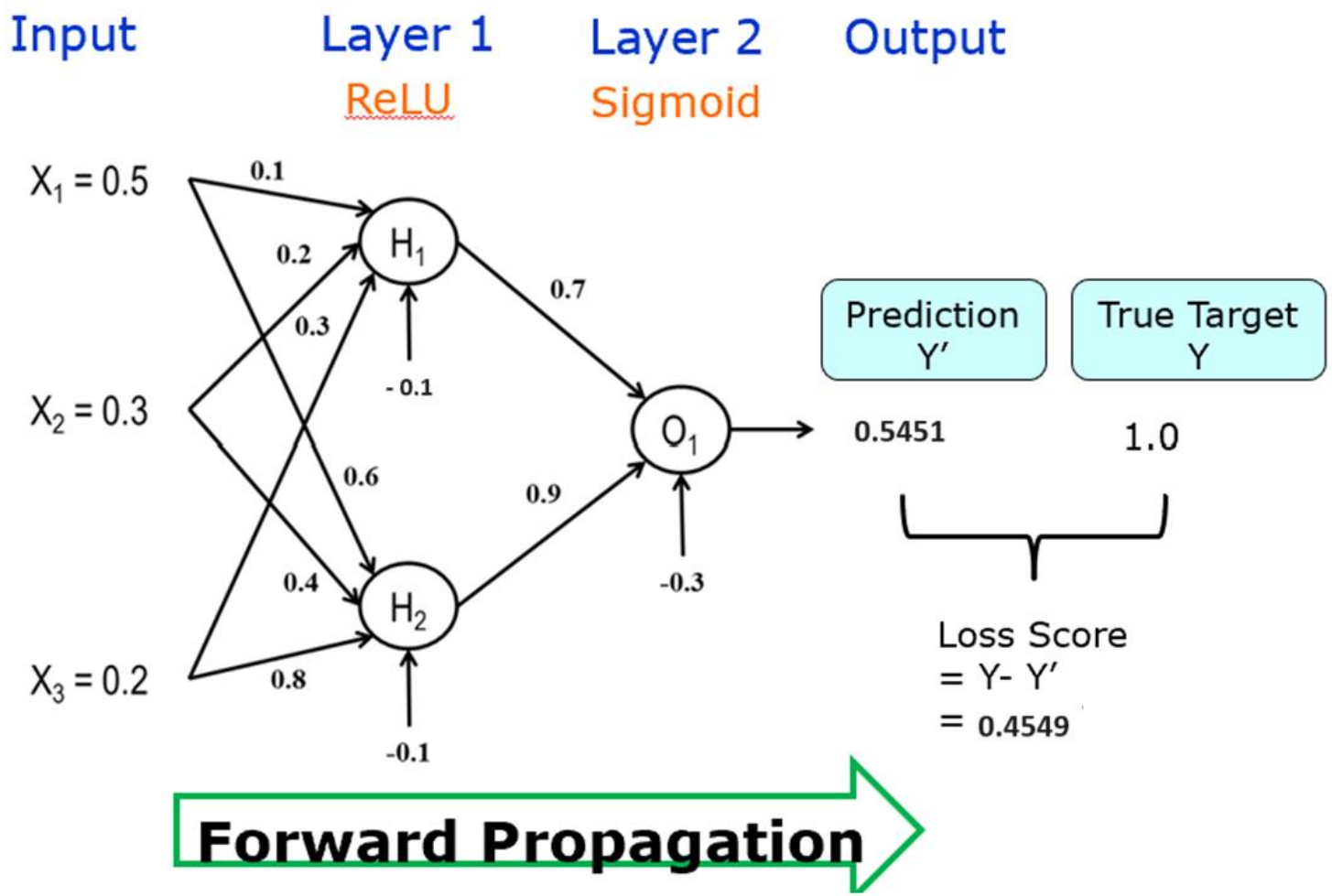
$$y = f(w_1x_1 + w_2x_2 + \cdots + w_nx_n + b)$$

Multilayer Perceptron (MLP) Networks

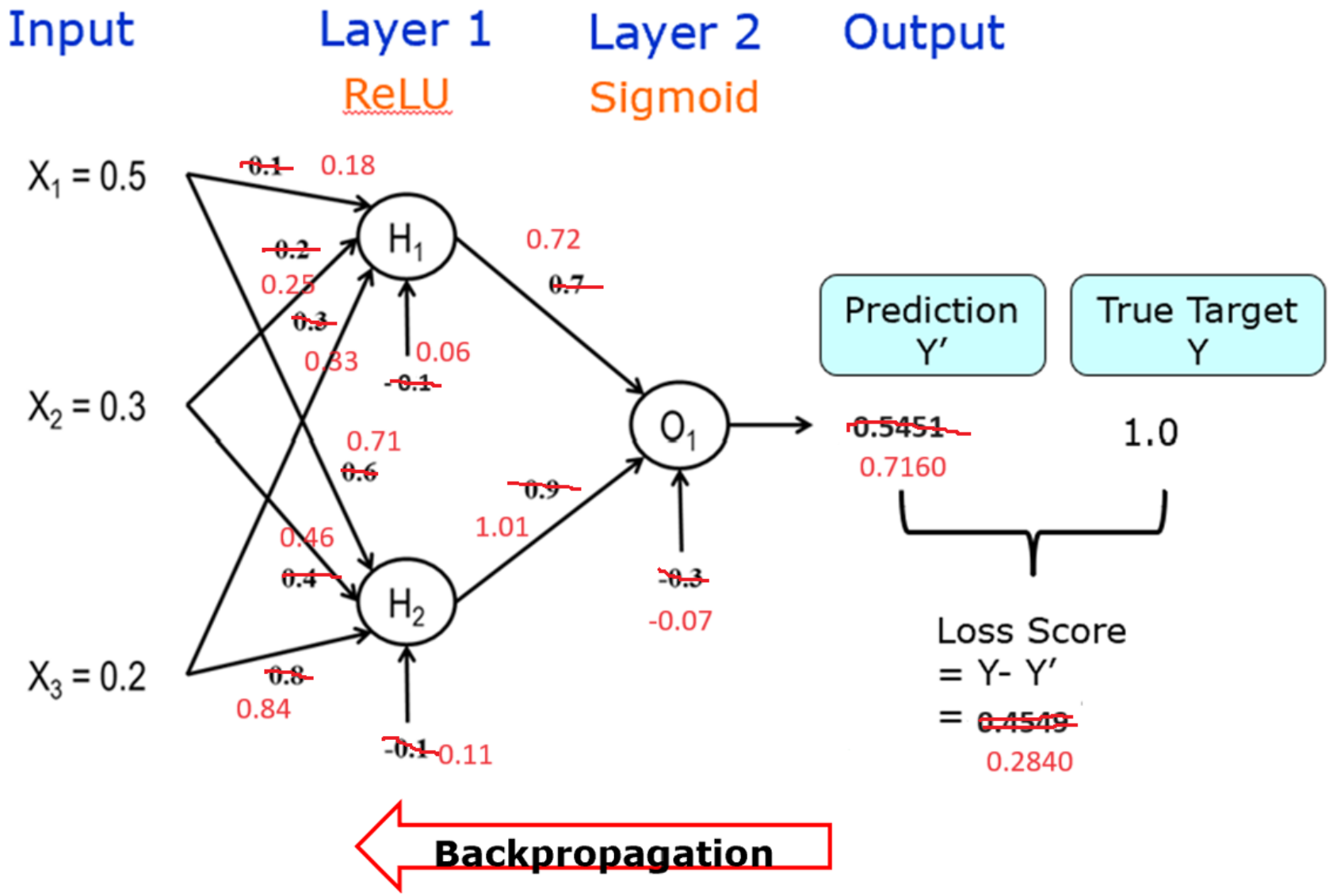
- By connecting the basic elements, neurons, through multilayer structure, Multilayer Perceptron (MLP) Network is built.



A Simple MLP


Input
Layer 1
ReLU
Layer 2
Sigmoid
Output
$X_1 = 0.5$
$X_2 = 0.3$
$X_3 = 0.2$
H_1
H_2
O_1
0.1
0.2
0.3
0.4
0.6
0.8
0.7
0.9
- 0.1
-0.1
-0.3
Prediction
Y'
True Target
Y
0.5451
1.0
Loss Score
= Y- Y'
= 0.4549
Forward Propagation

A Simple MLP


Input
Layer 1
ReLU
Layer 2
Sigmoid
Output
X1 = 0.5
X2 = 0.3
X3 = 0.2
H1
H2
O1
0.1
0.18
0.2
0.25
0.3
0.33
0.06
-0.1
0.72
0.7
0.71
0.6
0.9
1.01
0.46
0.4
-0.3
-0.07
0.8
0.84
-0.1
0.11
Prediction
Y'
True Target
Y
0.5451
0.7160
1.0
Loss Score
= Y- Y'
= 0.4549
0.2840
Backpropagation

- Learning through backpropagation learning algorithm
  - 1st iteration:
    - Take all the examples (data records)
    - Feedforward operation
    - Compute error (actual label vs. the predicted output of the network)
    - Propagate error backwards and update all connection weights
  - 2nd iteration:
    - Repeat operation in 1st iteration with all the examples/data records
  - Continue till stopping criteria is met
- Stopping criteria can be:
  - Total number of iteration is executed
  - A preset performance metric is achieved

Characteristics of ANNs

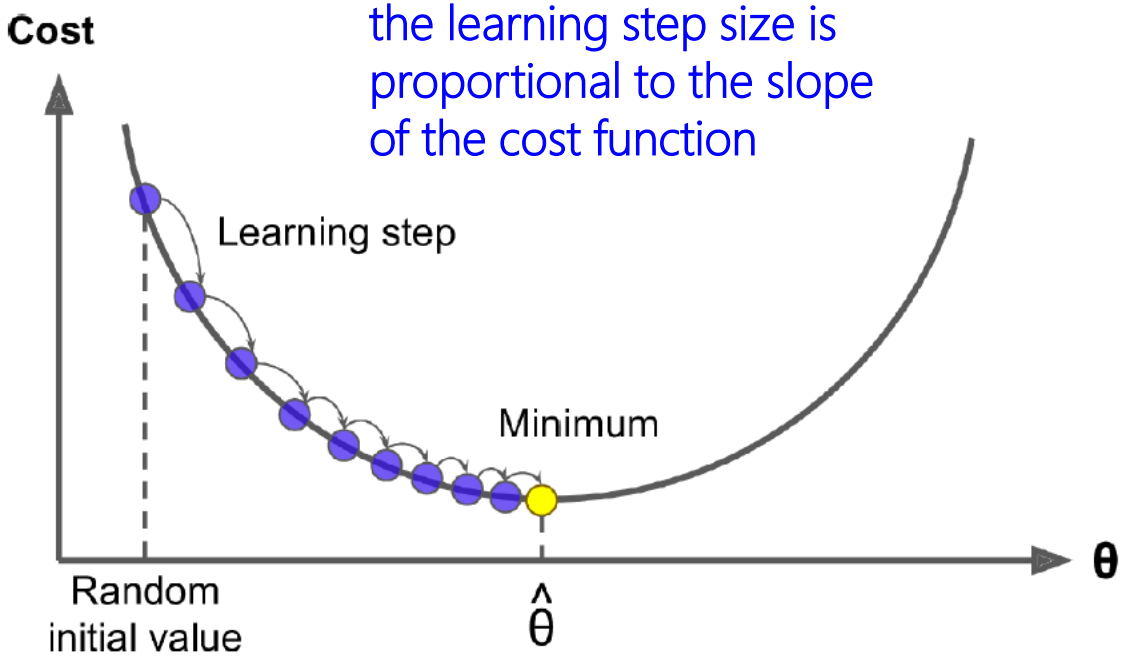
- Compared to decision trees, ANNs is often criticized as a black box model where the results (the connection weights) cannot be explained
- Therefore it is difficult to apply expert knowledge while using ANNs
- Neuron is an autonomous computational unit where its output is only affected by the inputs and weights connected to the neuron
- All input features are often normalized into a small range, e.g. [0, 1], [-3, +3]
- Can only handle numerical data

Characteristics of ANNs

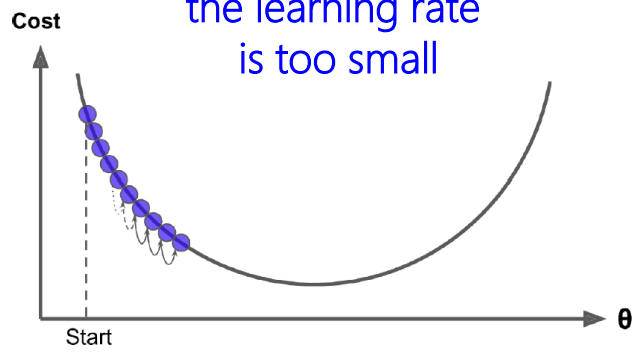
- Many parameters of ANN model can be defined and therefore the performance is somewhat unpredictable
- Backpropagation learning algorithm is a complex and time consuming process
- Nevertheless, ANNs work very well in both classification and regression task and is said to be able to model any complex nonlinear function given sufficient hidden neurons.

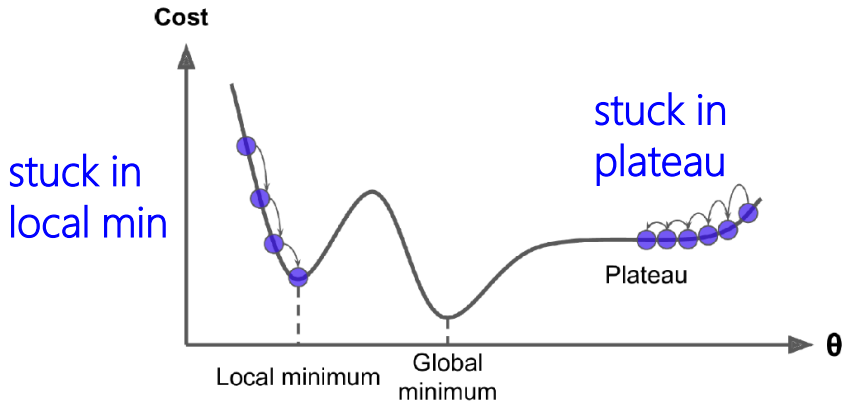
Gradient Descent

the learning step size is proportional to the slope of the cost function

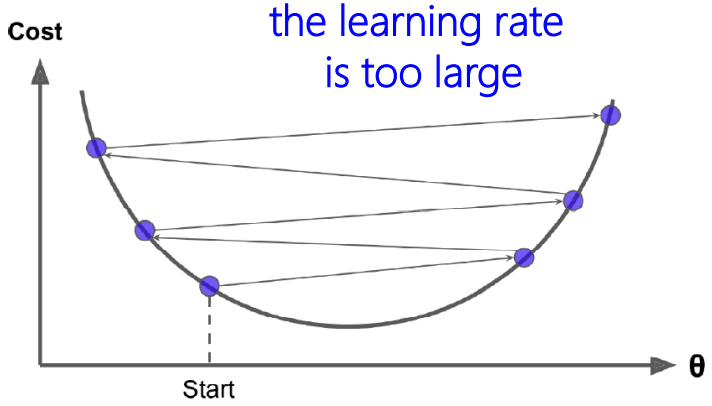


the learning rate is too small

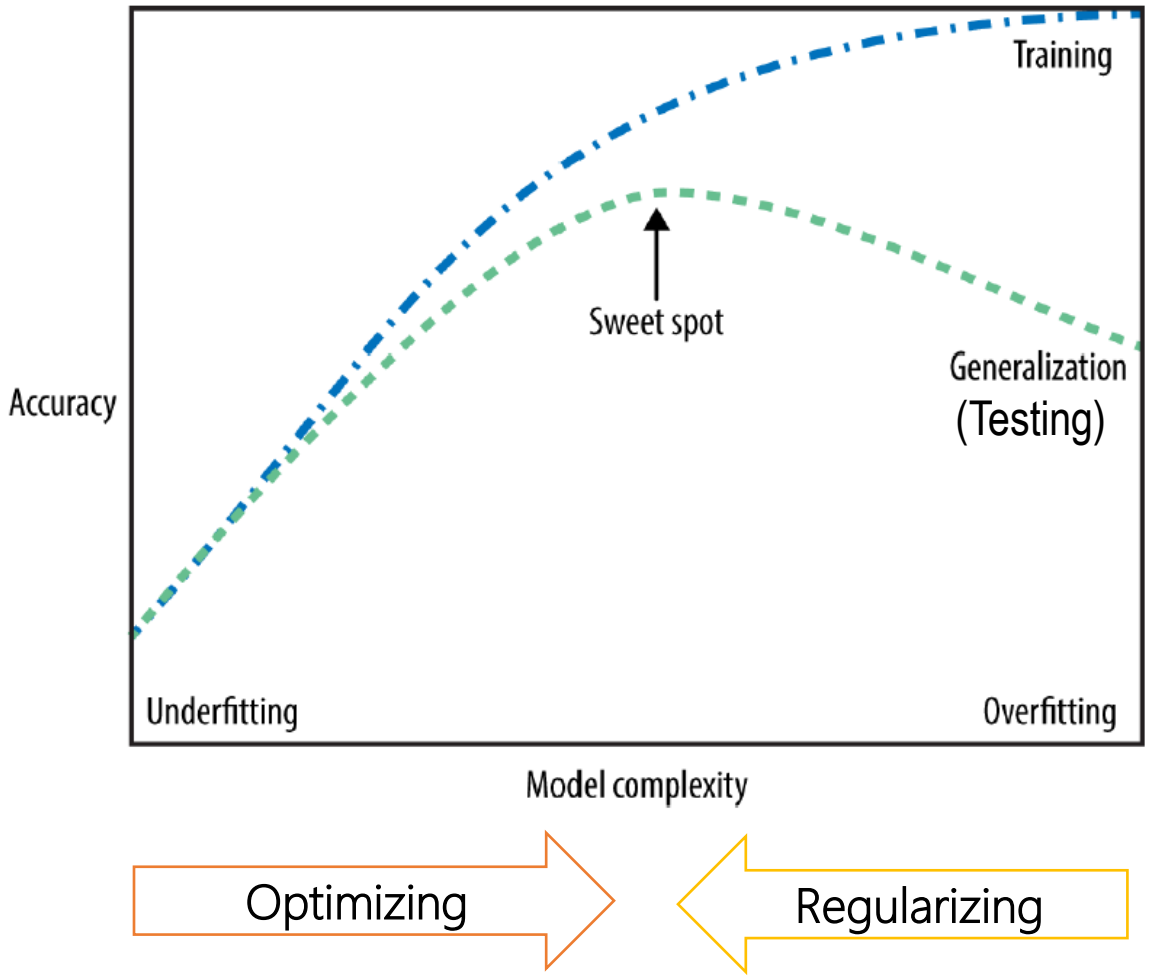




the learning rate is too large



Generalization, Overfitting and Underfitting

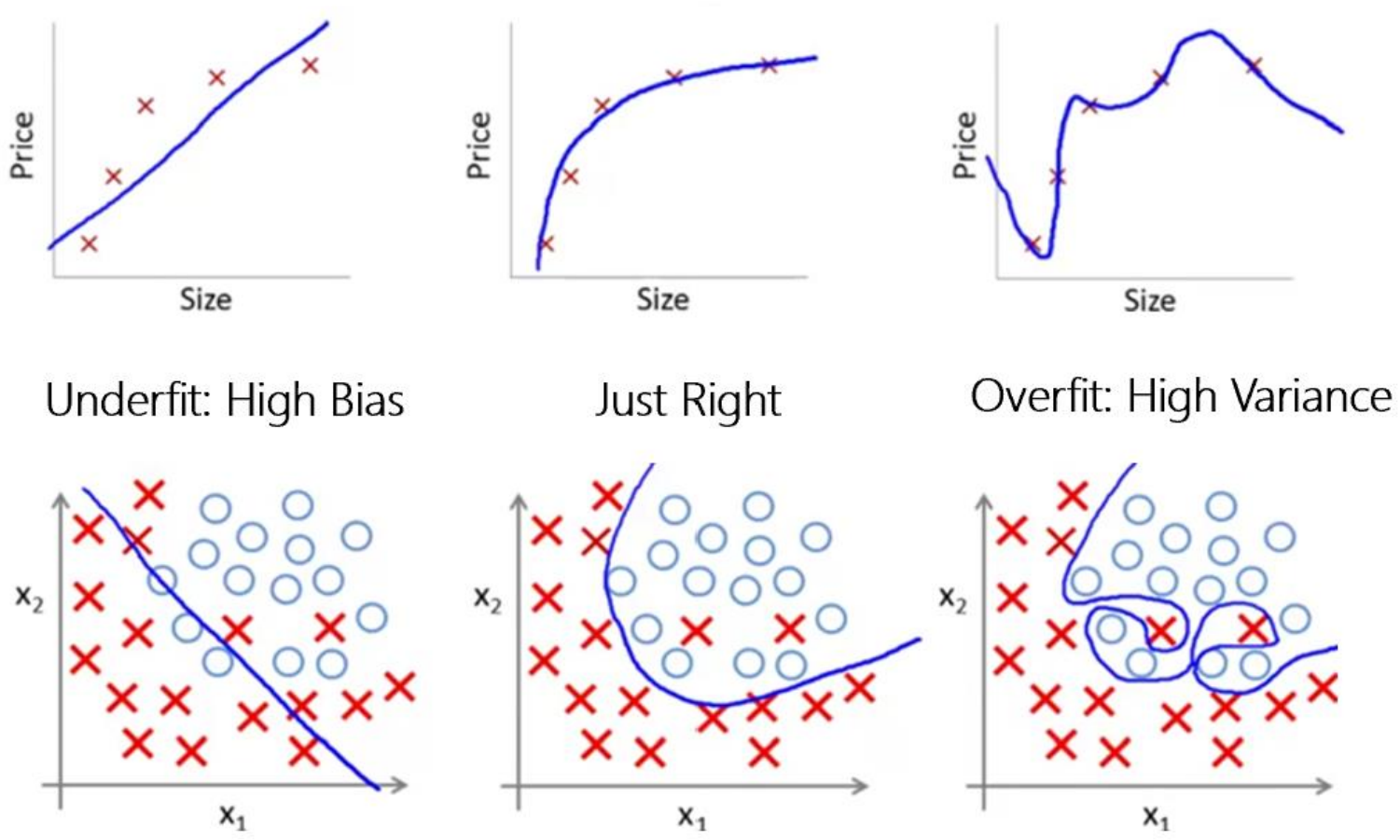


Balancing Optimization and Generalization

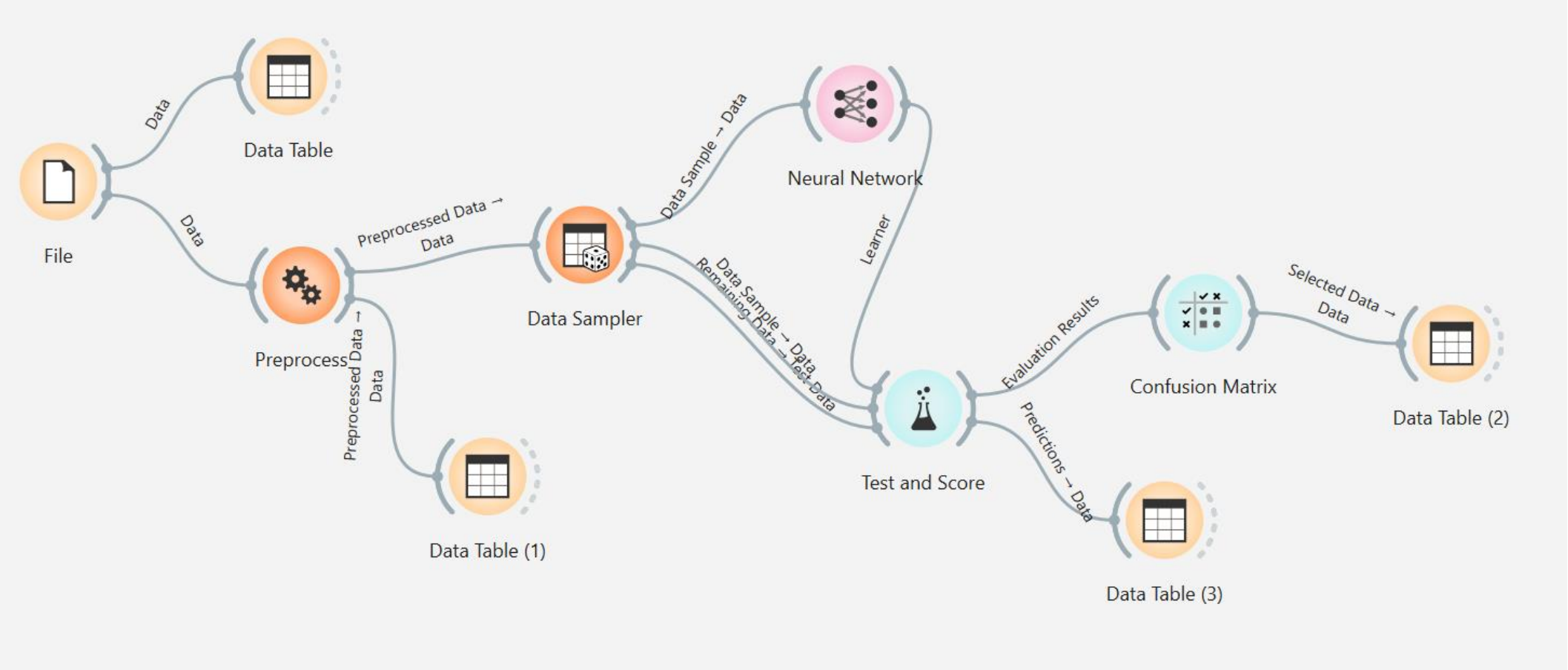
Tradeoff of Model Complexity against Training and Testing accuracy

The Bias-Variance Tradeoff

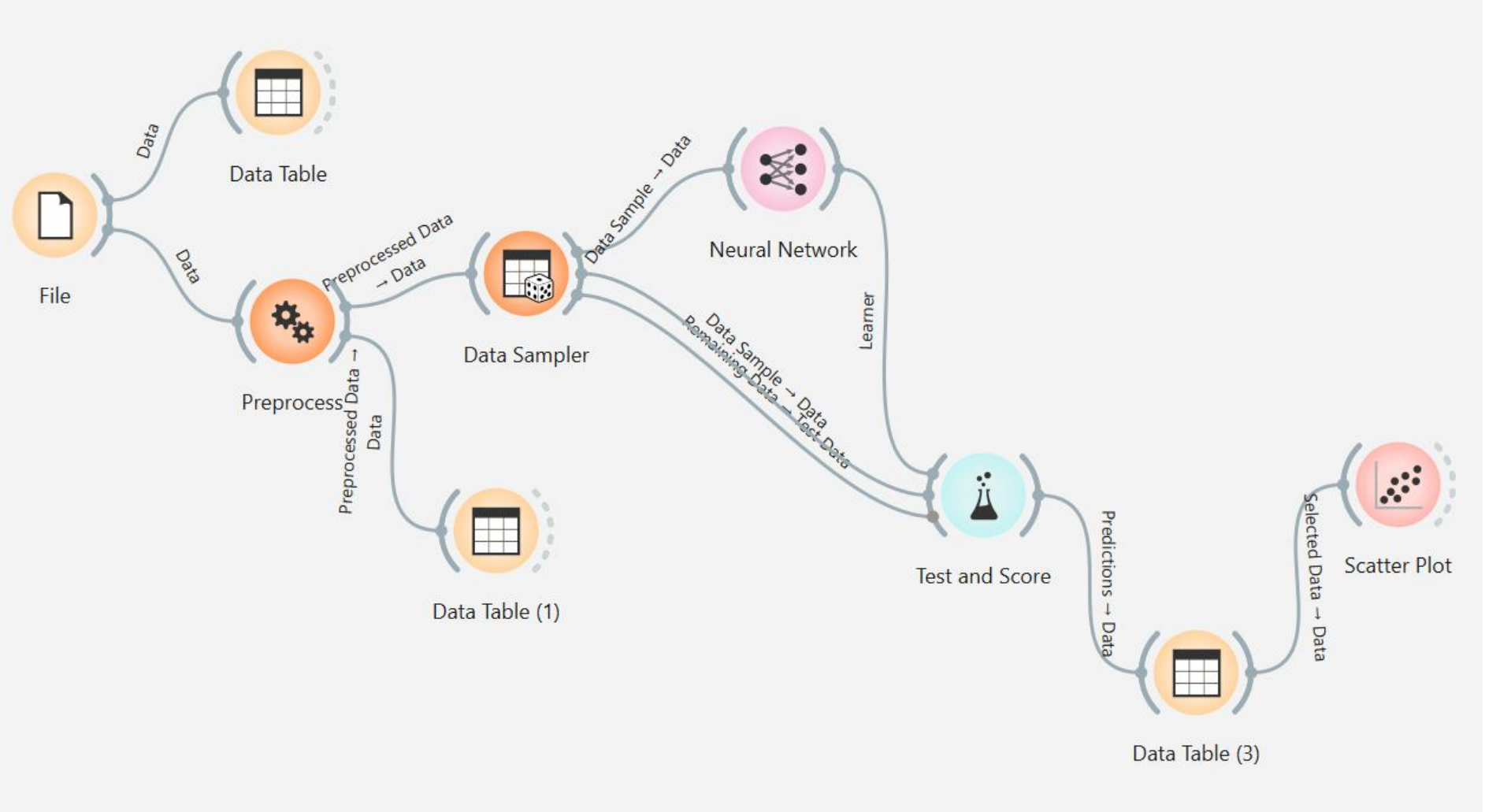
- Model Error
  - Bias
  - Variance
  - Noise



LAB: NEURAL NETWORK CLASSIFIER



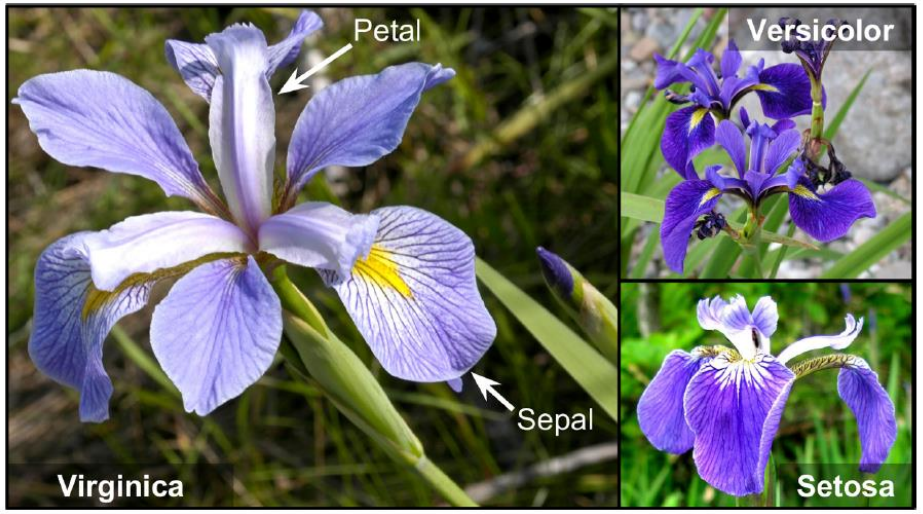
LAB: NEURAL NETWORK REGRESSOR



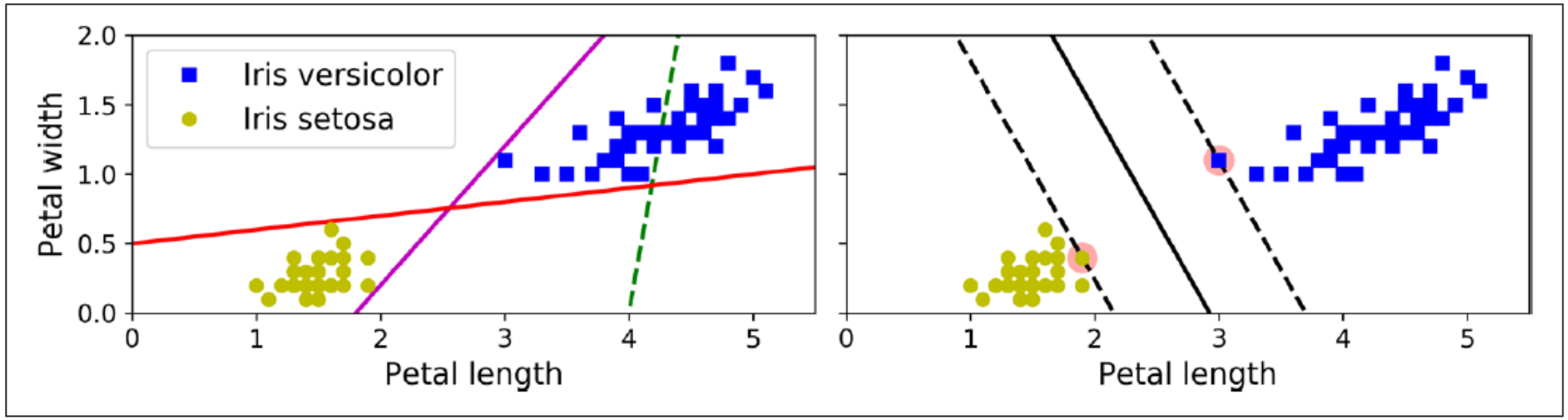
5. Support Vector Machine

SVM Model

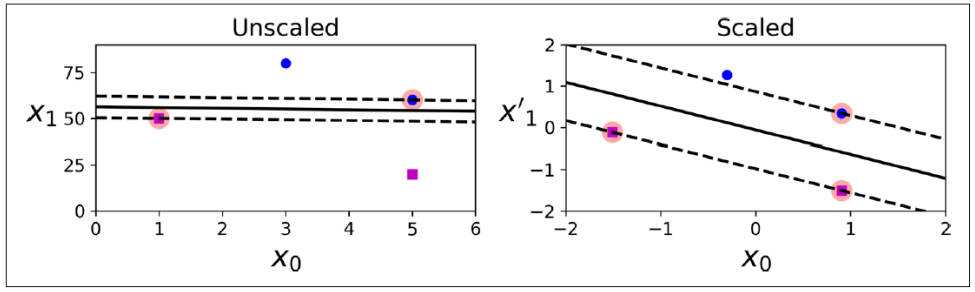
- One of the most popular machine learning model, powerful and versatile
- Capable of performing linear/non-linear, classification/regression tasks but particularly suited for classification tasks
  - SVM Classification:
    - Linear SVM Model
    - Kernel Trick
    - Nonlinear SVM Model
  - SVM Regression



Linear SVM Classification

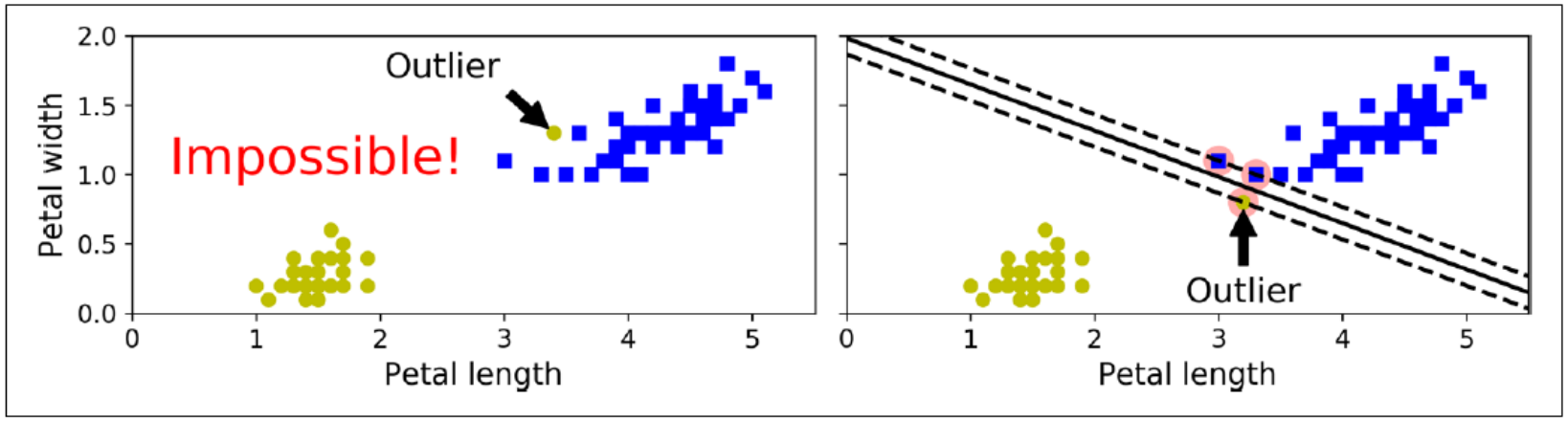


- Fitting the widest possible street between the two classes (large margin classification)
- All instances must be off the street and on the right side (i.e. No Margin Violations)??



Sensitive to Feature Scales

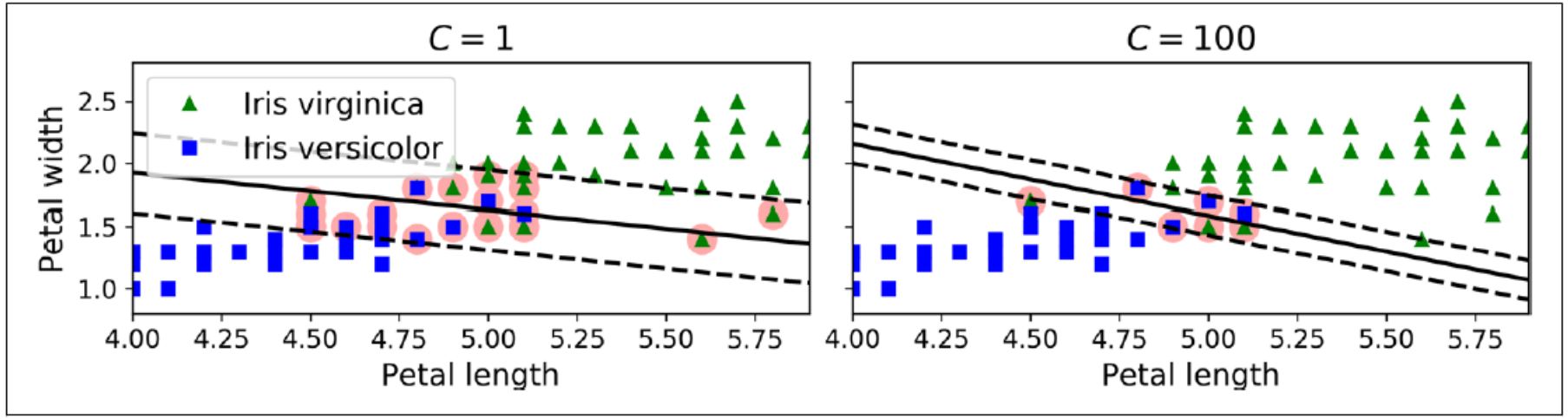
Use C to control the margin violations



Hard Margin Classification

- no margin violations
- sensitive to outliers

Soft Margin Classification



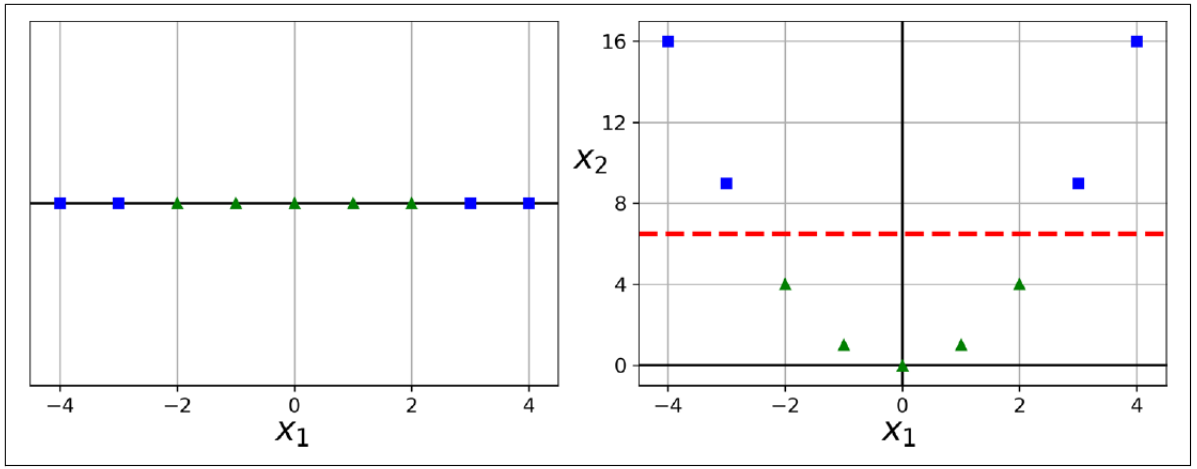
Large Margin Violations (wide street)

Fewer Margin Violations (narrow street)

Kernel Tricks

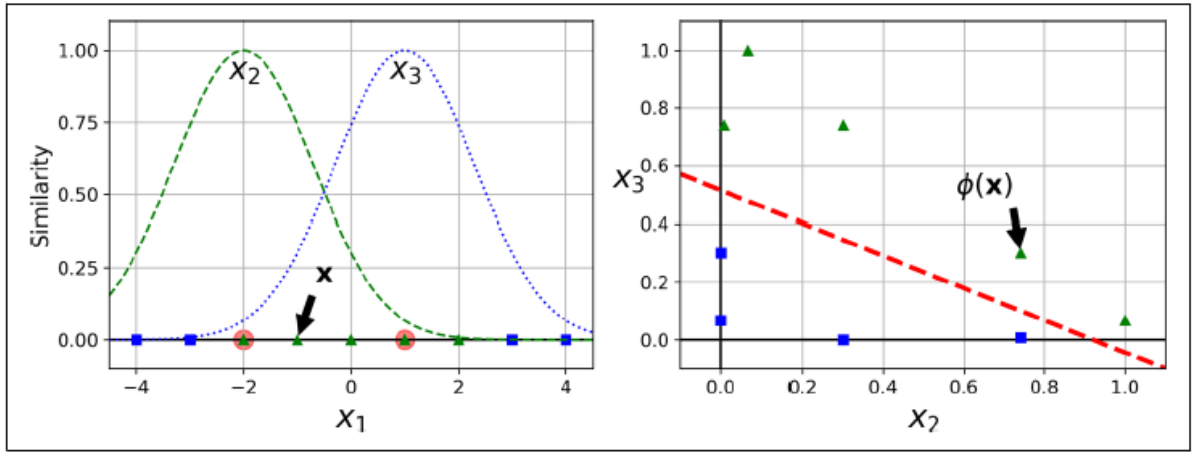
Feature Transformation using Polynomial Function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$$



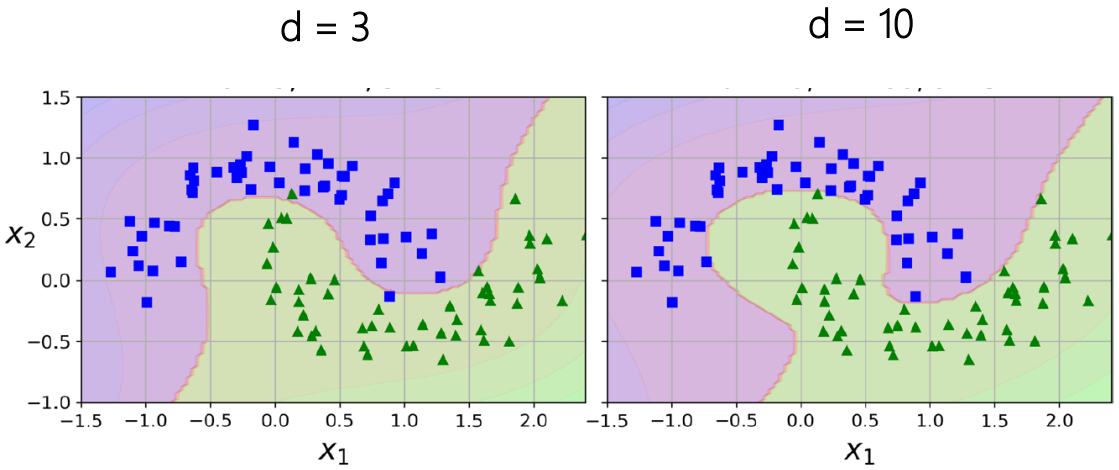
Feature Transformation using Gaussian Radial Basis Function (RBF)

$$\phi_\gamma(\mathbf{x}, \ell) = \exp\left(-\gamma \| \mathbf{x} - \ell \|^2\right)$$

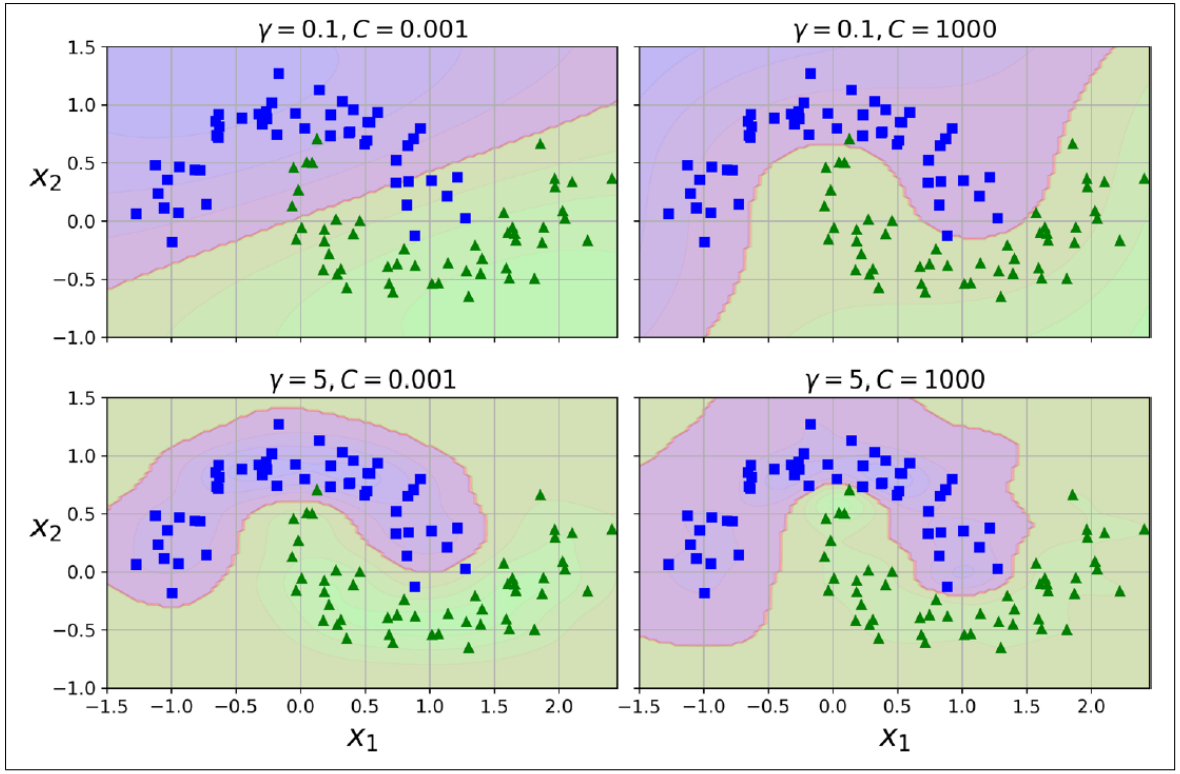


Nonlinear SVM Classification

Polynomial Kernel

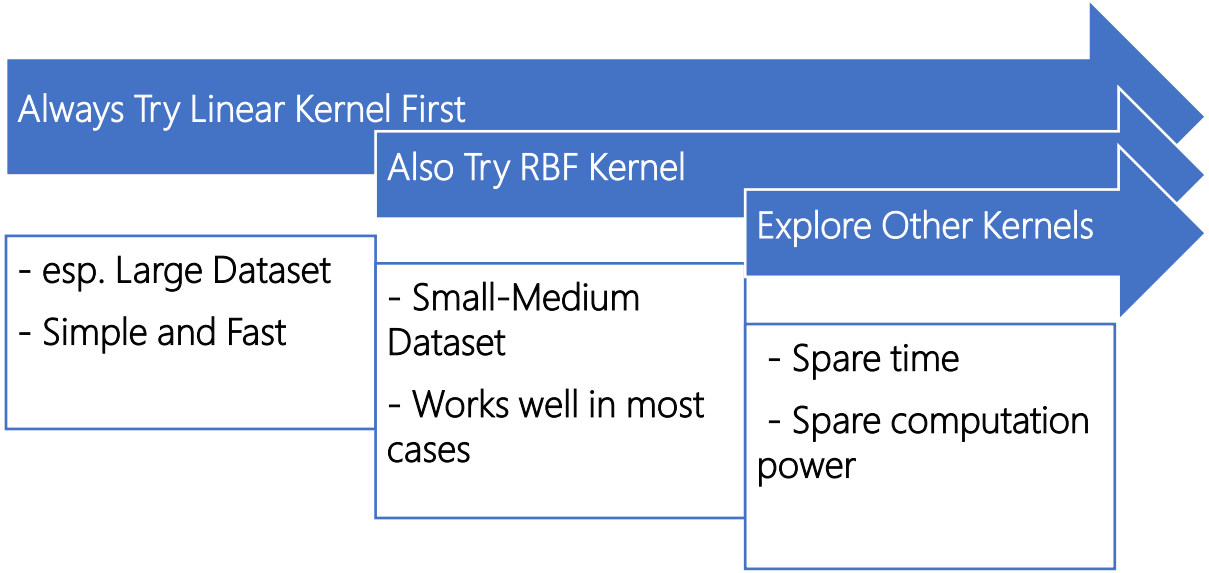


RBF Kernel



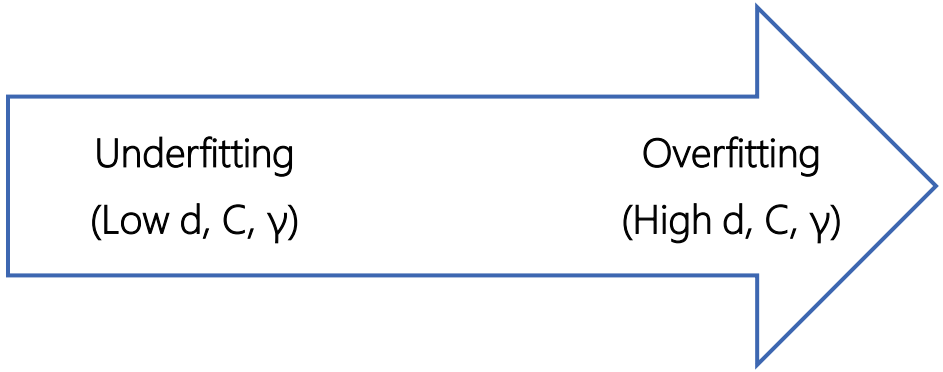
Tuning SVM Models

Choose from different kernels

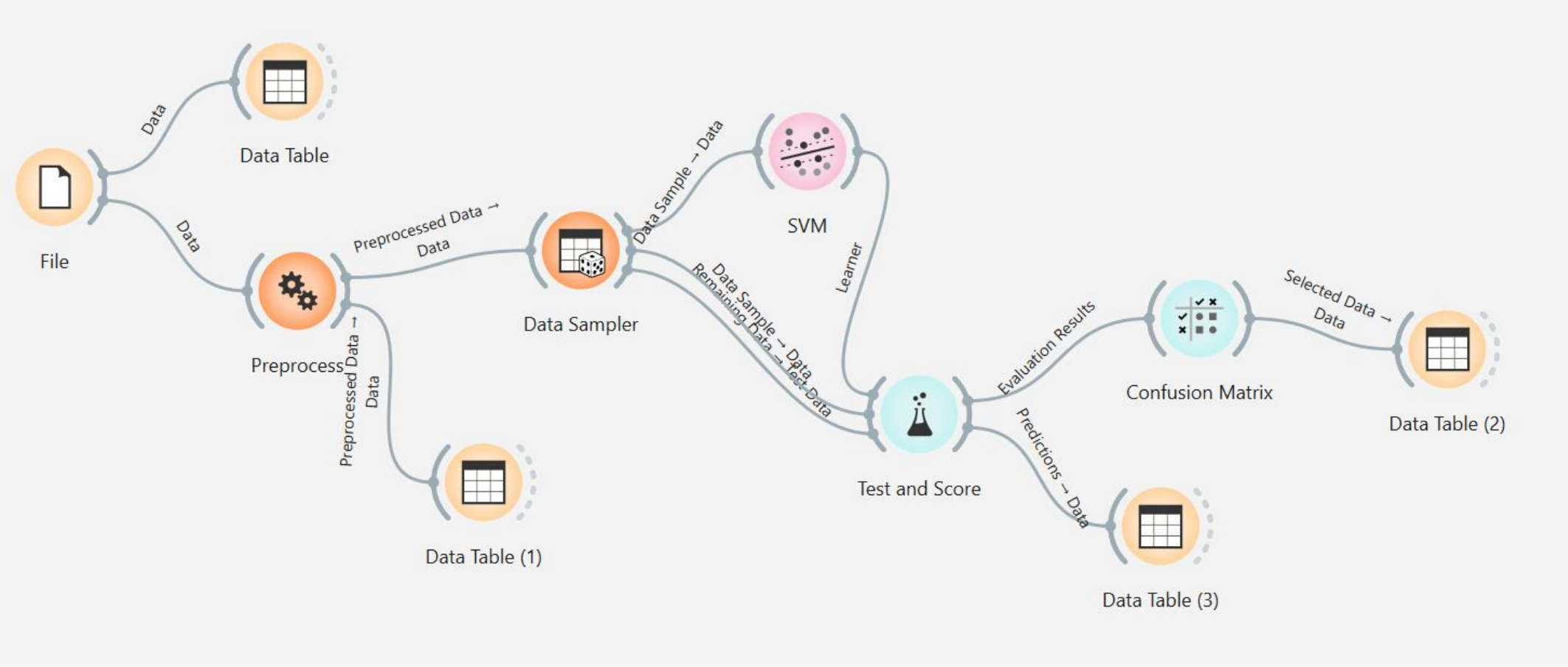


Tuning Hyperparameters

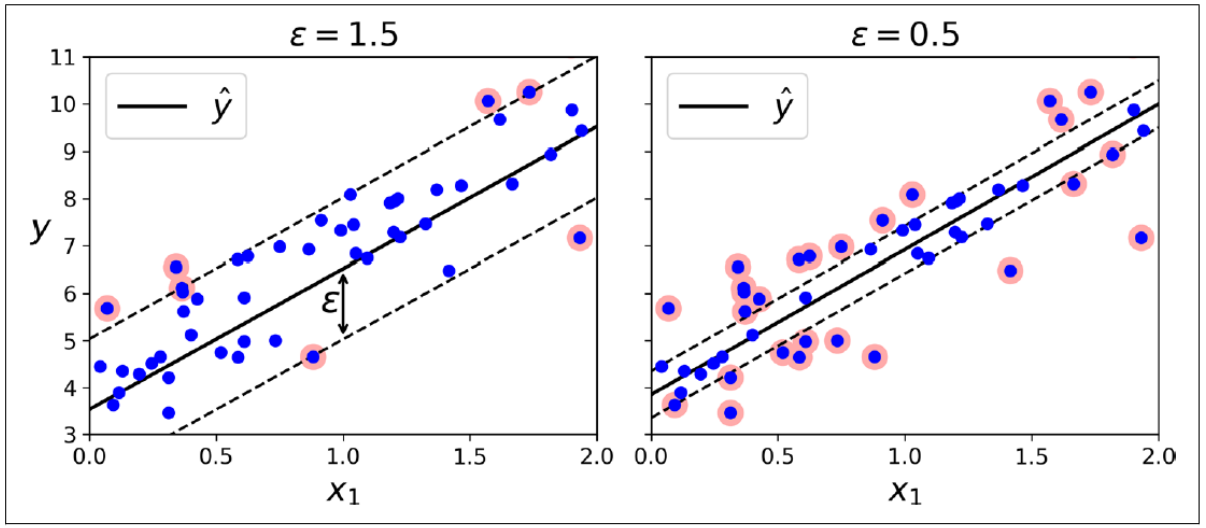
- First do a very coarse search and then a finer search
- Having a good sense of what each hyperparameter does helps on searching in the right direction



LAB: SVM CLASSIFIER

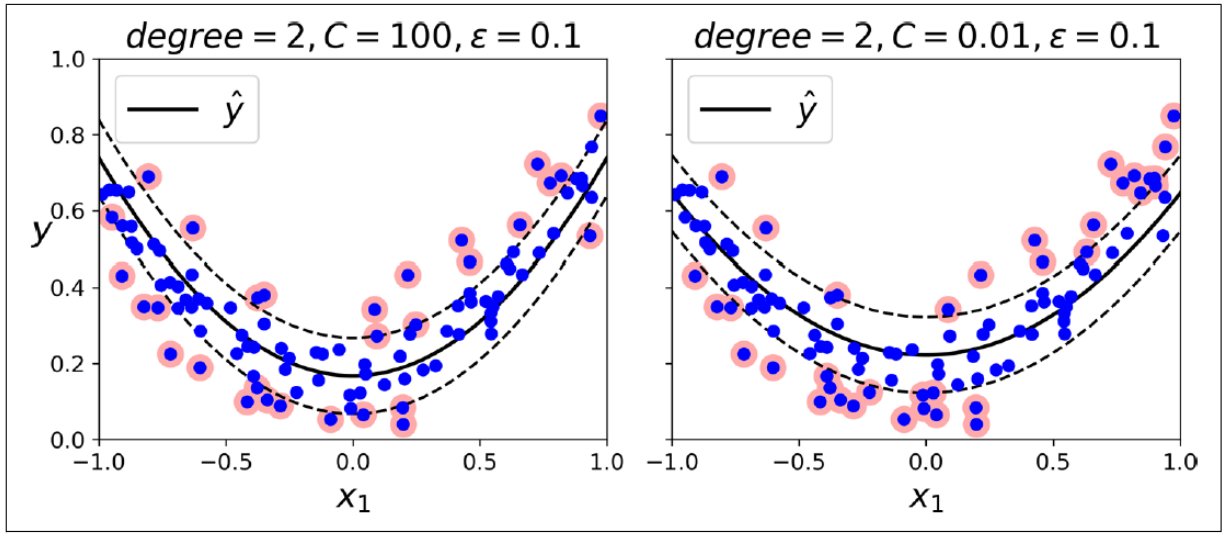


SVM Regression

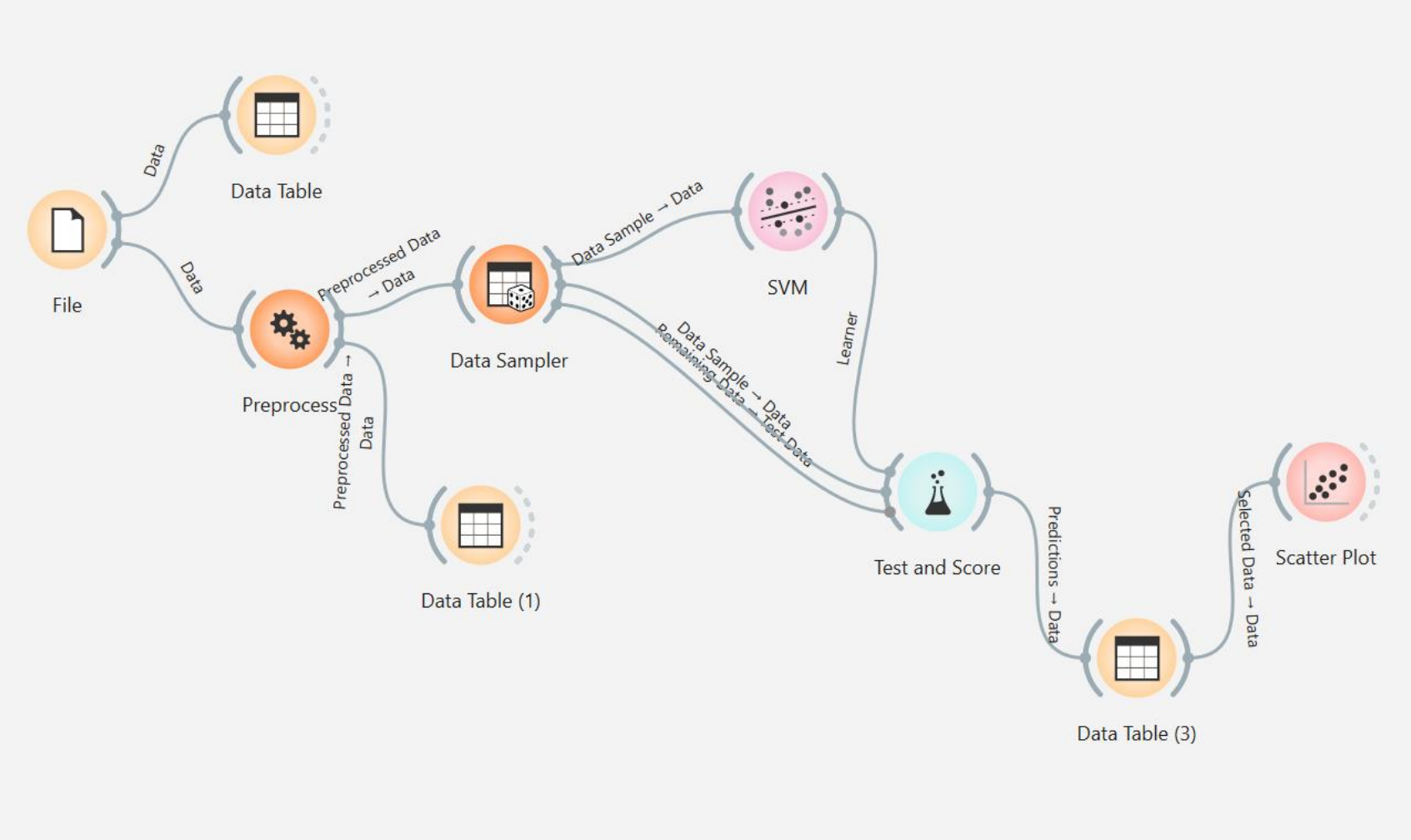


Fit as many instances as possible on the street while limiting margin violations (i.e. instances off the street)

The width of a street is controlled by hyperparameter ϵ



LAB: SVM REGRESSOR



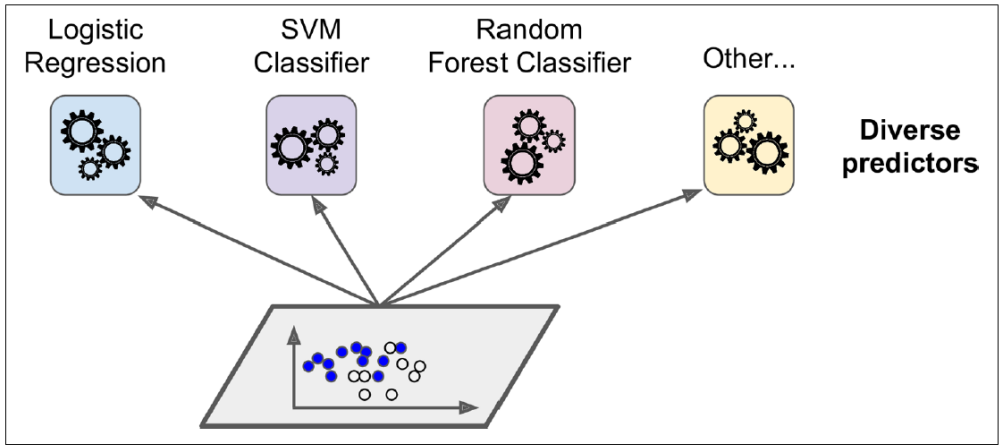
6. Ensemble Learning and Random Forest

What is Ensemble?

- Wisdom of Crowd
  - Aggregate the predictions of a group of predictors, you will get better predictions than with the best individual predictor
- Voting Classifier
- Bagging (Bootstrap Aggregating) Classifier & Regressor
- Random Forest Classifier & Regressor
  - an ensemble of Decision Trees via bagging method
- Boosting Classifier & Regressor

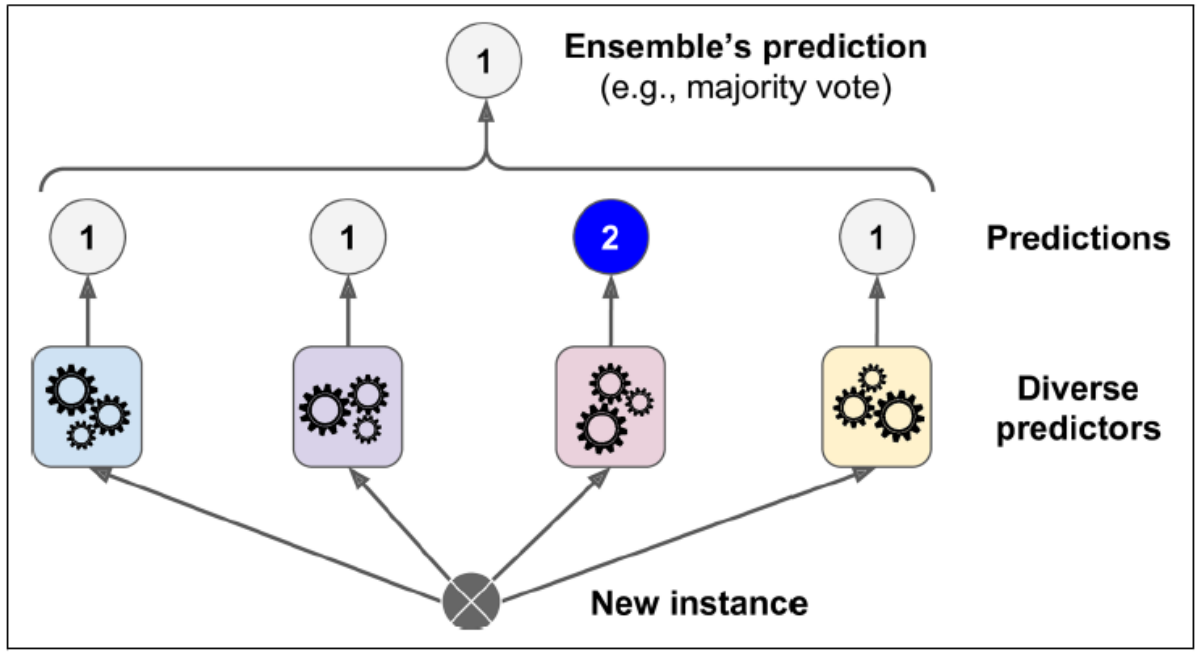
Voting Classifier

Training Diverse Classifiers



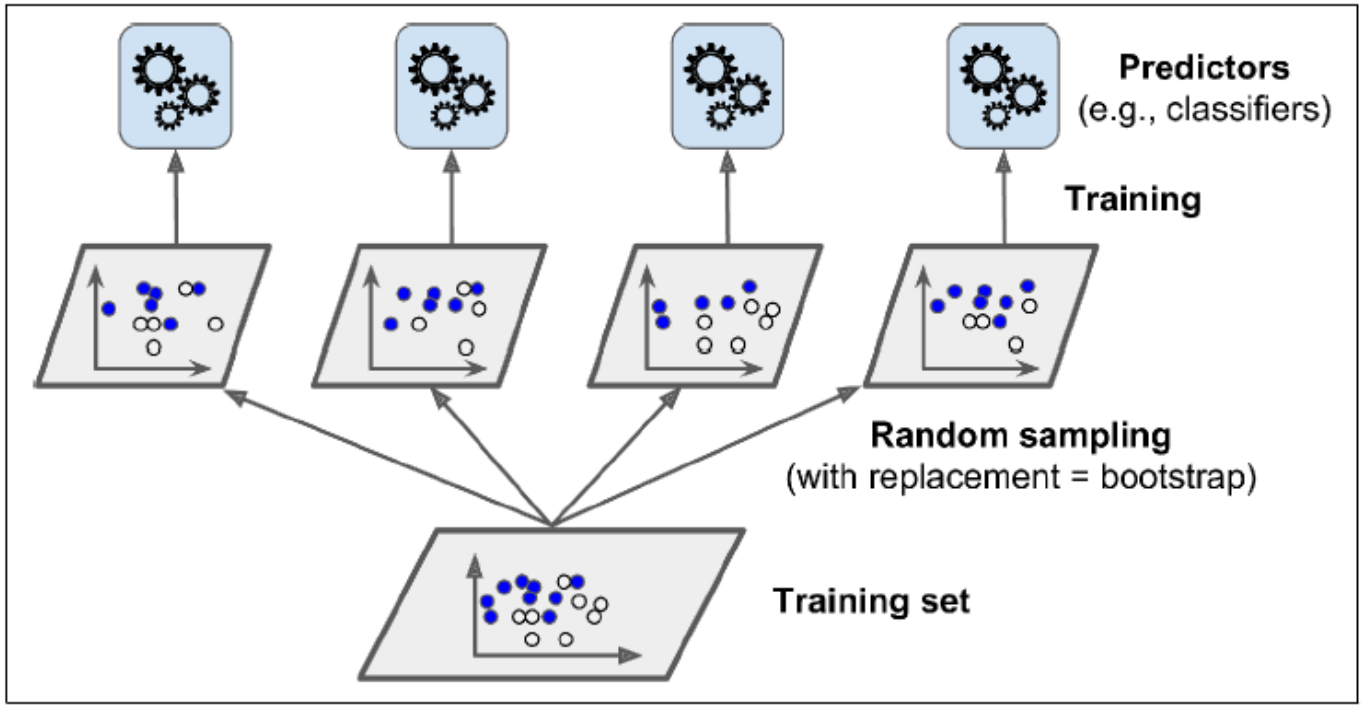
Soft Voting Classifier
predict the class with the highest class probability, averaged over all the individual classifiers

Hard Voting Classifier Predictions



Bagging: Bootstrap Aggregating

Training several predictors on different random samples of the training set

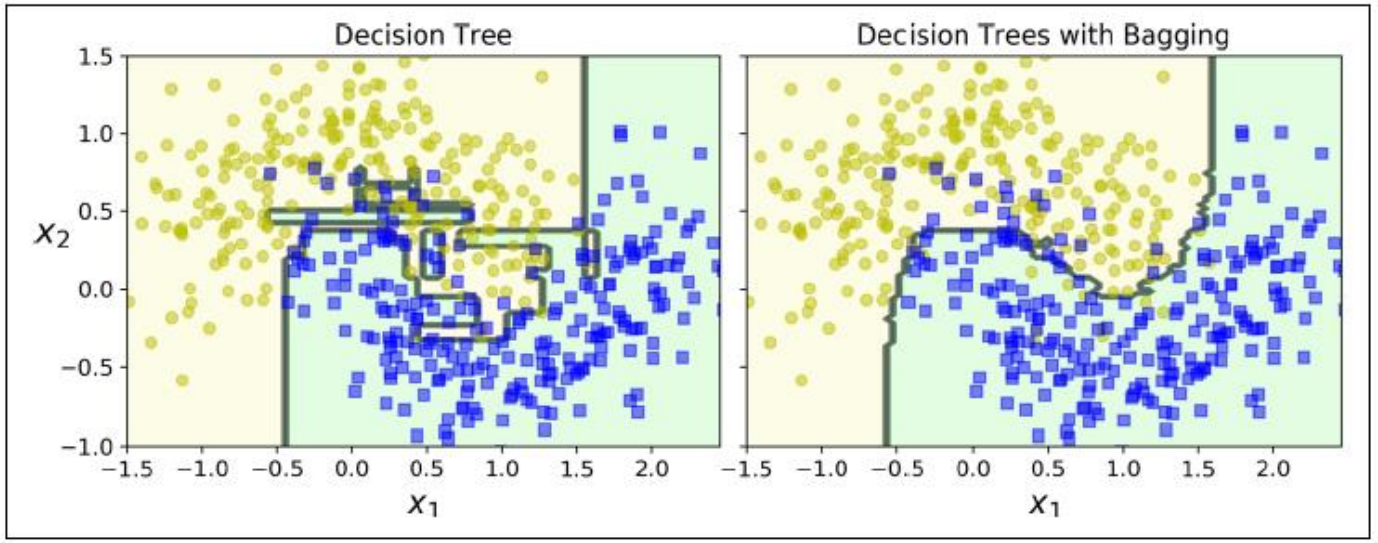


Bootstrap Sampling

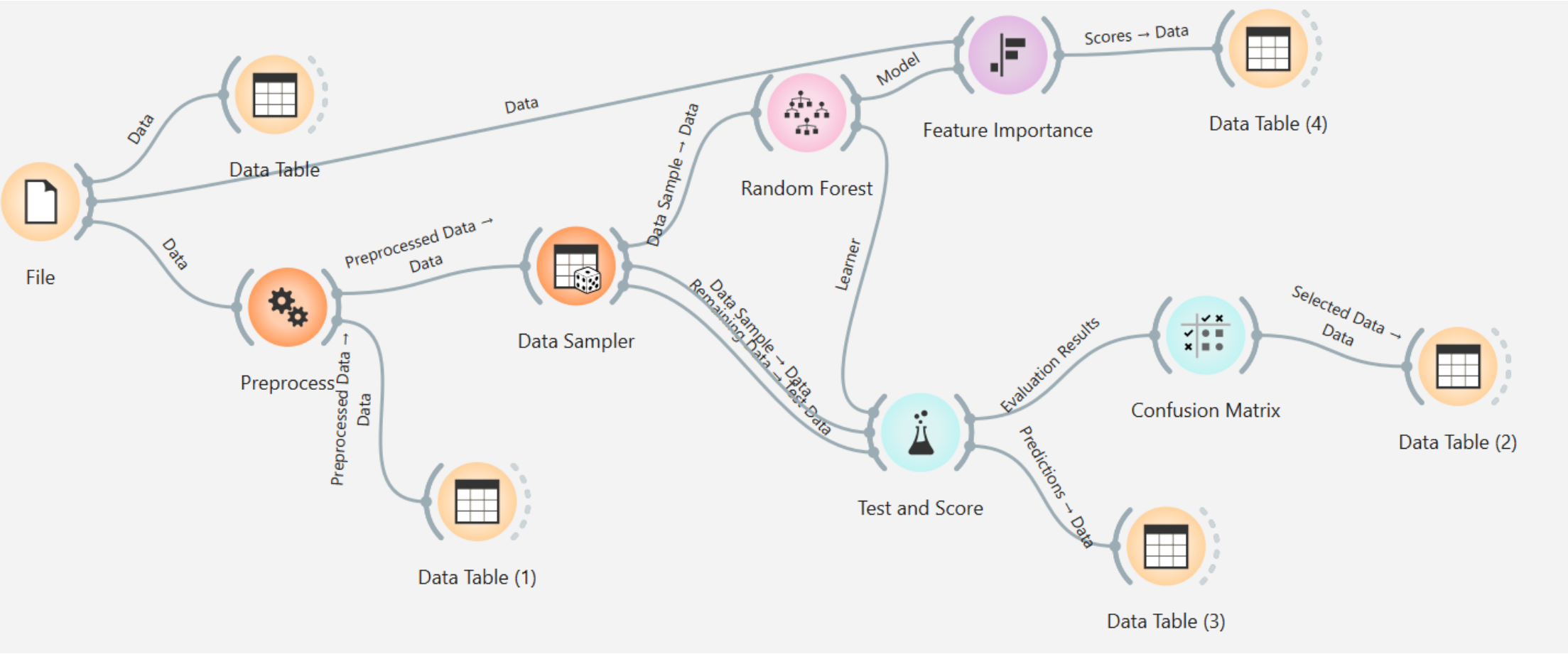
Sample ID	Bootstrap Sample 1	Bootstrap Sample 2	Bootstrap Sample 3	Bootstrap Sample 4
1	8	1	7	3
2	7	6	7	5
3	4	4	5	7
4	7	2	8	9
5	4	3	8	1
6	2	3	1	2
7	6	6	3	3
8	10	3	9	5
9	10	9	1	9
10	9	10	1	1

Random Forest

- An ensemble of Decision Trees, generally trained via the bagging method
- Typically with max_samples set to the size of the training set
- Introduces extra randomness while growing the trees
  - searches for the best feature among a random subset of features
- Result in great tree diversity
- A more general model

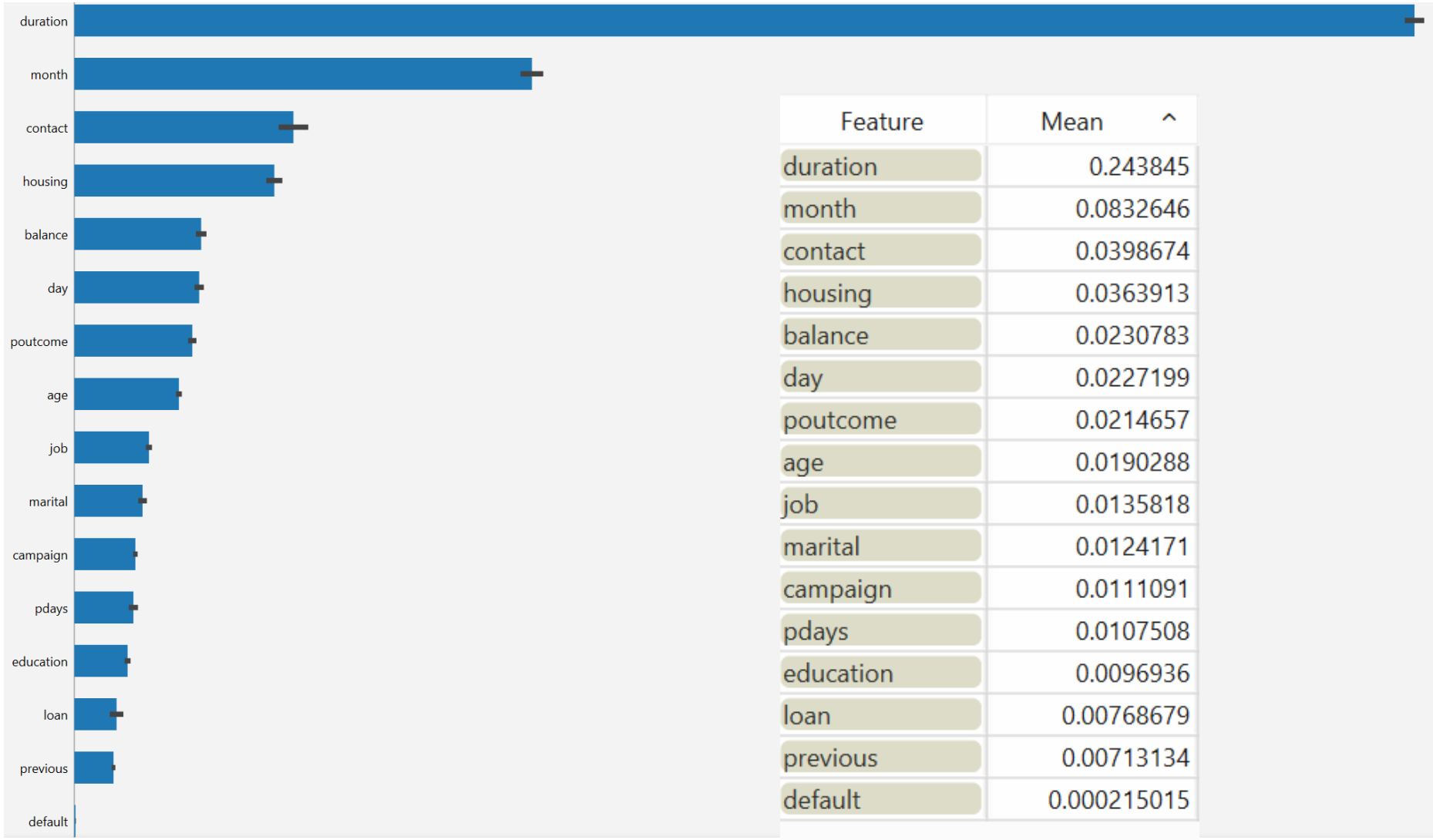


LAB: RANDOM FOREST CLASSIFIER



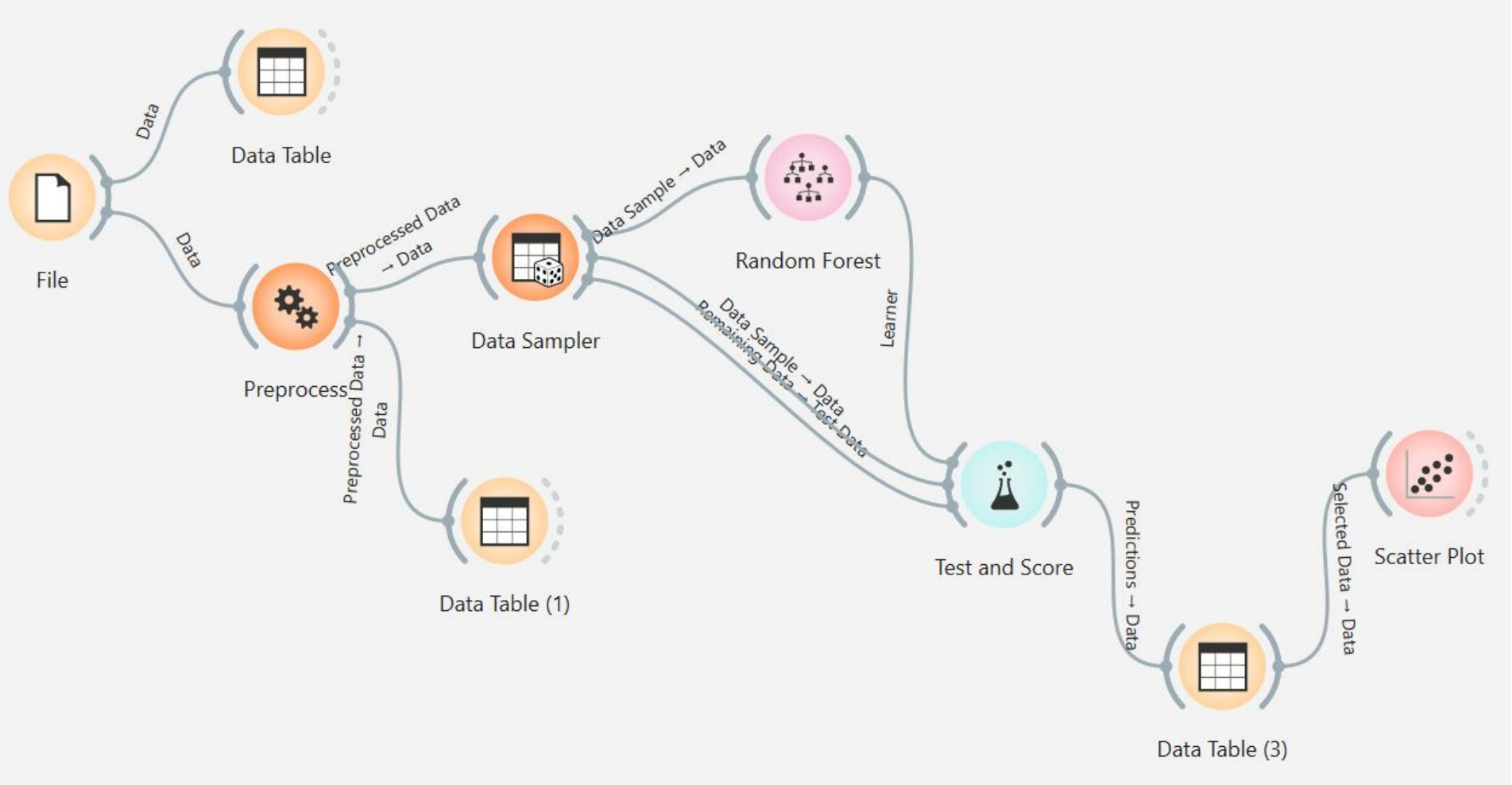
Feature Importance of Random Forest

Bank Marketing Campaign



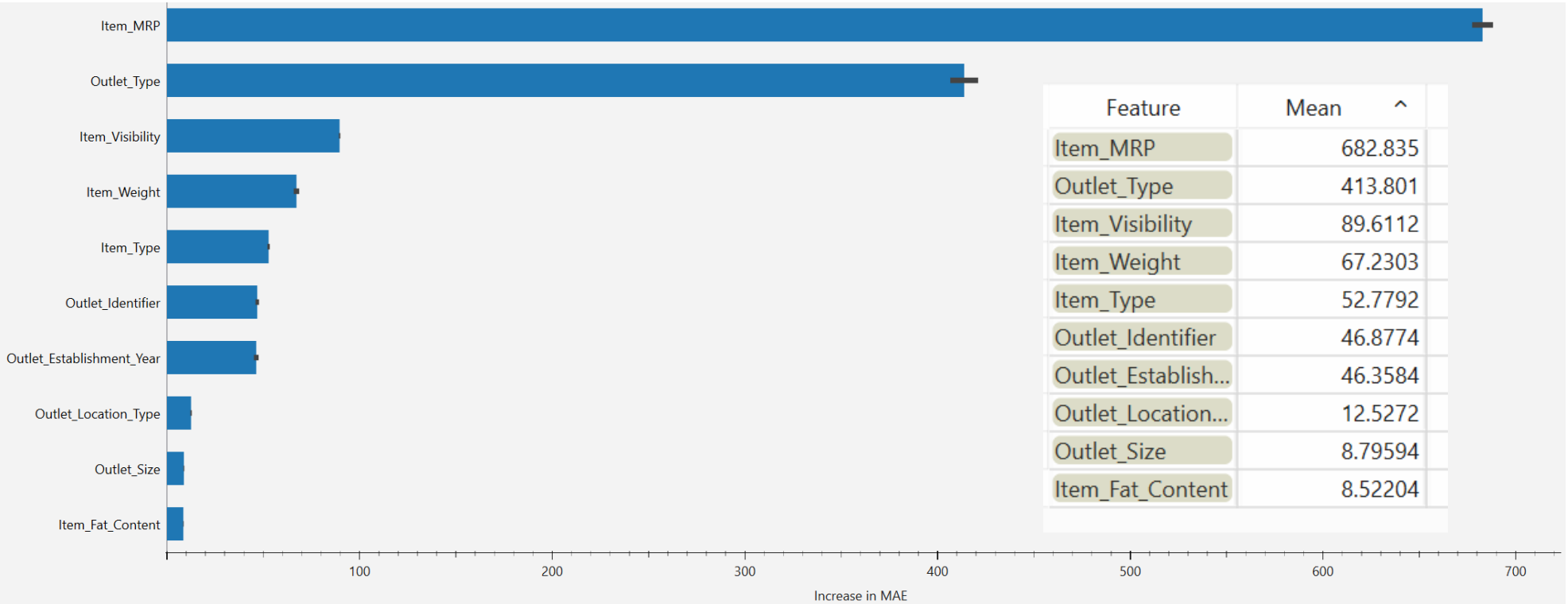
Feature	Mean
duration	0.243845
month	0.0832646
contact	0.0398674
housing	0.0363913
balance	0.0230783
day	0.0227199
poutcome	0.0214657
age	0.0190288
job	0.0135818
marital	0.0124171
campaign	0.0111091
pdays	0.0107508
education	0.0096936
loan	0.00768679
previous	0.00713134
default	0.000215015

LAB: RANDOM FOREST REGRESSOR



Feature Importance of Random Forest

Supermarket Sales Forecast



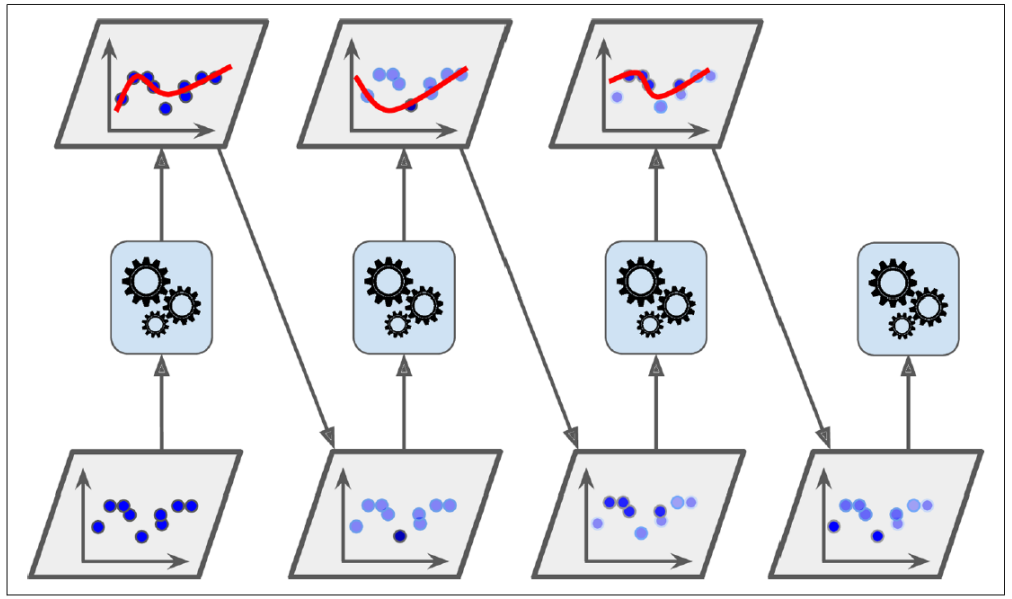
Feature	Mean
Item_MRP	682.835
Outlet_Type	413.801
Item_Visibility	89.6112
Item_Weight	67.2303
Item_Type	52.7792
Outlet_Identifier	46.8774
Outlet_Establish...	46.3584
Outlet_Location...	12.5272
Outlet_Size	8.79594
Item_Fat_Content	8.52204

Boosting

- Combine several weak learners into a strong learner
- General idea is to train predictors sequentially, each trying to correct its predecessor
- AdaBoost and Gradient Boosting
  - XGBoost (Extreme Gradient Boosting): Extremely fast, scalable and portable
- If overfitting to the training set
  - Reduce the number of estimators/predictors
  - more strongly regularizing the base estimator

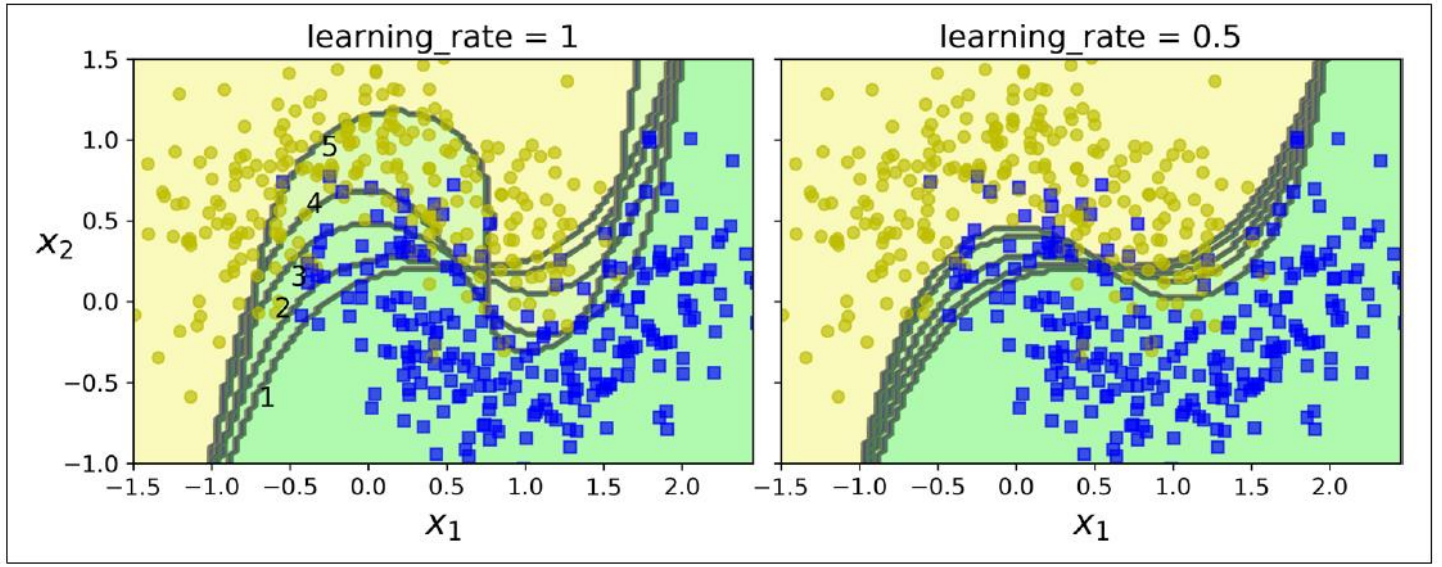
AdaBoost

- Train a base classifier
- Use it to make predication on training set
- Increases the relative weight of misclassified training instances
- Then train a 2nd classifier, using the updated weights, and again makes predictions, updates the training instance weights, and so on



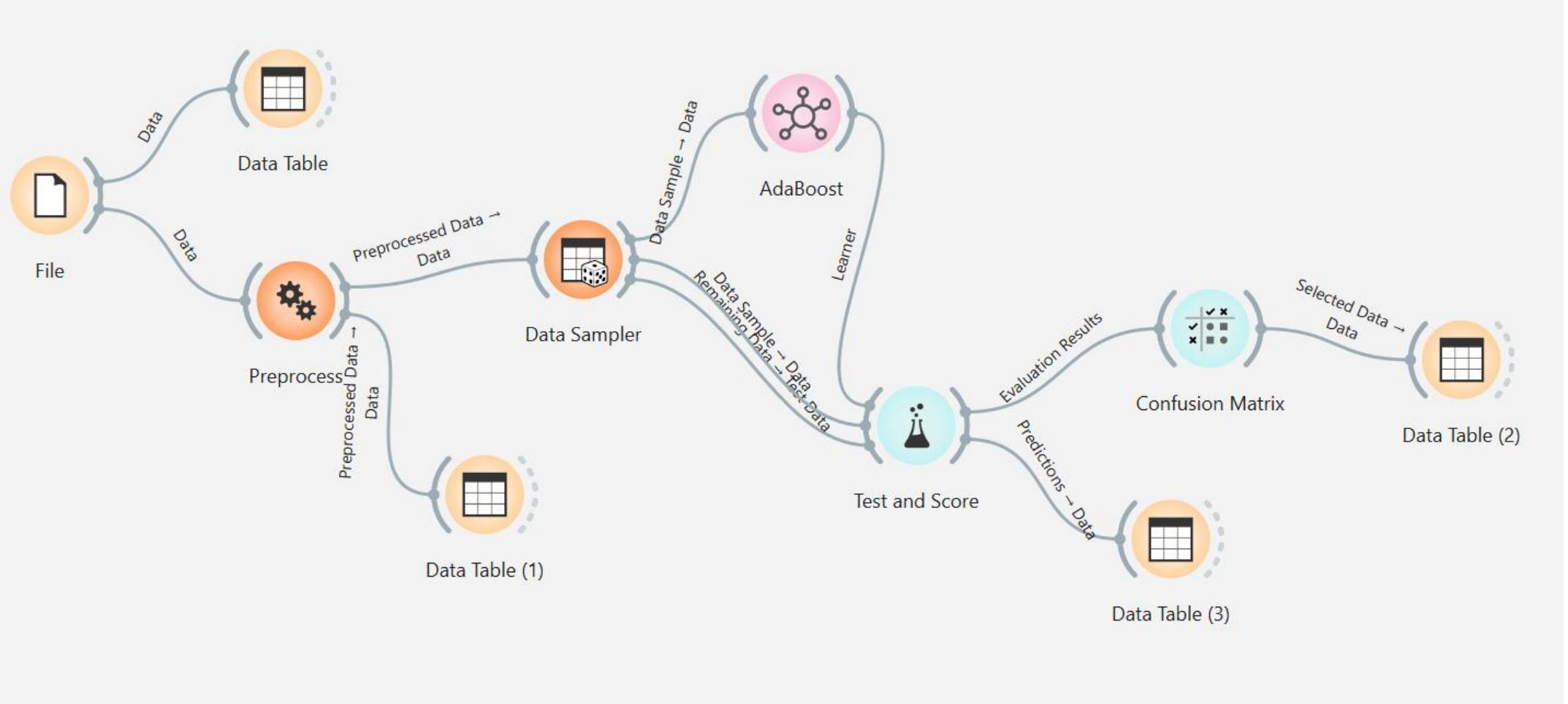
- After all predictors are trained, the ensemble makes predictions like bagging, except that predictors have different weights depending on their overall accuracy on the weighted training set.

In AdaBoost, we use learning_rate to control how fast/slow the misclassified instance weights are boosted….

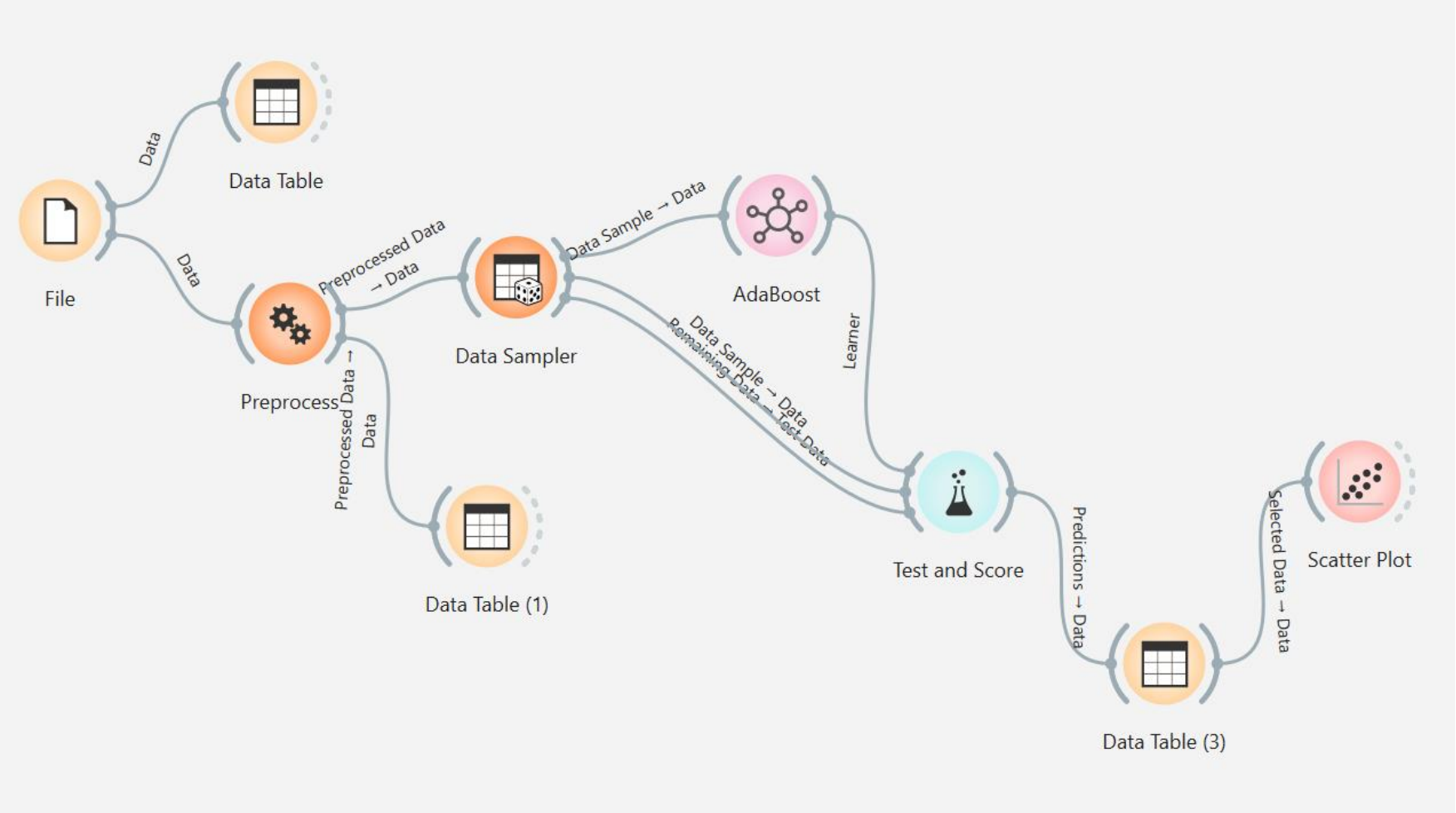


Decision boundaries of consecutive predictors in AdaBoost, with different learning_rate

LAB: ADA BOOST CLASSIFIER



LAB: ADA BOOST REGRESSOR

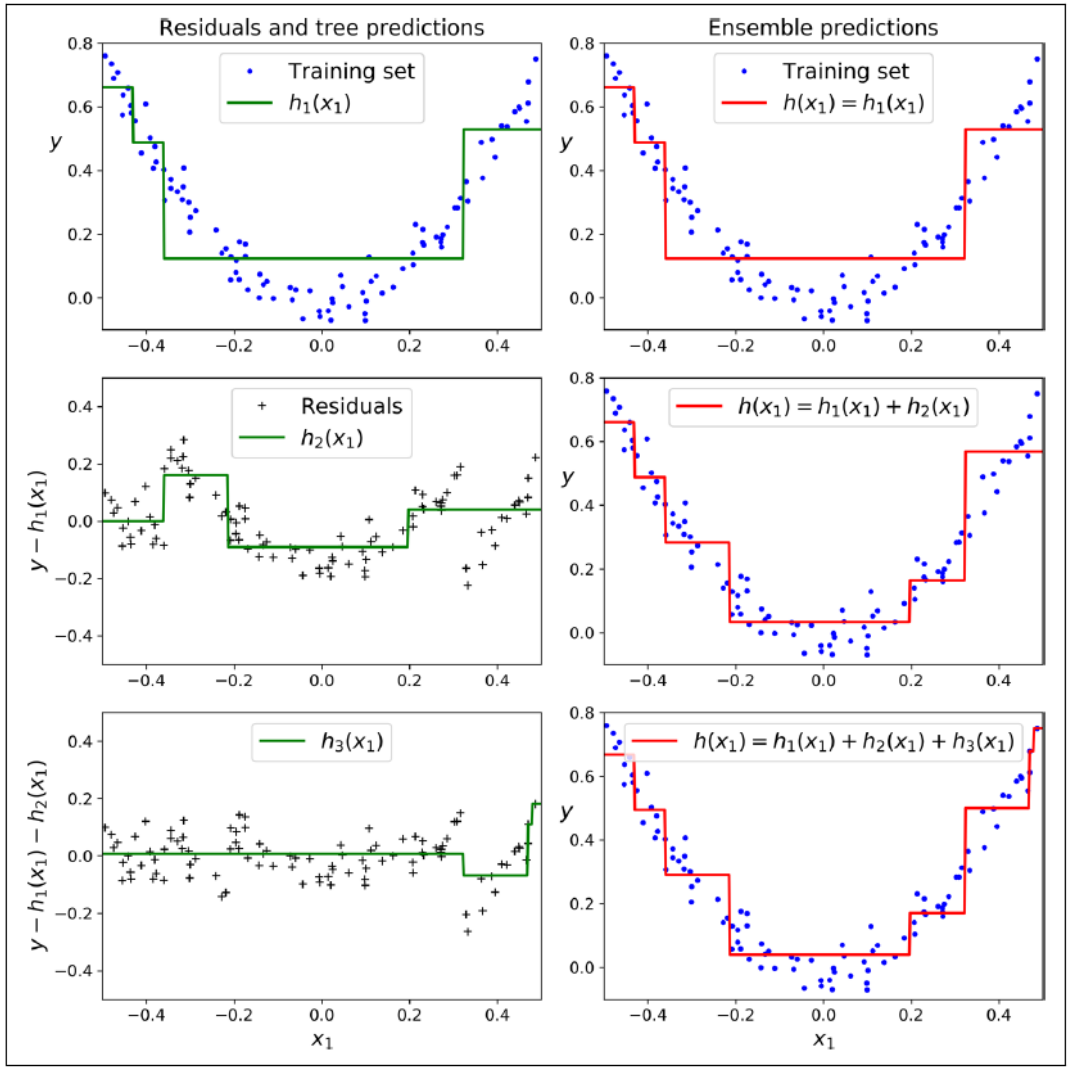


Gradient Boosting

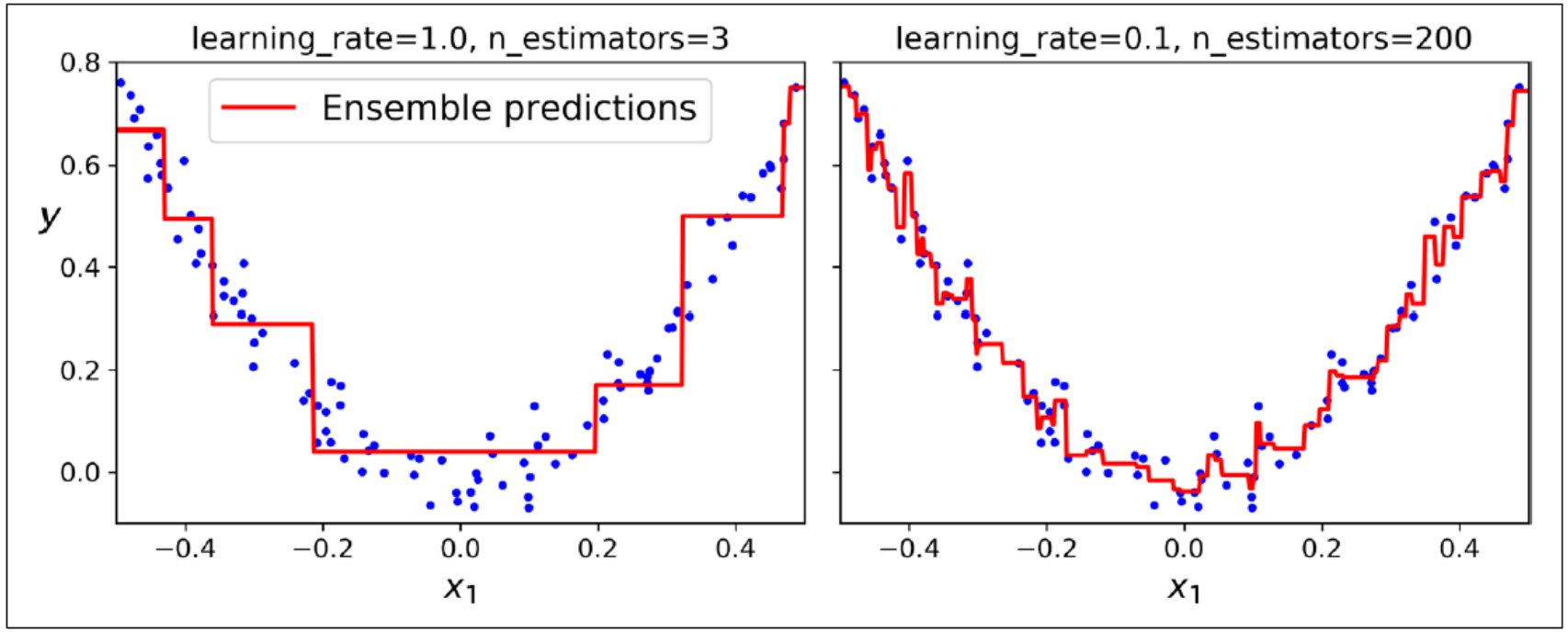
1st predictor is trained normally

2nd predictor is trained on 1st predictor's residuals

3rd predictor is trained on 2nd predictor's residuals



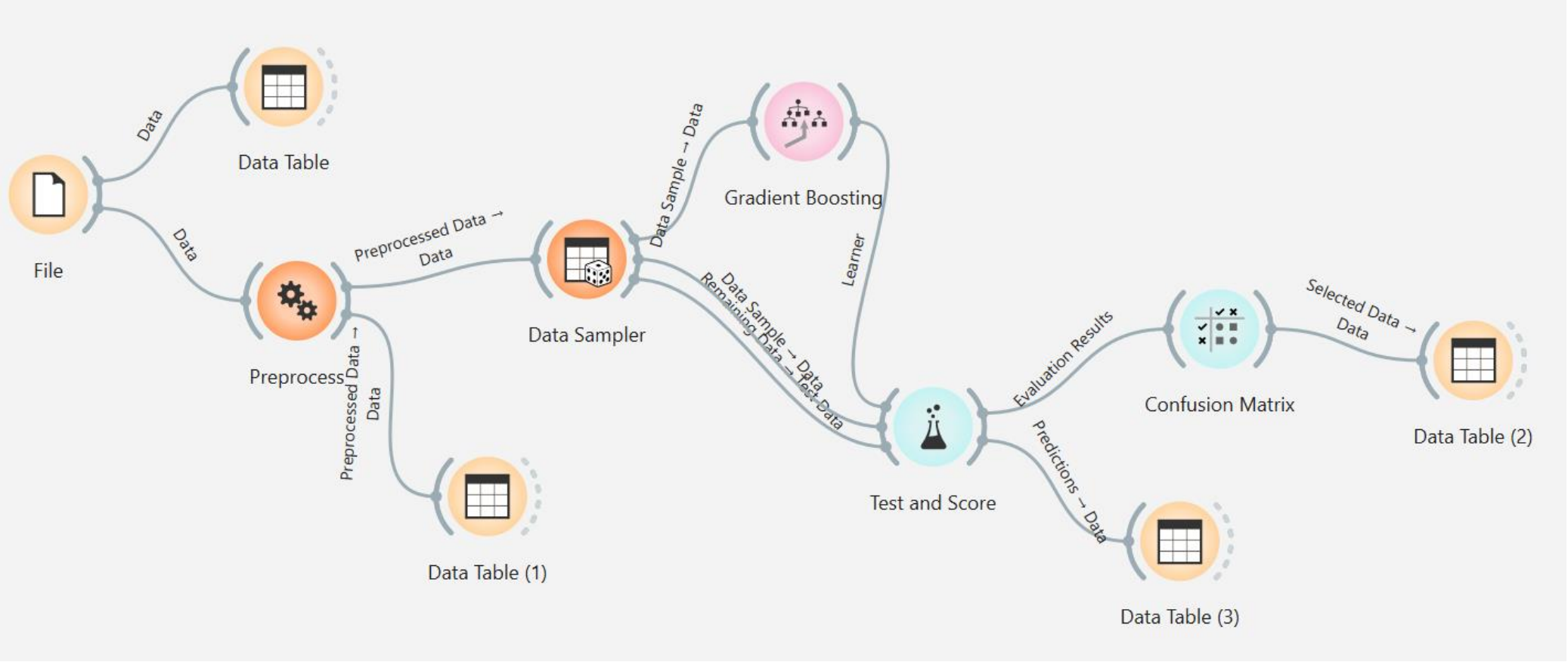
In Gradient Boosting, the learning_rate hyperparameter scales the contribution of each tree. If you set it to a low value, you will need more trees in the ensemble to fit the training set.



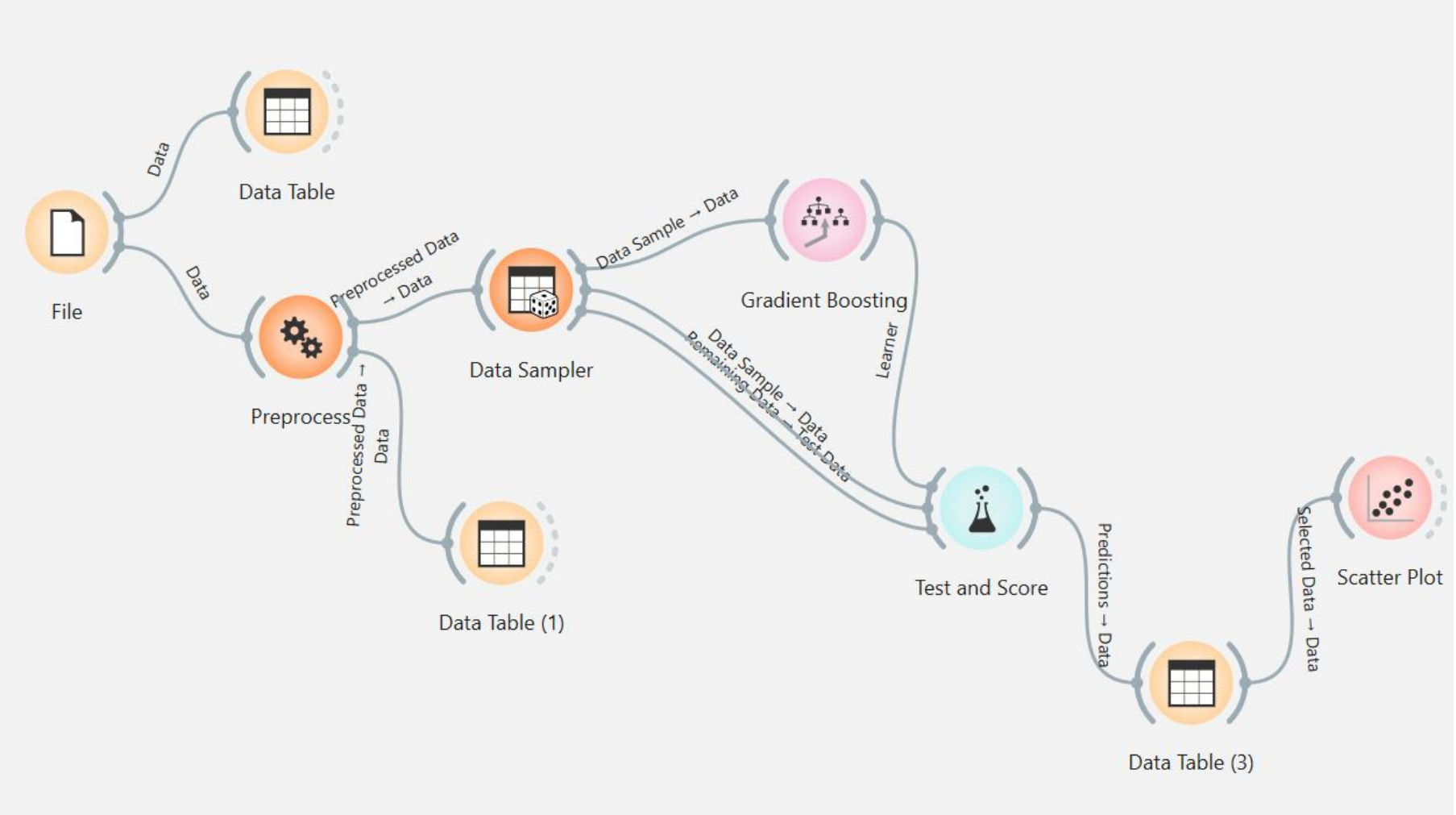
Not Enough Trees

Too Many Trees

LAB: GRADIENT BOOSTING CLASSIFIER



LAB: GRADIENT BOOSTING REGRESSOR



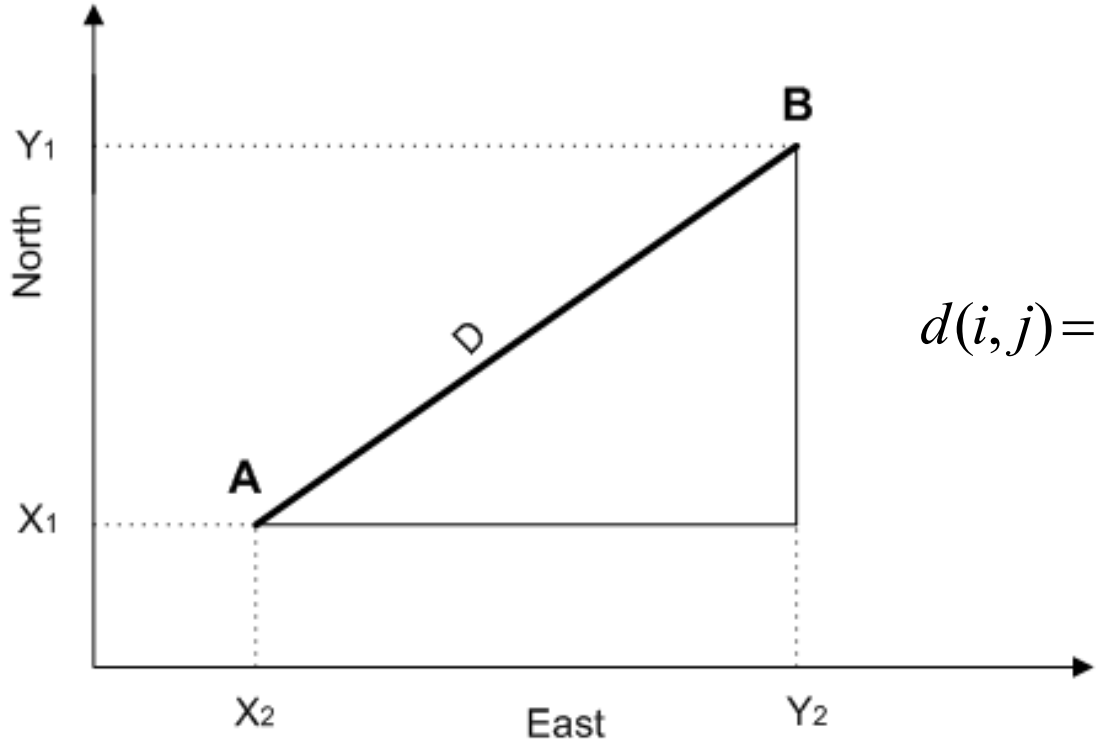
Part III: Unsupervised Learning

Clustering: grouping similar entities together

Basic elements in a Clustering solution

1. A sensible measure of similarity
2. An effective clustering algorithm
3. A good metric to evaluate cluster quality.

Clustering – Measure of Proximity



$$d(i,j)=\sqrt{(|x_{i_1}-x_{j_1}|^2+|x_{i_2}-x_{j_2}|^2+...+|x_{i_p}-x_{j_p}|^2)}$$

Euclidean distance D between two points A and B

- Preferred mode of measurement
- Measures direct distance between two points

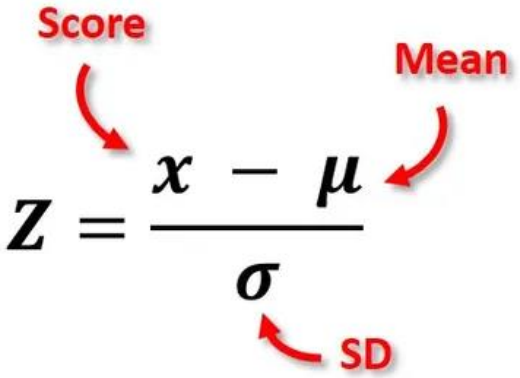
Clustering – Measure of Proximity

- Feature / Attribute scaling:

1. Min-max / Division-by-range normalization:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

2. z-score

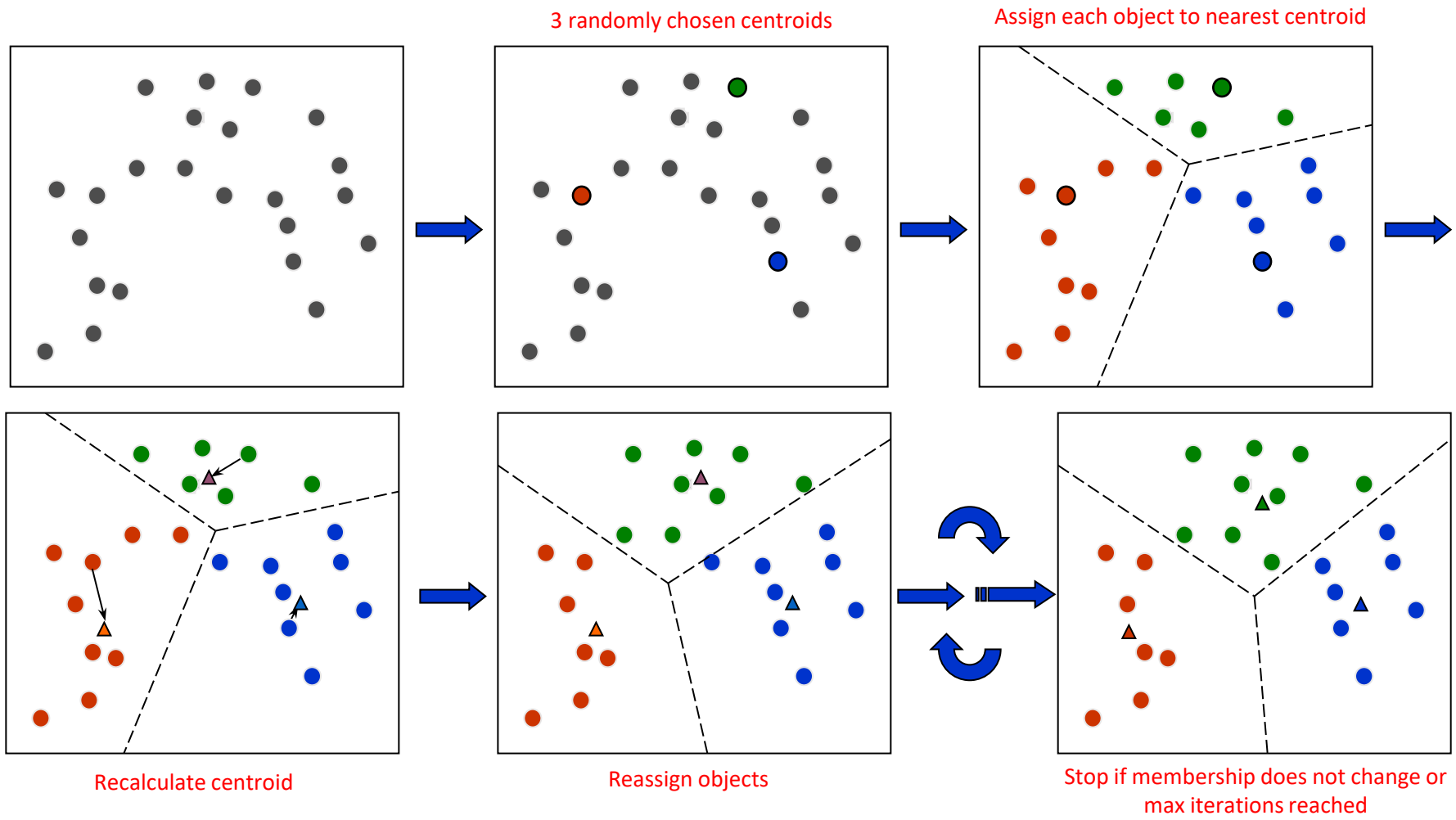


1. K-Means Clustering

k-means Clustering

- Steps to perform k-means clustering:
  1. Define the number of clusters (k).
  2. Choose k data points randomly to serve as the initial centroids for the k clusters.
  3. Assign each data point to the cluster represented by its nearest centroid.
  4. Find a new centroid for each cluster by calculating the mean vector of its members.
  5. Undo the memberships of all data objects. Repeat steps 3 to 5 until cluster membership no longer changes.

k-means Clustering - Illustration



k-means Clustering

- Strengths:
  - Algorithm is efficient and easy to implement

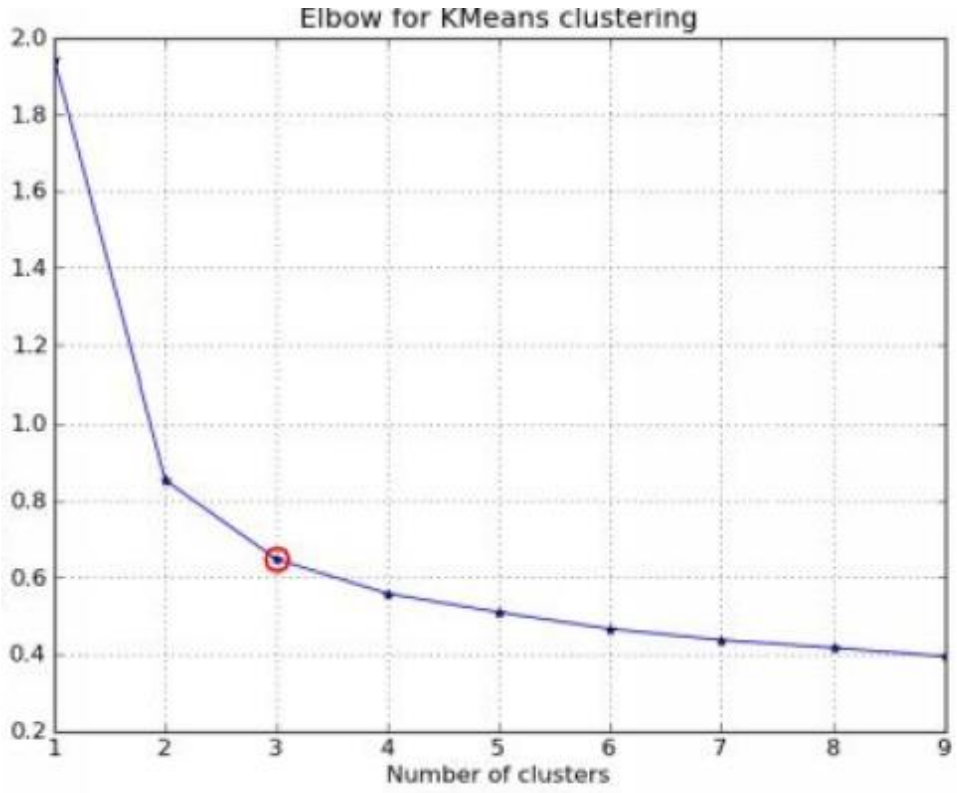
- Weaknesses:
  - May not know what the value of k should be beforehand
  - Sensitive to the choice of initial k centroids: clusters tend to converge to a local optimum solution
  - Sensitive to noise

k-means Clustering

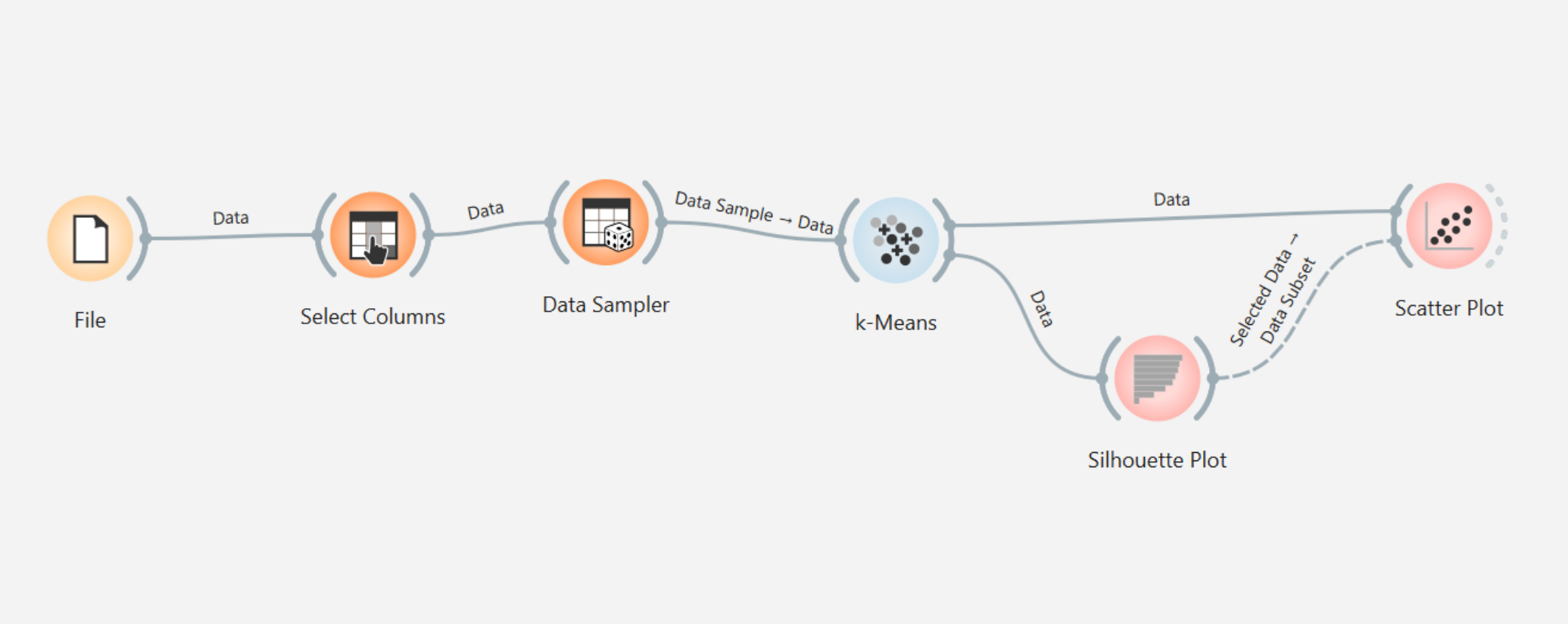
- Determine the value of k using Elbow Method
    - Sum of Squared Errors (SSE) at each number of clusters is calculated and graphed
    - Look for a change of slope from steep to shallow (an elbow) to determine the optimal number of clusters
    - This method is inexact, but still potentially helpful

k-means Clustering

- Elbow Method
  - SSE measures the distances from each data object to the cluster centroid, take a squared value on the distances and then sum them up together.
  - We can see the larger K the lower SSE, but beyond certain point (e.g. K=3) adding more clusters will not reduce SSE significant, therefore K=3 is the desired K.



LAB: K MEANS CLUSTERING



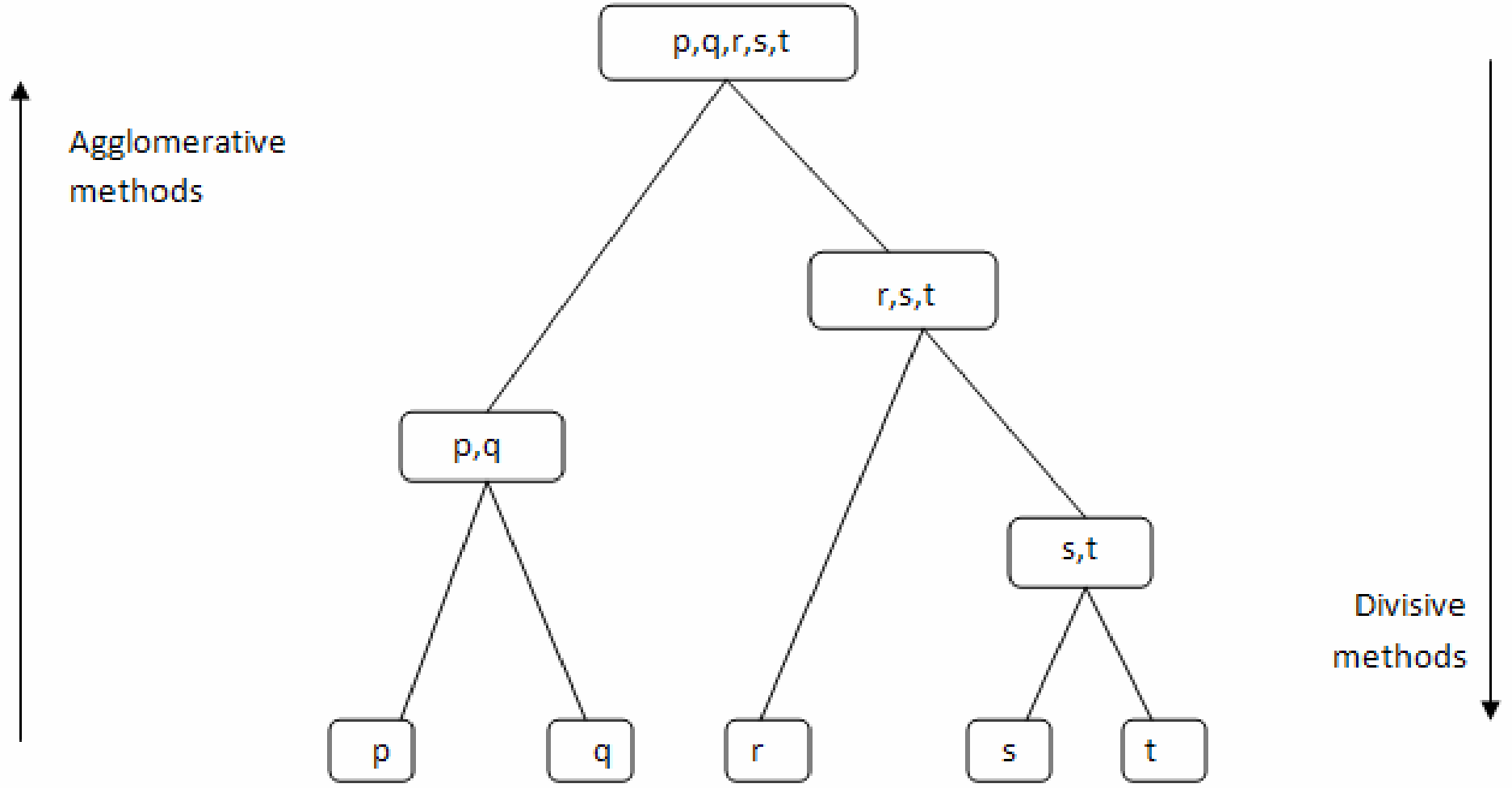
2. Hierarchical Clustering

Hierarchical Clustering

- Divisive clustering algorithm
  - Begin with one large cluster and proceed to split into smaller clusters of items that are most dissimilar

- Agglomerative clustering algorithm
  - Start with each data point as a cluster, and with each step, combine clusters until there is only one large cluster

Hierarchical Clustering



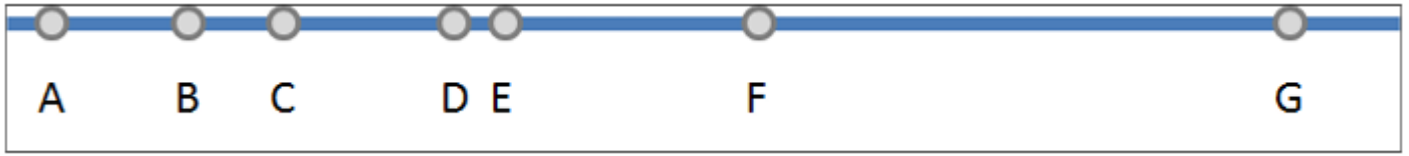
Agglomerative Clustering

- Steps to perform Agglomerative clustering:

1. Let each data point be a cluster.
2. Compute the proximity matrix of all clusters.
3. Merge the two clusters that are the most similar.
4. Repeat steps 2 and 3 until only a single cluster remains.

Agglomerative Clustering

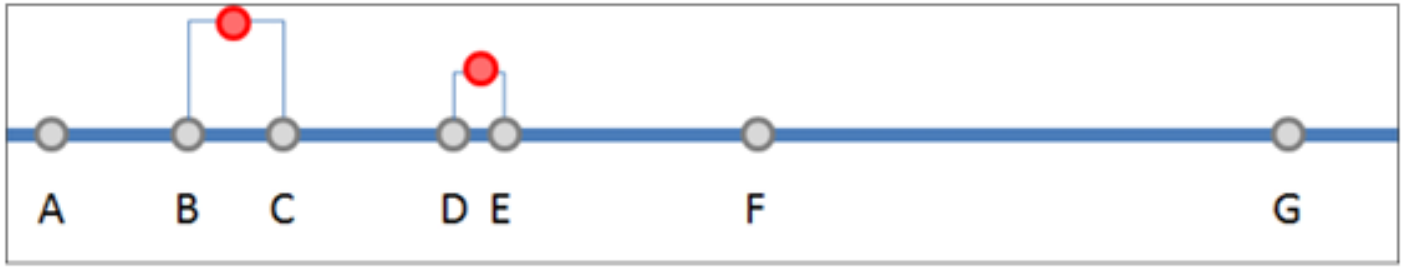
Visualise 7 data points in a straight line



Calculate distance between each point. D & E are closest, so group into a cluster

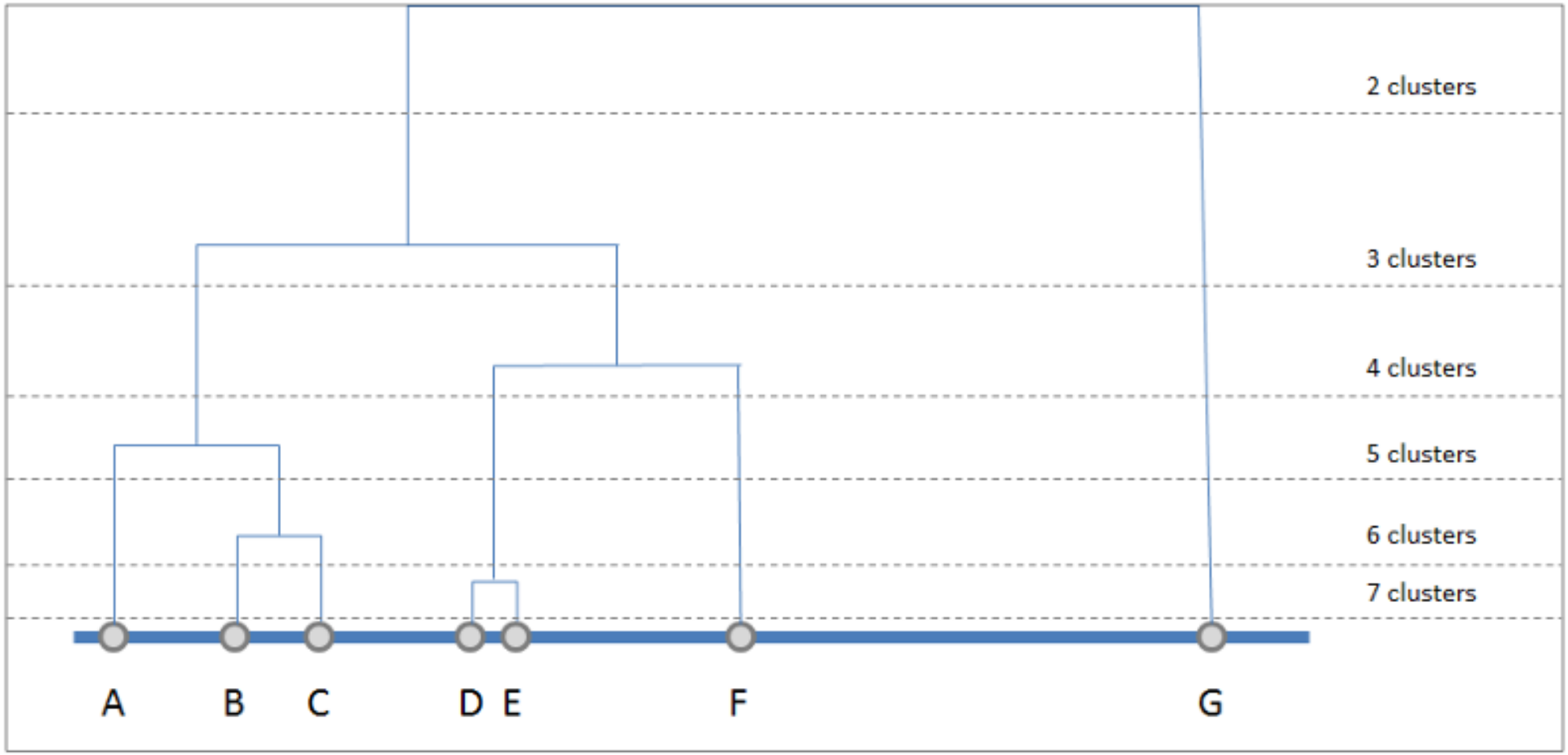


Substitute D & E for their mean point. Look for next 2 pts with smallest distance



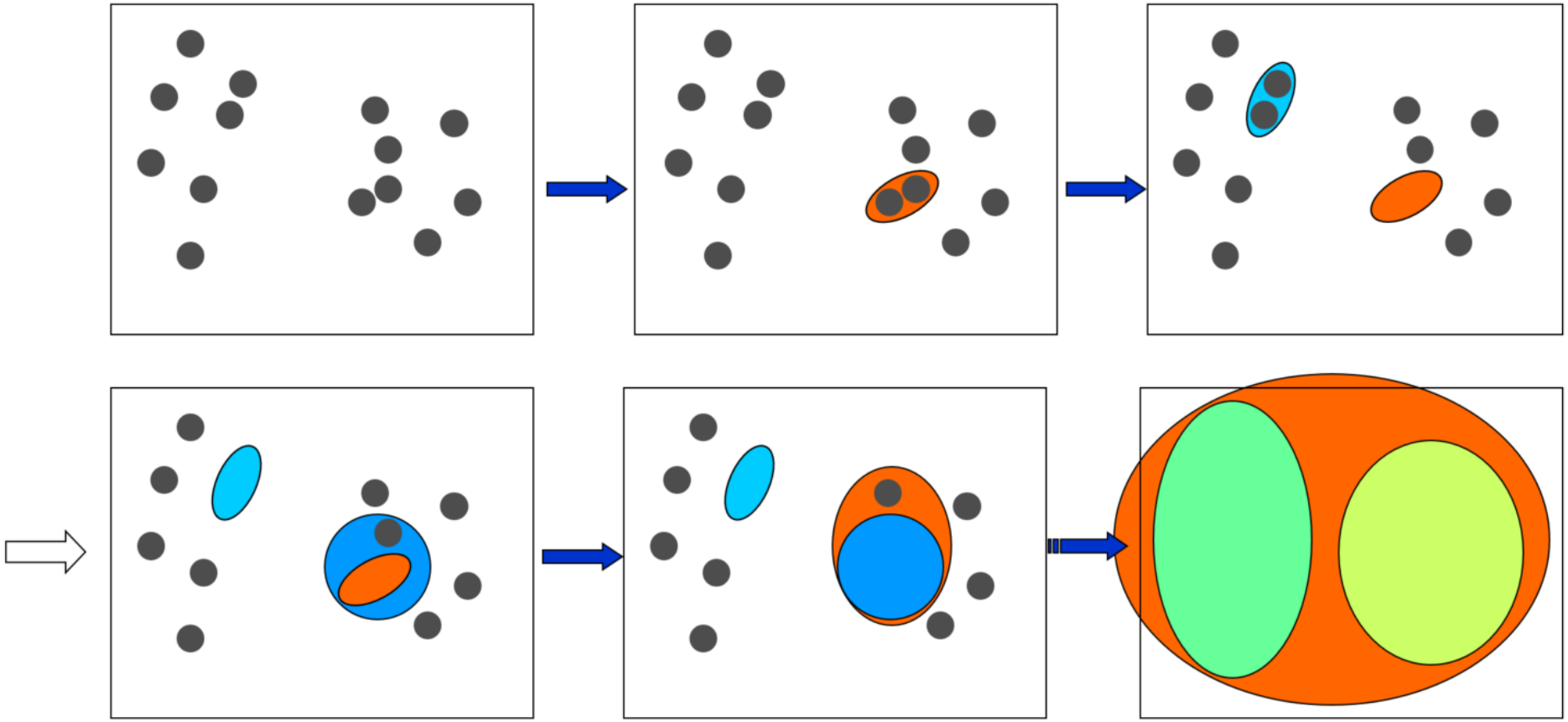
Agglomerative Clustering

Continue iterating until all observations in the dataset are grouped



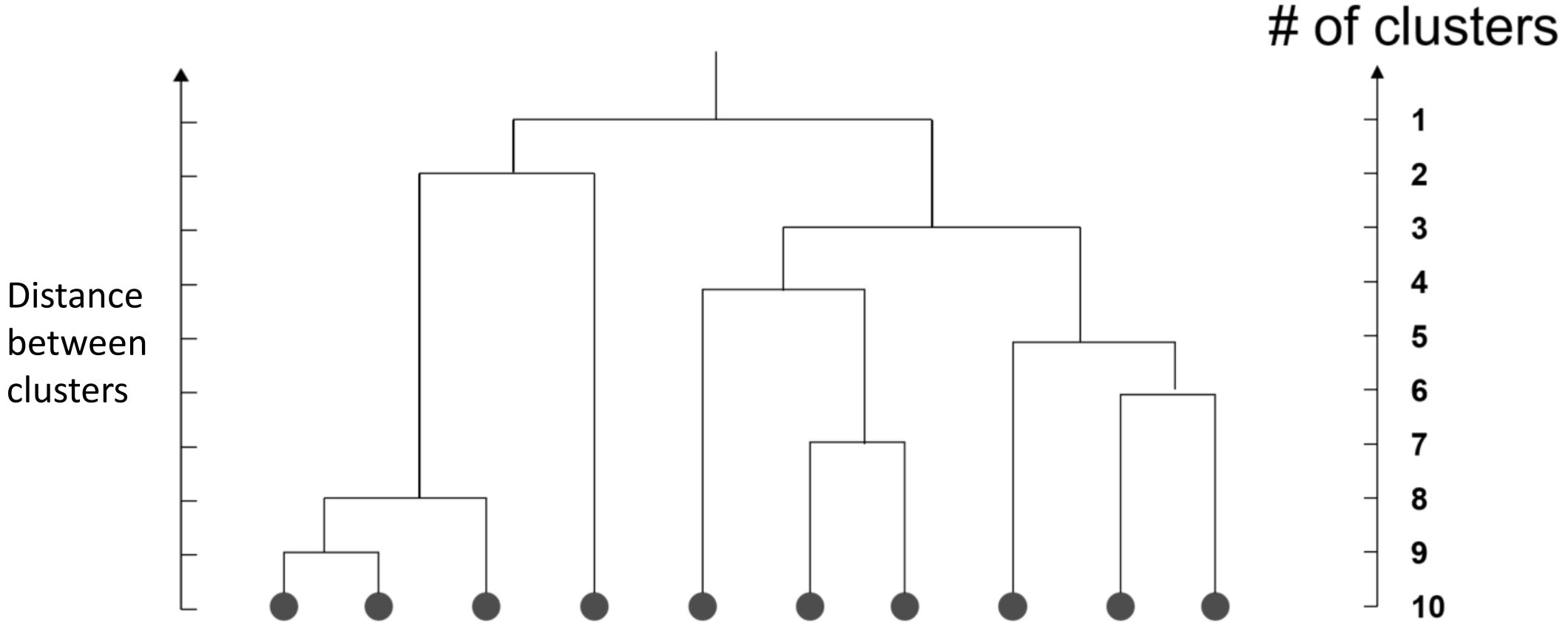
Agglomerative Clustering

- Illustration of Agglomerative Clustering:



Agglomerative Clustering

- Illustration of a dendrogram:



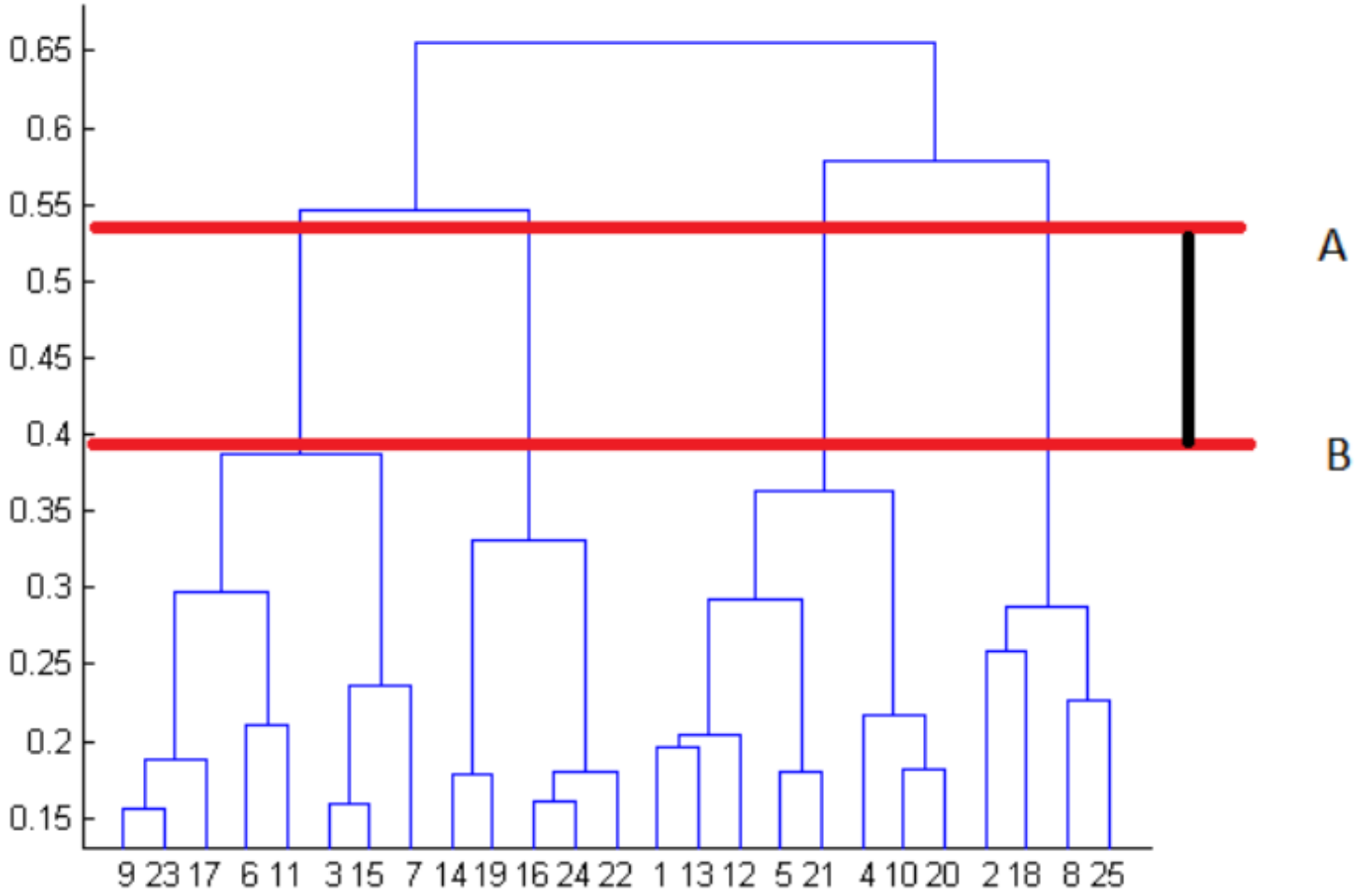
Agglomerative Clustering

- The optimal number of clusters is determined by the number of vertical lines cut by a horizontal line in a dendrogram, with the vertical lines traversing the maximum distance vertically without intersecting a cluster

- However, there is still a lot of subjectivity involved in the choice of clusters and the process may need to be repeated a few times or combined with other clustering methods to get the best output.

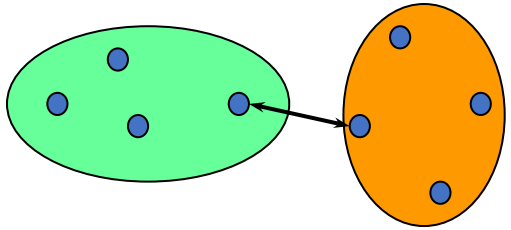
Agglomerative Clustering

- Example of identifying optimal number of clusters

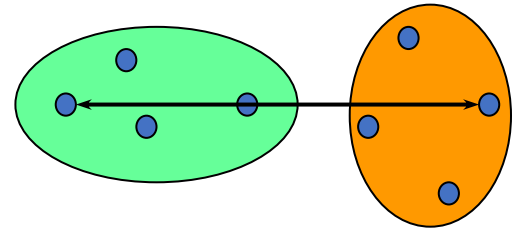


Agglomerative Schemes

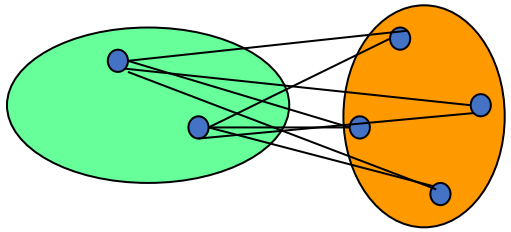
- Single link:
  - Distance between two closest points



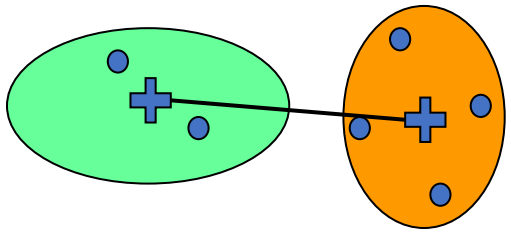
- Complete link:
  - Distance between two farthest points



- Group average:
  - Average of all pair-wise distances



- Centroids:
  - Distance between the centroids



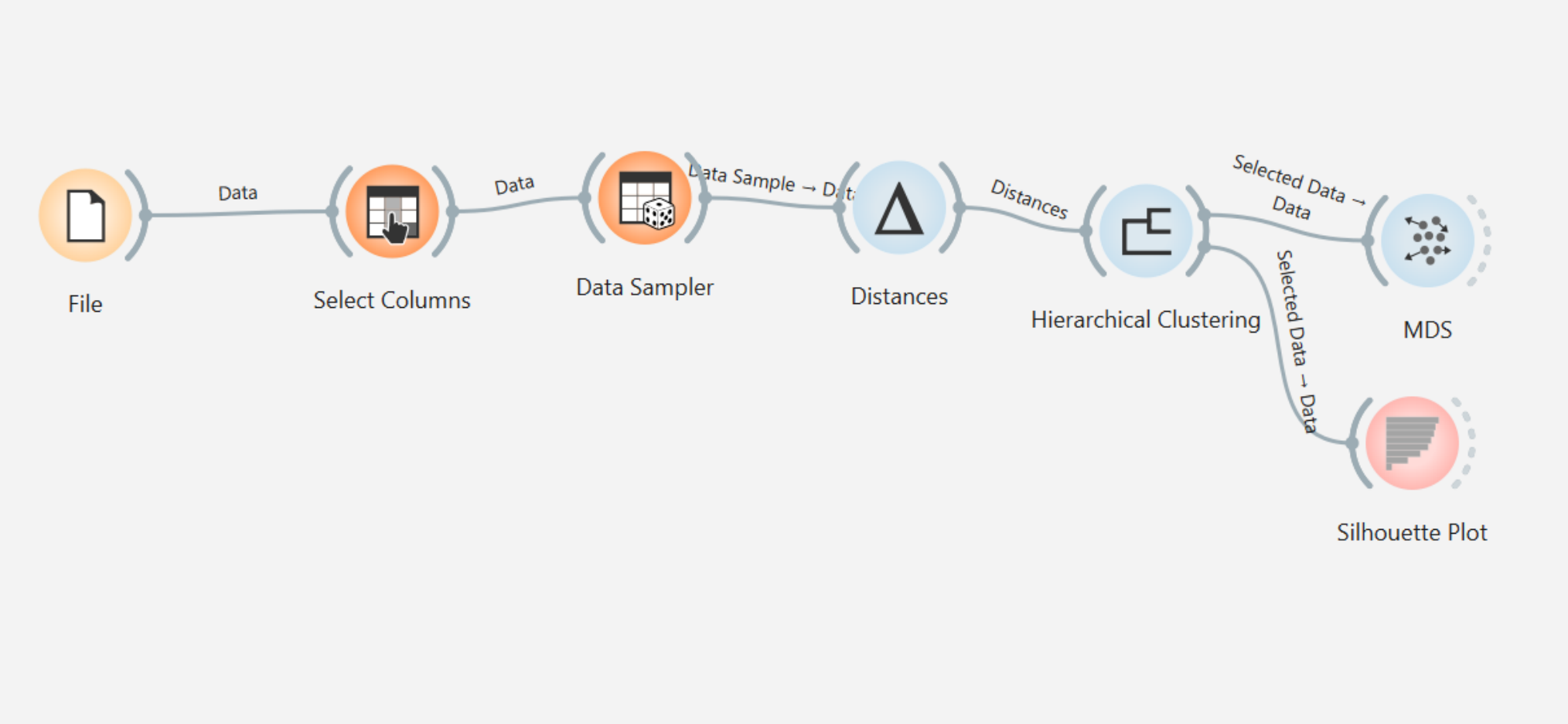
Ward (scikit-learn default):
picks up the two clusters to merge such that the variances (i.e. SSE) within these two clusters increases the least

Agglomerative Clustering

- Strengths
  - Deterministic results
  - Multiple possible versions of clustering
  - No need to specify the value of a k beforehand

- Weaknesses
  - Does not scale up for large data sets
  - Cannot undo membership like the K-means

LAB: HIERARCHICAL CLUSTERING

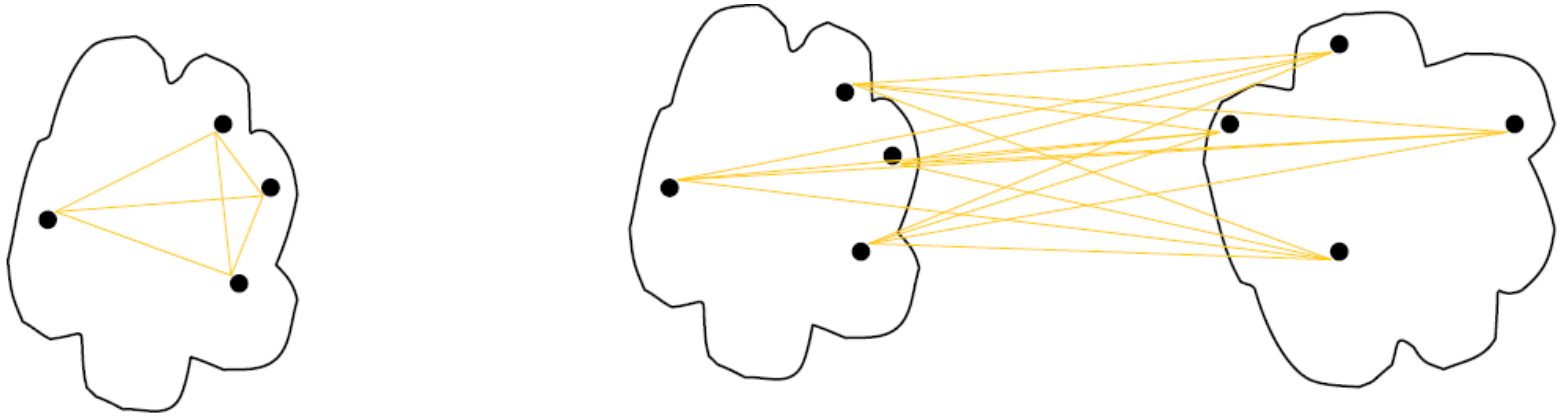


3. Clustering Evaluation

Cluster Evaluation

- Cohesion
  - The distances between points within a cluster

- Separation
  - The distances between points and the nearest cluster



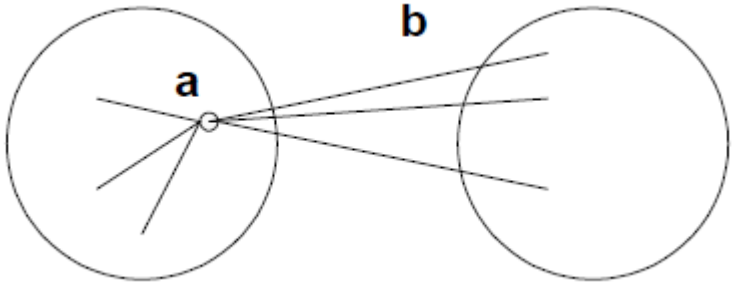
cohesion

separation

Custer Evaluation: Silhouette Score

- Calculation of Silhouette Coefficient
  - For each point / sample compute two scores:
    - cohesion (a) : The mean distance between a point and all other points in the same cluster
    - separation (b) : The mean distance between a point and all other points in the nearest cluster
  - Silhouette Coefficient (s)

$$s = \frac{b - a}{max(a, b)}$$



Cluster Evaluation

- Interpretation of Silhouette Coefficient
  - The best value is 1 and the worst value is -1
  - Values near 0 indicate overlapping clusters
    - s=0 if the cluster only has one point
  - Negative values generally indicate that a point has been assigned to the wrong cluster, as a different cluster is more similar (i.e. shorter distance)

- Silhouette Score
  - The mean Silhouette Coefficient over all points / samples

Custer Evaluation: Example

SubjectID	Body Height	Body Weight	Cluster Tag	a	b	Silhouette Coefficient
s1	125	61	2	24.50	55.67	0.56
s2	178	90	1	9.50	68.67	0.86
s3	180	83	1	10.00	66.00	0.85
s4	167	85	1	12.50	57.33	0.78
s5	135	40	2	19.50	61.00	0.68
s6	120	35	2	21.00	75.33	0.72

Silhouette Score

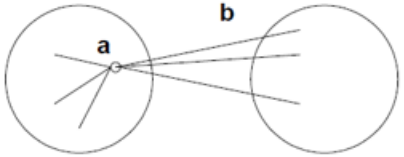
0.742259854



Step 1: Generate Distance Matrix

	s1	s2	s3	s4	s5	s6
s1	0					
s2	60	0				
s3	59	7	0			
s4	48	12	13	0		
s5	23	66	62	55	0	
s6	26	80	77	69	16	0

Step 2: For each point/sample compute two scores (a and b) in the Table on the left.



Step 3: Compute Silhouette Coefficient (s) and Silhouette Score (i.e. the average of the Sihouette Coefficients.

$$s = \frac{b - a}{max(a, b)}$$

Cluster 1 (with higher Sihouette values) is a better quality cluster than Cluster 2.

Cluster Evaluation: Summary

- Cluster Quality:
  Good-quality clusters are as different from each other as possible and contain data objects that are as similar as possible (Silhouette Coefficients closer to 1)

- Cluster Interpretability:
  Clustering results that are difficult to interpret have little practical use. Data points that belong to a specific cluster should have features in common. These should help in the interpretation of the cluster

References

Aurélien Géron (2019). *Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow*, 2nd edition.

Andreas C. Müller & Sarah Guido (2019). *Introduction to Machine Learning with Python*.

Machine Learning (Coursera Course) https://www.coursera.org/learn/machine-learning

Kaggle https://www.kaggle.com/

THE END